Are Frontier VLM Agents Ready to Be Robot Generalists?

An Empirical Study with the Embodied Agent Arena

Haojian Huang[1,3], **Pukun Zhao**[3], **Zexi Li**[2,3], **Yehang Zhang**[1,3], **Yangkai Wei**[3], **Wenqian Li**[3], **Han Yang**[3], **Kaiwen Zhou**[3], **Ying-Cong Chen**[1,3,†], **Yinchuan Li**[3,†]

[1] HKUST (GZ) [2] The Chinese University of Hong Kong [3] Knowin AI

Abstract. A robot generalist must turn scene understanding into actions that satisfy a complete goal. How far do frontier vision-language models (VLMs) meet this requirement? We introduce Embodied Agent Arena to evaluate ten VLM agents across Geometry, Spatial Reasoning, Affordance, Task Planning, and Manipulation. Its 1,000 cases combine 32 established sources with GeoProbe, our 168-case benchmark for camera, motion, depth, and scale estimation. A minimal harness leaves task decisions to the model through source-native interfaces. Our results reveal complementary model strengths but a shared gap between local competence and complete goal satisfaction. Better geometric estimates and region overlap do not establish reliable interaction, while successful early actions can fail to compose into a complete procedure. These failures are costly: unsuccessful attempts consume most planning tokens. Execution traces further show that agents often resume code execution after an error without completing the task. Across the evaluated tasks, the central bottleneck is sustaining progress through the state changes that a complete goal requires.

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1 Introduction

A robot generalist must turn scene understanding into actions that satisfy a complete goal. Locating an object, selecting a usable contact, and executing a grasp are intermediate achievements: the object must still reach its destination in the required state. Vision-language models (VLMs) increasingly address spatial relations (Ma et al., 2024, Yang et al., 2024) and action-relevant grounding (Wang et al., 2025a, Song et al., 2024). Learned visuomotor policies connect observations to robot actions (Kim et al., 2024, Black et al., 2024), while language-model controllers select skills or generate executable programs (Ahn et al., 2022, Liang et al., 2022). Perception support and visual action rehearsal further extend these systems (Fu et al., 2026, Guo et al., 2026, Li et al., 2026b, Zhang et al., 2026a). As execution competence expands, the central question becomes: *Are frontier VLM agents ready to be robot generalists?*

Answering this question requires identifying which requirements remain unsatisfied after an agent has made useful progress. A trajectory can closely follow a reference yet miss its endpoint; reaching a seat and issuing a sitting action need not establish the required sitting

[†] Corresponding authors.

Figure 1. Overview of Embodied Agent Arena. Each row illustrates six tasks in one capability domain, spanning camera and scene estimation, spatial relations, functional grounding, goal-directed planning, and robot manipulation. Fixed-input probes test estimates and judgments; interactive episodes test goal completion. Spatial-tracing overlays show reference paths and a 3D target region.

state. Average accuracy obscures these decisive conditions, while a terminal failure alone hides the capabilities already demonstrated. Measuring local performance alongside complete outcomes can therefore explain more than either measure in isolation: it distinguishes an inaccurate estimate from an execution that fails at a specific goal condition.

We introduce Embodied Agent Arena, which evaluates ten VLM agents across Geometry, Spatial Reasoning, Affordance, Task Planning, and Manipulation (Figure 1). The arena combines established benchmarks with GeoProbe, which varies camera motion, object motion, and scale references to diagnose geometric estimation. A minimal harness preserves source interfaces while leaving task decisions to the model. Across the five domains, metric precision, functional grounding, and native goal completion expose different requirements of embodied generalism. Within interactive tasks, recorded actions and feedback reveal whether further computation advances the environment toward the goal.

The results reveal complementary capabilities whose strengths do not consistently extend to complete execution. Models succeed on partly different tasks, yet their combined successes leave shared failures in interaction and state change. These unresolved tasks are also expensive: unsuccessful episodes absorb most Planning tokens. Matched traces distinguish useful computation from stalled execution. Hundreds of actions may achieve no subgoal, whereas repairing an early code error can restore progress and complete the task. The central diagnostic is therefore what each additional interaction accomplishes, alongside what the model can perceive or predict.

Our contributions are:

- **An evaluation centered on complete task requirements.** We distinguish local precision and grounding from complete task success across five domains.
- **A unified arena with a new geometric benchmark, GeoProbe.** The 1,000-case arena combines 32 established sources with 168 GeoProbe cases, retaining native operations and success criteria in a shared model-directed evaluation framework.
- **A diagnosis of the gap between local competence and complete execution.** Ten-model comparisons reveal complementary successes and shared failures; task traces locate missed endpoints, unsatisfied interaction states, and costly repetition without progress.

2 Related Work

Embodied evaluation has developed along two complementary directions. Control suites measure execution and task transfer (James et al., 2019, Mees et al., 2021, Liu et al., 2023). Varying environments and physical deployment broaden this evaluation to robustness and multiple dimensions of manipulation competence (Chen et al., 2025a, 2026). Visual diagnostics isolate spatial inference (Ma et al., 2024, Yang et al., 2024) and interaction grounding (Song et al., 2024, Qian & Fouhey, 2023). Scene and task question answering (Majumdar et al., 2024, Sermanet et al., 2023) extends this diagnostic view to embodied decisions (Li et al., 2024b, Qi et al., 2025). Scene-linked assessment connects grounding, relational inference, and action to distinguish stage competence from error propagation (Ruan et al., 2026). As agents undertake longer interactions, final success leaves important aspects of execution unresolved: adherence to constraints (Park et al., 2026), use of relevant history (Luo et al., 2026), and recognition of progress and recovery (Liu et al., 2026). These developments motivate measuring decisions within an episode alongside complete outcomes.

The transition from judgment to action depends on where control is represented. Learned policies map observations and instructions directly to actions (Brohan et al., 2023, Zitkovich et al., 2023), with broader task coverage motivating generalist policies (Ghosh et al., 2024, Kim et al., 2024, Black et al., 2024). Language-mediated control instead composes supplied skills (Ahn et al., 2022, Huang et al., 2023b) or executable programs (Liang et al., 2022, Singh et al., 2023); hybrid coordination selects between modular skills and learned policies from the current state (Zhou et al., 2026a). Explicit programs make dependencies (Fu et al., 2026, Du et al., 2025), geometric feedback (Huang et al., 2023a), and reusable interfaces (Li et al., 2026a) available for model-directed composition. Evaluation consequently shifts from the availability of individual actions to the controller's ability to coordinate a complete procedure.

Composition also requires evidence that remains useful as the scene changes. Visual grounding can fail under physical stress or misleading background cues (Li et al., 2025, Wu et al., 2026, Huang et al., 2026a), motivating methods that repair context and reason over evolving spatial evidence (Huang et al., 2026c, Zhao et al., 2026b,a). Demonstrations supply task context (Jiang et al., 2023, Huang et al., 2026b); retained histories preserve state (Zhang et al., 2026b); verification checks whether execution meets its conditions (Guo et al., 2024, Huang et al., 2026d). This support changes what the overall system can accomplish (Wang et al., 2026), making the supplied interfaces and feedback part of the evaluation design.

Broader benchmarks bring perception and goal-directed interaction into one evaluation (Yang et al., 2025a, Cheng et al., 2025), extending visual probes toward navigation (Ni et al., 2025,

Table 1. Evaluation coverage, diagnostic analysis, and sample scale. Fixed probes use supplied observations; region probes score image grounding; manipulation requires arm/gripper control; memory spans sessions. Calib./pose tests camera calibration or relative pose; Forms denotes task-type/condition breakdowns. Scale counts QA items or task episodes; NR: aggregate count not reported (Appendix G).

	Sample scale	Evaluation coverage					Diagnostic analysis	
Benchmark	Instances	Fixed	Region	Nav.	Manip.	Memory	Calib./pose	Forms
OpenEQA (2024)	1,636	✓	✗	✓	✗	✗	✗	✓
VLABench (2024)	NR	✓	✗	✗	✓	✗	✗	✓
BEAR (2025)	4,469	✓	✓	✗	✓	✗	✗	✓
EmbodiedBench (2025a)	1,128	✗	✗	✓	✓	✗	✗	✓
EmbodiedEval (2025)	328	✗	✗	✓	✗	✗	✗	✓
Embodied Arena (2025)	NR	✓	✓	✓	✗	✗	✗	✓
CaP-X (2026)	700/tier	✗	✗	✓	✓	✗	✗	✓
WorldLines (2026b)	331	✓	✗	✗	✗	✓	✗	✓
Hard/Easy Vision (2026)	NR	✓	✓	✓	✗	✗	✗	✓
Embodied Agent Arena	**1,000**	✓	✓	✓	✓	✗	✓	✓

✓ included; ✗ not evaluated. BEAR manipulation: separate BEAR-Agent online validation.

Rasheed et al., 2026) and diverse manipulation tasks (Zhang et al., 2024). Table 1 compares their coverage, diagnostics, and scale. Embodied Agent Arena spans metric geometry, spatial inference, functional grounding, planning, and manipulation, with GeoProbe isolating camera, motion, and scale requirements. Its local metrics and execution records complement terminal success by identifying the specific requirements that remain unresolved.

3 The Embodied Agent Arena

3.1 Benchmark Construction and Capability Domains

From local requirements to complete goals. The arena organizes 1,000 cases from 32 established sources and GeoProbe into five capability domains (Figures 1–2; Appendix A). Each case specifies observations, a query or goal, available operations, and scoring. Geometry, Spatial Reasoning, and Affordance isolate what the agent must estimate or ground; Task Planning and Manipulation test whether decisions produce the required environment state. GeoProbe adds 168 cases that vary camera motion, object motion, and scale references in controlled Blender scenes, complemented by real images. Separating these factors makes it possible to distinguish a failure to infer the scene from a failure to complete an interaction.

Geometry: metric precision and feasibility. Agents estimate intrinsics, pose, depth, dimensions, or displacement from images. InFlux and Map-free test calibration and relative pose (Liang et al., 2025, Arnold et al., 2022); GeoProbe varies camera motion and scale references. TraceSpatial-Bench adds paths on 56 scenes (Zhou et al., 2026b), pairing trajectory error with initiation, arrival, and collision constraints. These complementary measures distinguish a close geometric estimate from one that satisfies the task's spatial requirements.

Spatial Reasoning: relations and reference frames. Agents infer relations, viewpoints, configurations, and temporal order from images or video (Ma et al., 2024, Wang et al., 2025b, Yang et al., 2024). Comparing object distances, imagining another observer's view, and inferring

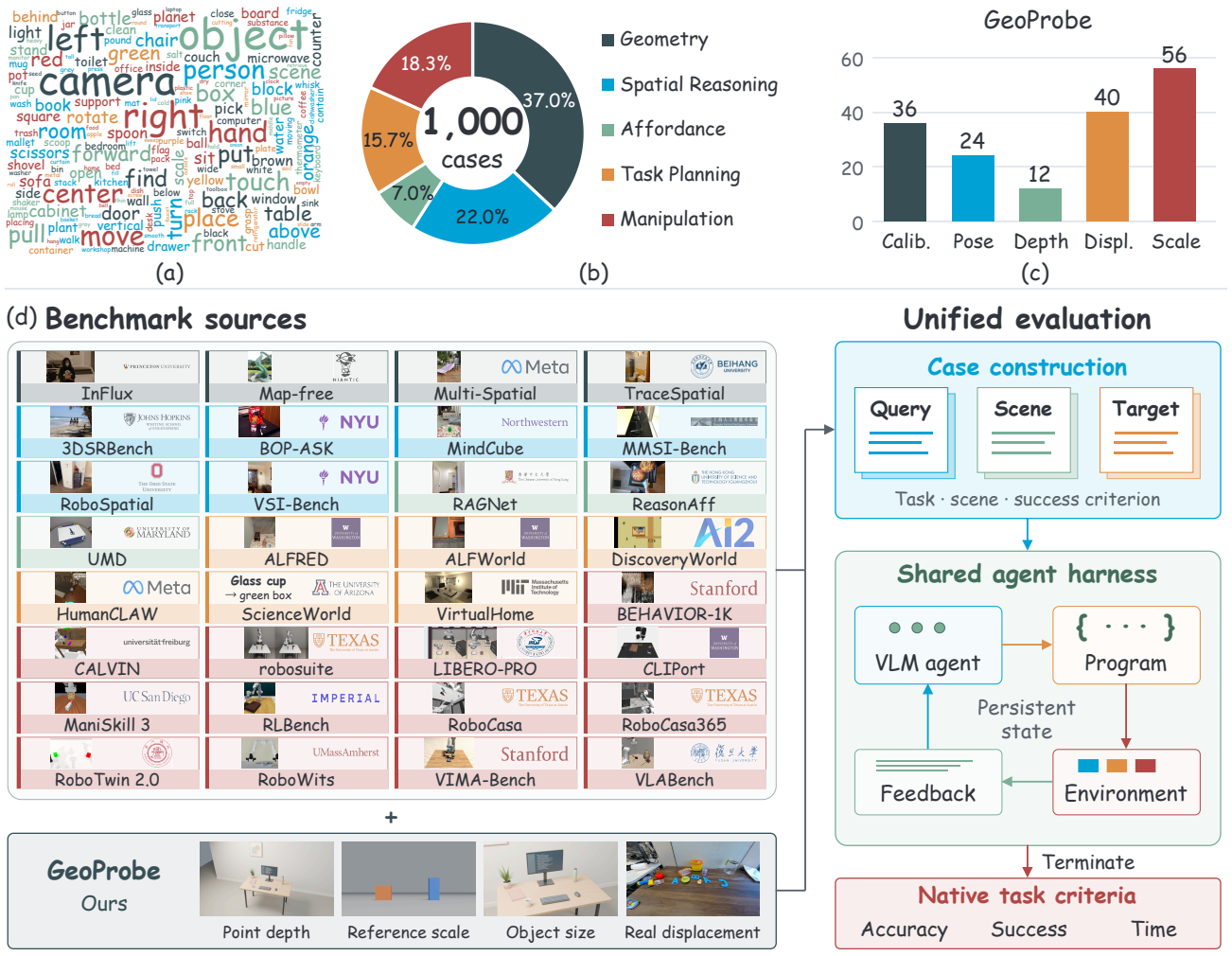


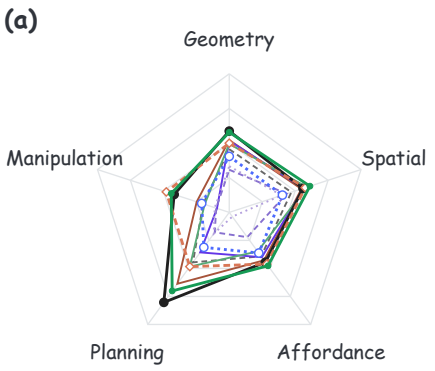
Figure 2. Construction of the 1,000-case arena. (a) Representative task vocabulary across sources. (b) Five-domain composition, including 112 spatial-tracing cases in Geometry. (c) GeoProbe estimation targets. (d) Source queries, scenes, and targets connect model interaction to termination and scoring. Colors identify domains; logos denote source institutions.

camera motion require different reference-frame transformations. Questions specify the relevant frame and expect a category, quantity, or point set.

Affordance: contact and functional extent. Agents ground an intended action through a contact point or region. UMD tests tool functions such as cutting and scooping (Myers et al., 2015); instruction- and action-conditioned 3DOI queries test interaction contacts (Qian & Fouhey, 2023). Point validity measures whether a contact lies in the functional region. Region overlap measures how much of that region the model recovers, using native boxes and, in a separate experiment, direct polygons.

Task Planning: satisfying all goal conditions. Agents compose source-provided operations into household, navigation, or scientific procedures (Shridhar et al., 2019, Wang et al., 2022), using visual, textual, or symbolic feedback. Compound goals require several conditions to hold, such as heating an object and placing it at the destination. For terminal state s_T and required predicates $g_1, \ldots, g_K$, complete success is

$$\text{Success}(s_T, g) = \mathbf{1}\left[\bigwedge_{k=1}^{K} g_k(s_T)\right]. \tag{1}$$



(b)

Domain indices · 0–100

	Geo.	Spatial	Afford.	Plan.	Manip.
Astra	58.7	56.0	44.4	80.3	41.8
Sol	45.6	47.2	39.2	44.6	19.8
Fable	52.2	54.8	42.8	63.7	24.7
Q-Max	51.6	56.4	40.0	35.7	9.3
Gemini 3.8	49.4	59.0	34.8	48.4	18.7
Q-397B	30.8	43.6	22.1	17.8	9.9
Q-27B	33.3	39.8	3.8	19.1	11.0
Opus 5.5	49.9	57.6	46.1	48.4	47.8
DS-1017	40.4	40.4	36.3	31.2	20.9
Gemini 4	57.8	61.2	47.7	70.1	44.0

Figure 3. Complementary capability profiles across ten agents. (a) Thicker lines highlight Astra, Opus 5.5, DeepSeek-1017, and Gemini 4 Argon. Row labels in (b) identify model colors in both panels. (b) Domain indices on fixed 0–100 scales (Appendix C.6), each summarizing its own metrics. Affordance combines box–mask IoU and contact validity; Figure 6 examines polygons. Manipulation uses the recorded schedules and budgets in Appendix E.1.

Intermediate scores identify which predicates in Equation (1) remain unsatisfied.

Manipulation: executing and preserving progress. Agents execute robot motions or skills for placement, insertion, articulation, and coordinated handling (James et al., 2019, Chen et al., 2025a). Language-conditioned variants change the target relation or object (Zhang et al., 2024); sequential tasks require preserving progress across ordered goals. Native task success records completion, while condition-level scores expose achieved relations and control events.

3.2 Unified Agent Harness

Task decisions remain with the model. The harness provides persistent interaction through each source's observations and operations (Figure 2d). Models interpret observations, select operations, and construct action sequences from feedback. We use *native capability* to mean these model-directed decisions through the supplied source interfaces, including their helpers. The baseline adds no perception or control modules beyond those interfaces. Geometry, Spatial Reasoning, Affordance, and Planning use a Python runtime; Manipulation follows its native agent loop. Persistent variables let the agent build on earlier computations, while images produced during a round enter the next model request. Appendix B.1 specifies the interfaces; the supplementary review study adds specialist measurements (Section 4.4) (Zhang et al., 2025, Huang et al., 2026d).

Termination is distinct from success. Fixed-input submissions commit predictions. Interactive episodes end on agent termination, an environment terminal condition, or budget exhaustion. The evaluator then checks the submitted output or environment state independently of the agent's declaration. Declaring completion therefore cannot substitute for satisfying the goal. Appendix B.2 specifies the round, token, action, and time limits.

3.3 Evaluation Protocol and Metrics

Completion and precision. Spatial Reasoning and Planning report Pass@1; categorical spatial accuracy is also reported separately. Manipulation pairs native success with consecutive goals and task-specific condition or control-evidence scores. Geometry and numerical spatial

Table 2. Performance across five domains. Box–mask and Polygon IoU use separate outputs on the same 70 cases. Planning covers 157 cases; Manipulation covers 182. Repeated metrics show mean ± run SD (five runs; Fable: three); the last three columns use one run. Best means are bold; Manipulation execution settings are detailed in Appendix E.1.

	Geometry		Spatial Reasoning		Affordance		Task Planning	Manipulation
Agent	Rot.↓ (°)	Trans.↓ (cm)	Pass@1↑ (%)	AbsRel↓	Box–mask↑ IoU	Polygon↑ IoU	Pass@1↑ (%)	Task SR↑ (%)
Astra	**17.2**±6.7	106.9±36.0	69.4±1.0	0.214±0.016	0.286±0.010	**0.379**	**80.3**	41.8
Sol	27.0±1.5	151.9±6.2	60.7±2.5	0.273±0.016	0.264±0.010	0.325	44.6	19.8
Fable	24.2±8.3	116.6±42.6	62.0±4.6	0.185±0.011	0.284±0.013	**0.328**	63.7	24.7
Qwen-Max	20.5±3.5	138.1±10.2	68.9±1.1	0.206±0.033	0.272±0.017	0.307	35.7	9.3
Gemini 3.8	23.7±5.6	138.0±41.7	65.4±1.3	**0.161**±0.028	0.251±0.081	0.291	48.4	18.7
Qwen-397B	38.4±1.1	249.8±10.2	50.6±1.8	0.251±0.027	0.116±0.014	0.092	17.8	9.9
Qwen-27B	38.4±3.6	213.7±12.6	48.0±0.9	0.288±0.023	0.013±0.006	0.051	19.1	11.0
Opus 5.5	23.4±3.7	135.8±12.7	67.5±5.1	0.184±0.040	0.311±0.020	0.397	48.4	**47.8**
DeepSeek-1017	30.2±3.9	182.0±29.0	51.6±1.5	0.309±0.035	0.217±0.011	0.249	31.2	20.9
Gemini 4 Argon	18.5±1.8	**105.6**±9.6	**71.6**±0.7	0.169±0.023	**0.356**±0.009	**0.441**	70.1	44.0

queries use MAE, RMSE, AbsRel/MAPE, displacement error, or pose error as appropriate to the target. Tracing pairs trajectory RMSE with offline 3D constraint satisfaction.

Functional grounding. The main table reports box–mask IoU for native rectangles and polygon–mask IoU for direct polygons on the same cases. Both compare rasterized predictions with the reference mask. Contact validity, box–box IoU, and strict box success provide complementary diagnostics in the appendix.

Variation and resource use. Repeated metrics report the equally weighted mean and sample standard deviation (SD) of run-level scores. Continuous scores average valid predictions within each run; binary success rates retain all assigned cases. Tokens, action calls, model rounds, and elapsed time describe different resources consumed before termination. Section 4.1 specifies repetitions, and Appendix B.3 defines aggregation and accounting.

4 Experiments

Capability profiles and shared failures motivate task-level diagnostics, followed by execution traces and feedback comparisons.

4.1 Experimental Setup

Models. We evaluate Astra, Sol, Fable, Opus 5.5, Qwen-Max, Gemini 3.8 Flash, Gemini 4 Argon, Qwen-397B, Qwen-27B, and DeepSeek-V4.1-Flash-1017 (DeepSeek-1017). Appendix B.2 gives full identifiers and configurations. DeepSeek-1017 uses the `deepseek-flash` endpoint updated on 17 October 2026.

Repetitions and execution conditions. Non-tracing Geometry, Spatial Reasoning, and point-and-box Affordance use five runs per case for nine models and three for Fable. We report the mean and sample SD across run-level scores. Tracing, direct polygons, Planning, and Manipulation each use one episode per assignment. Polygon IoU covers the same 70 Affordance cases for all ten agents (Appendix C.5). Planning retains all 157 assignments, and Manipulation all 182 binary assignments; execution errors count as unsuccessful. All ten agents are evaluated on the same assigned tasks through source-native interfaces, with RGB observations and native success criteria. Manipulation retains each evaluation batch's recorded observation schedule

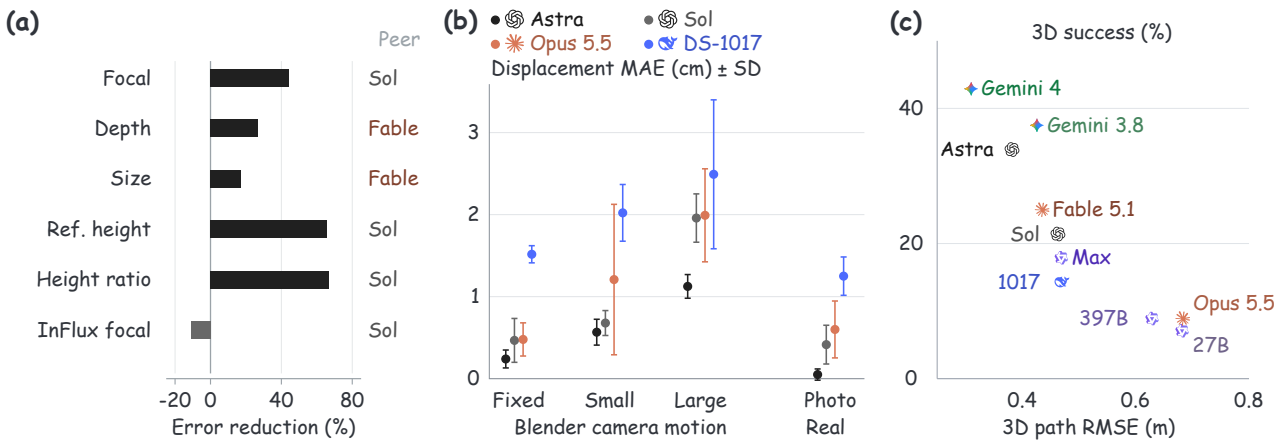


Figure 4. Geometric precision and task completion. (a) Astra's error reduction relative to the lowest-error original peer; the first five targets are GeoProbe. (b) Object-displacement error under three Blender camera conditions and separate real photos; mean ± sample SD of five run means. (c) Single-run 3D tracing error versus joint success for all ten agents. Logos mark measured coordinates; adjacent labels distinguish model versions. Details: Appendix D.1.

and execution ceilings; Appendix E.1 distinguishes these settings. Repetition details and later model configurations appear in Appendices C.3, C.4, and J.

4.2 Main Results

Capability breadth. ① *The leader changes with the requirement.* Astra leads rotation estimation and Planning, Gemini 4 Argon leads Spatial Pass@1 and both region-overlap metrics, and Opus 5.5 records the highest Manipulation success (Table 2; Figure 3). Even within Spatial Reasoning, the lowest numerical AbsRel belongs to Gemini 3.8 (0.161), while Gemini 4 Argon leads Pass@1 (71.6%). Passing a source test and minimizing numerical error reveal different aspects of competence. ② *Similar spatial scores conceal different planning outcomes.* Qwen-Max and Astra differ by only 0.5 percentage points on Spatial Pass@1, yet complete 35.7% and 80.3% of Planning tasks. Selecting a model by a perceptual proxy can therefore miss a much larger difference in its ability to execute a procedure.

Complementarity and shared failures. ① *Model diversity expands observed coverage.* The union of ten models' successes covers 142 of 157 Planning tasks, adding 16 beyond Astra's 126 (Figure 7). This post hoc union summarizes complementary recorded solutions. ② *Broad coverage can remain fragile.* Of those 142 tasks, 39 are completed by only one or two models; only eight are solved by all ten. Success therefore often depends on which model is chosen, even among tasks with a recorded solution. The 15 universally unsolved tasks form a separate frontier. Model selection targets complementary strengths; shared failures instead require improvements to task execution. Average rankings obscure this distinction.

The cost of unfinished tasks. ① *Failures dominate token use.* Unsuccessful episodes consume 55.5–86.9% of each model's recorded Planning tokens (Appendix I). Even Astra's 31 failures account for 55.5% of its tokens, despite representing fewer than one fifth of its tasks. Gemini 4 Argon shows the same concentration: 47 unsuccessful episodes consume 64.0% of its tokens. ② *Additional interaction needs a progress criterion.* An Astra failure uses 5.1 times as many tokens as a success on average. The execution traces below distinguish attempts that advance the next subgoal from those that repeat ineffective operations (Section 4.4). Completion rate and inference expenditure together expose the cost of leaving a goal unresolved.

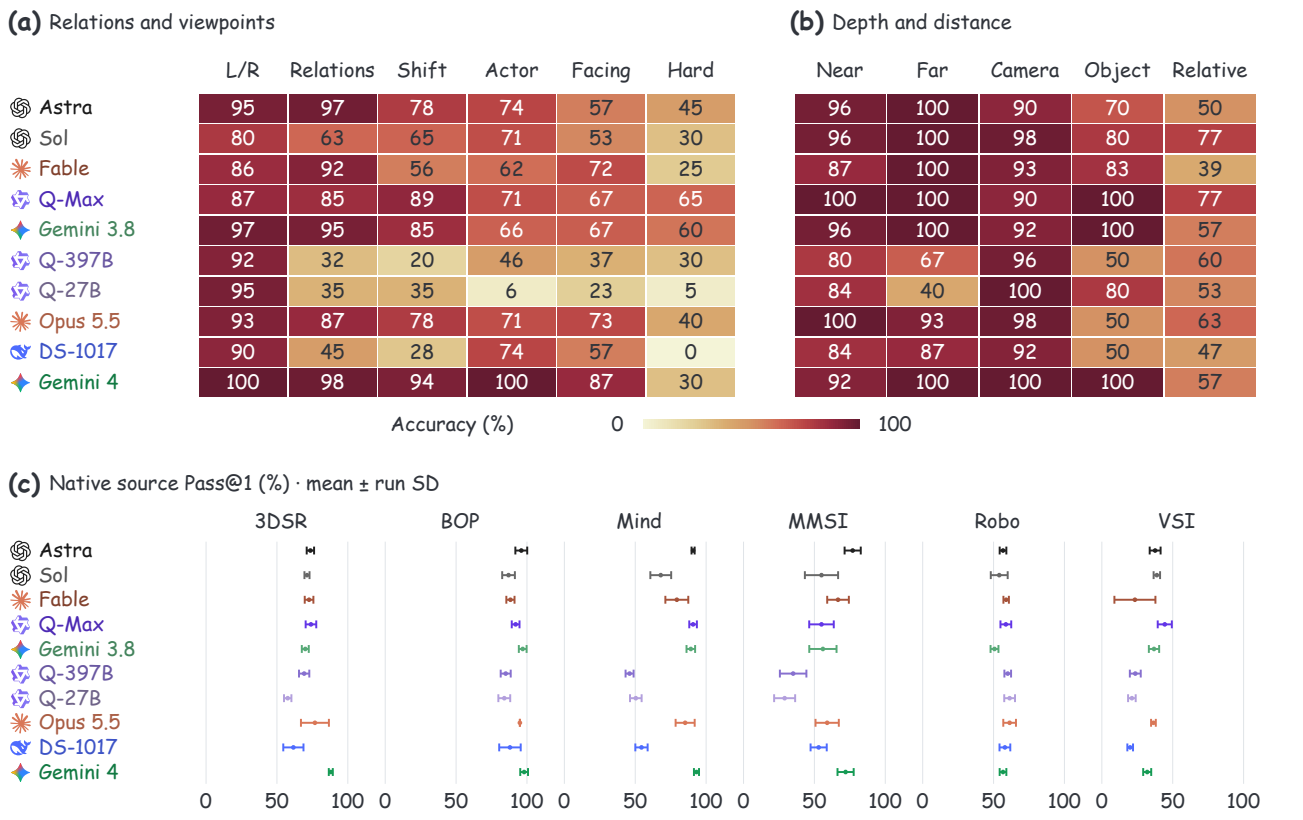


Figure 5. Spatial strengths change with the required relation. (a–b) All ten models on the same relation/viewpoint and depth/distance question groups, respectively; cells show mean accuracy (%), with one shared color scale. L/R denotes left/right. Appendix D.2 gives task definitions and denominators. (c) Native source Pass@1, including numerical questions: means and sample SD over five runs (three for Fable). 3DSR, Mind, Robo, and VSI abbreviate 3DSRBench, MindCube, RoboSpatial-Home, and VSI-Bench. Gemini 3.8 and Gemini 4 denote Gemini 3.8 Flash and Gemini 4 Argon throughout.

4.3 From Local Competence to Complete Tasks

Figures 4–8 follow the requirements of an embodied task: estimate the scene, identify an interaction region, and carry the resulting procedure to completion. The diagnostics locate where a strong intermediate result ceases to predict the final outcome.

Geometry: precision at the point of use. ① *Correct ordering can coexist with poor calibration.* Astra orders depths correctly in 96.3% of GeoProbe cases, yet oracle scale alignment reduces depth AbsRel from 0.104 to 0.038. Camera motion introduces another ambiguity: displacement MAE rises from 0.24 to 1.13 cm between fixed and large-shift cameras (Figure 4a–b). Relative geometry, metric scale, and camera compensation remain distinct requirements. ② *A close path can still miss its goal.* Gemini 4 Argon has the lowest 3D trajectory RMSE, 0.310 m, but only 24/56 paths satisfy all constraints (panel c). For Astra, initiation or arrival fails in 36 of its 37 unsuccessful traces. Mean path error averages over the local violations that determine whether a trajectory is usable.

Spatial Reasoning: rankings depend on the relation. ① *Viewpoint gains do not extend to every spatial operation.* Gemini 4 Argon reaches 93.8% on camera shifts and 86.7% on object-facing judgments, compared with Astra's 78.5% and 56.7%. On hard video directions, however, Gemini 4 scores 30.0%, behind Gemini 3.8's 60.0% and Qwen-Max's 65.0% (Figure 5a). The complete ten-model comparison reveals reversals hidden by the aggregate ranking. ② *The observation context matters as much as the question type.* Gemini 4 scores 100% on 3DSRBench's

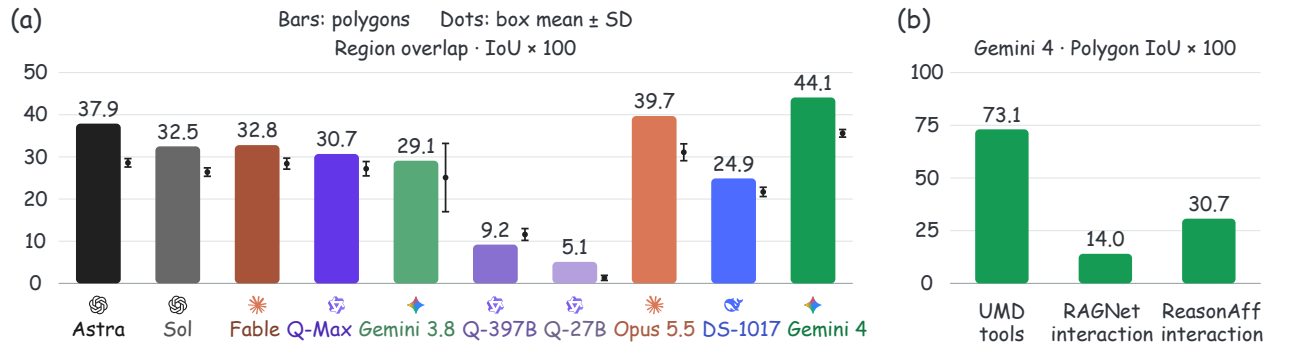


Figure 6. More expressive outlines improve overlap, but interaction grounding remains difficult. (a) Direct-polygon IoU (bars) and box–mask IoU (dots, with run SD) on the same 70 cases for all ten models. Boxes use five runs, or three for Fable; polygons use one separate run. Invalid or absent polygon outputs score zero. (b) Gemini 4 Argon, the aggregate leader, scores much higher on tools than on interaction-conditioned scenes. Appendix C.5 and Figure A1 detail source and contact results.

camera-relative and inter-object distance groups, but 56.7% on VSI nearest-object selection, versus 76.7% for Sol and Qwen-Max (panel b). Its native source scores likewise span 98.0% on BOP-ASK to 32.0% on VSI-Bench (panel c). Spatial competence must be characterized by the relation, reference frame, and source conditions under which it is measured.

Affordance: shape and target identity. ① *Flexible output geometry improves overlap.* Direct polygons outperform native boxes in aggregate mask IoU for nine of ten agents. Gemini 4 Argon leads both representations, rising from 0.356 to 0.441 (Figure 6a). ② *Interaction meaning remains the harder requirement.* Gemini 4's polygon IoU reaches 73.1% on UMD tools but only 14.0% on RAGNet interaction scenes (panel b); Astra, Opus 5.5, and DeepSeek-1017 show the same ordering. Matched cases separate an imprecise outline of the correct part from a precise outline of the wrong object (Figure A8). The former calls for better boundaries; the latter requires a different interpretation of the requested action. Neither contact validity nor region overlap alone establishes successful physical interaction.

Task Planning: approach does not complete an interaction. ① *Shared failures concentrate at specific requirements.* The 15 universally unsolved tasks comprise nine HumanCLAW cases and two each from ALFRED, ALFWorld, and ScienceWorld (Figure 7). They coexist with complementary goal-family strengths, such as Astra's ALFRED heating advantage and Fable's stronger ScienceWorld thermometer use (Figure A2). Changing the model can recover some procedures while leaving others unresolved. ② *Target proximity is an intermediate state.* All ten agents fail all eight HumanCLAW sitting tasks. Five Astra episodes reach the seat and issue a sitting action, yet none establishes the required sitting state; Gemini 4's two HumanCLAW successes are navigation tasks. With ceilings of 101 code rounds, 2M tokens, and 180 minutes (Table A6), these records locate the obstacle beyond finding and approaching the target, at completing the interaction.

Manipulation: subgoals must survive composition. ① *Early progress overstates full-sequence reliability.* Gemini 4 Argon reaches the first CALVIN goal in all ten episodes, but only five reach the third and four reach the fifth (Figure 8a). Astra averages 3.1 consecutive goals yet completes only three sequences. The declining counts along the same task chains expose a requirement that isolated-action success cannot measure. ② *Local skill does not ensure household completion.* Gemini 4 completes 8/10 RLBench tasks but only 2/10 RoboCasa tasks, which combine navigation, articulated fixtures, and object handling (Nasiriany et al., 2024) (Table A28). All ten agents fail the five BEHAVIOR-1K tasks. These outcomes motivate examining

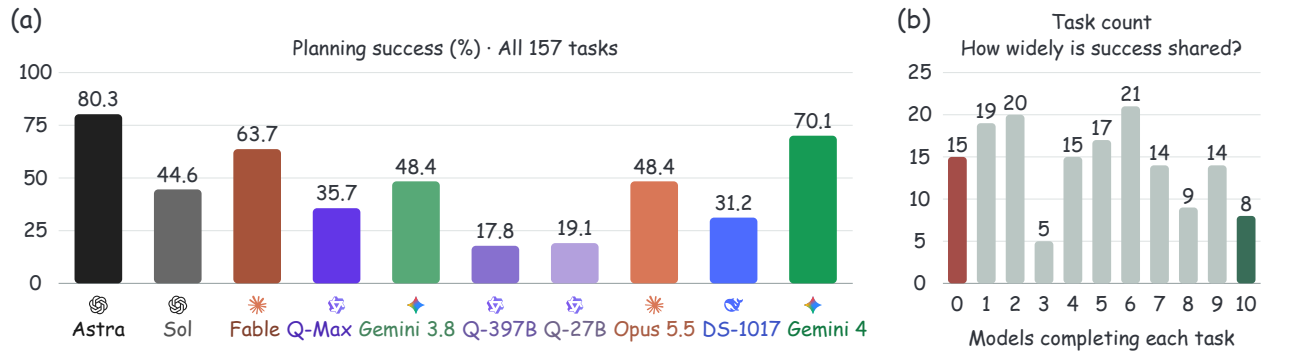


Figure 7. Complementary successes coexist with shared failures. (a) Planning success across all 157 assigned tasks, with one episode per model and task. (b) The same tasks grouped by how many of the ten models complete them: 15 defeat every model, while only eight are completed by all ten. The union of recorded successes covers 142 tasks, versus Astra's 126; this is retrospective coverage, not an evaluated routing policy. Figure A2 and Appendix D.4 provide source and goal-family diagnostics.

whether feedback restores progress through the remaining conditions, rather than merely producing another executable action. Recorded observation schedules and execution ceilings remain specified in Appendix E.1.

4.4 Feedback and Recovery Must Restore Progress

Execution recovery and task recovery. ① *Restarting code is common; finishing afterward is rare.* Among 100 CALVIN episodes, 65 contain an execution error and 62 later execute code without an error, but only three subsequently complete all five goals (Figure 8b). These nested episode counts distinguish repairing the execution interface from recovering the complete task (Appendix D.5.1). ② *A useful correction must change the task outcome.* The fixed-input review study exhibits a related distinction: on its 46 cases, Qwen-27B's self-review yields higher mean box IoU than tool assistance, but only tool assistance produces a box above the success threshold (Table A34). A better intermediate score can still miss the success criterion.

Feedback that advances the next state. ① *An early error need not determine the outcome.* Opus 5.5 completes 48/105 Planning episodes containing execution errors; Gemini 4 Argon completes 16/28. In the matched Archaeology Dating case, both recover from early code errors and finish in seven and eight rounds, respectively (Figure A13). ② *Repeated activity can leave the environment unchanged.* DeepSeek-1017 makes partial progress on that task but spends its last three rounds on code errors without advancing the environment. The informative recovery signal is therefore the next verified state change, together with the conditions still unmet. Appendices F–H connect these outcomes to recorded actions; Appendix I gives the complete Planning resource and recovery analysis.

Progress under resource limits. ① *Successful episodes can have very different costs.* On the same HumanCLAW chair-navigation task, Gemini 4 Argon completes in 17 rounds using 158k tokens, whereas DeepSeek-1017 uses 65 rounds and 938k tokens. Both reach the same goal, yet DeepSeek-1017 uses nearly six times as many tokens. ② *The binding budget determines whether recovery can continue.* In the BEHAVIOR-1K toolbox case, Gemini 4 Argon's next round is blocked by the 640k-token ceiling after 37.5 minutes of a 180-minute time allowance (Appendix I). More wall-clock time alone would not admit the blocked call. These cases separate the cost of successful execution from the budget needed to finish an incomplete task.

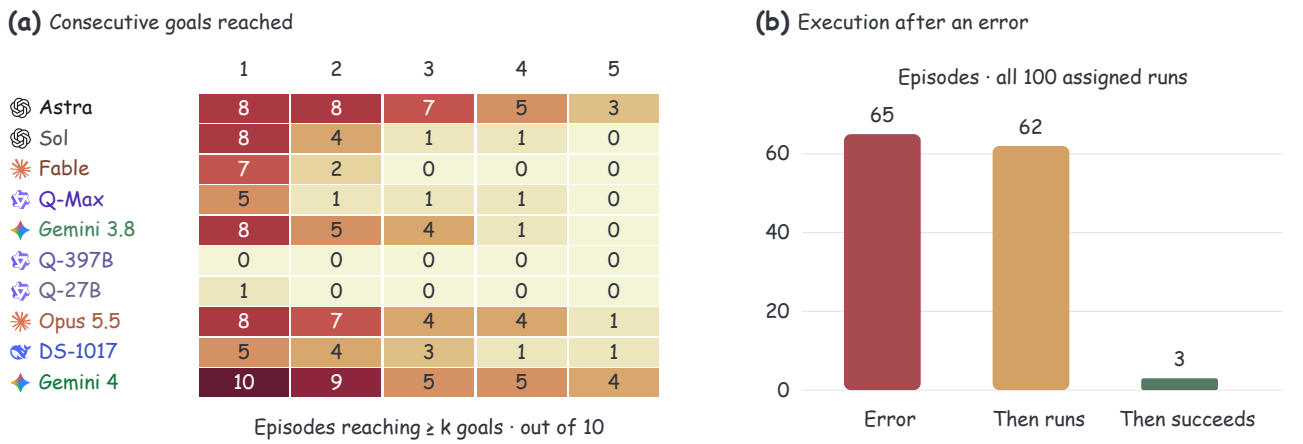


Figure 8. Resuming execution rarely restores complete task success. All 100 adopted CALVIN episodes: ten sequences per model. (a) Number of episodes reaching at least each consecutive goal; the fifth column is complete-sequence success. Unverified progress receives no credit. (b) Of 65 episodes with an execution error, 62 later execute code without an error, but only three subsequently pass the native task verifier. These are nested episode counts, not counts of error events. Appendix D.5.1 gives model-level counts, missing-prefix handling, and the recorded execution settings.

5 Conclusion

Embodied Agent Arena examines frontier VLM agents through the requirements of complete tasks. The ten-model comparison reveals substantial local competence alongside failures at decisive conditions: paths miss endpoints, target approach fails to establish an interaction state, and successful early actions leave a sequence unfinished. Broader capability and model diversity improve observed coverage while leaving a shared set of unsolved tasks.

Unsuccessful Planning episodes consume most recorded tokens, while matched traces show that more actions can yield less progress. Productive recovery changes a failed operation and enables the next required state transition. Progress toward embodied generalism therefore calls for measuring complete goal satisfaction together with the interaction it takes to achieve it. The arena makes both visible through task outcomes and execution records.

References

Michael Ahn, Anthony Brohan, Noah Brown, Yevgen Chebotar, Omar Cortes, Byron David, Chelsea Finn, Chuyuan Fu, et al. Do As I Can, Not As I Say: Grounding Language in Robotic Affordances. *arXiv preprint arXiv:2204.01691*, 2022.

Eduardo Arnold, Jamie Wynn, Sara Vicente, Guillermo Garcia-Hernando, Áron Monszpart, Victor Adrian Prisacariu, Daniyar Turmukhambetov, and Eric Brachmann. Map-free Visual Relocalization: Metric Pose Relative to a Single Image. *arXiv preprint arXiv:2210.05494*, 2022.

Vineet Bhat, Sungsu Kim, Valts Blukis, Greg Heinrich, Prashanth Krishnamurthy, Ramesh Karri, Stan Birchfield, Farshad Khorrami, et al. BOP-ASK: Object-Interaction Reasoning for Vision-Language Models. *arXiv preprint arXiv:2511.16857*, 2025.

Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch,

Lucy Xiaoyang Shi, James Tanner, Quan Vuong, Anna Walling, Haohuan Wang, and Ury Zhilinsky. π_0: A Vision-Language-Action Flow Model for General Robot Control. *arXiv preprint arXiv:2410.24164*, 2024.

Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Joseph Dabis, Chelsea Finn, Keerthana Gopalakrishnan, Karol Hausman, Alexander Herzog, Jasmine Hsu, Julian Ibarz, Brian Ichter, Alex Irpan, Tomas Jackson, Sally Jesmonth, Nikhil Joshi, Ryan Julian, Dmitry Kalashnikov, Yuheng Kuang, Isabel Leal, Kuang-Huei Lee, Sergey Levine, Yao Lu, Utsav Malla, Deeksha Manjunath, Igor Mordatch, Ofir Nachum, Carolina Parada, Jodilyn Peralta, Emily Perez, Karl Pertsch, Jornell Quiambao, Kanishka Rao, Michael S Ryoo, Grecia Salazar, Pannag R Sanketi, Kevin Sayed, Jaspiar Singh, Sumedh Sontakke, Austin Stone, Clayton Tan, Huong Tran, Vincent Vanhoucke, Steve Vega, Quan H Vuong, Fei Xia, Ted Xiao, Peng Xu, Sichun Xu, Tianhe Yu, and Brianna Zitkovich. RT-1: Robotics Transformer for Real-World Control at Scale. In *Proceedings of Robotics: Science and Systems*, 2023. doi: 10.15607/RSS.2023.XIX.025.

Nicolas Carion, Laura Gustafson, Yuan-Ting Hu, Shoubhik Debnath, Ronghang Hu, Didac Suris, Chaitanya Ryali, Kalyan Vasudev Alwala, Haitham Khedr, Andrew Huang, Jie Lei, Tengyu Ma, Baishan Guo, Arpit Kalla, Markus Marks, Joseph Greer, Meng Wang, Peize Sun, Roman Rädle, Triantafyllos Afouras, Effrosyni Mavroudi, Katherine Xu, Tsung-Han Wu, Yu Zhou, Liliane Momeni, Rishi Hazra, Shuangrui Ding, Sagar Vaze, Francois Porcher, Feng Li, Siyuan Li, Aishwarya Kamath, Ho Kei Cheng, Piotr Dollár, Nikhila Ravi, Kate Saenko, Pengchuan Zhang, and Christoph Feichtenhofer. SAM 3: Segment anything with concepts. *arXiv preprint arXiv:2511.16719*, 2025.

Tianxing Chen, Zanxin Chen, Baijun Chen, Zijian Cai, Yibin Liu, Zixuan Li, Qiwei Liang, Xianliang Lin, et al. RoboTwin 2.0: A Scalable Data Generator and Benchmark with Strong Domain Randomization for Robust Bimanual Robotic Manipulation. *arXiv preprint arXiv:2506.18088*, 2025a.

Zeren Chen, Xiaoya Lu, Zhijie Zheng, Pengrui Li, Lehan He, Yijin Zhou, Jing Shao, Bohan Zhuang, and Lu Sheng. Geometrically-constrained agent for spatial reasoning. *arXiv preprint arXiv:2511.22659*, 2025b. doi: 10.48550/arXiv.2511.22659.

Tianxing Chen, Yue Chen, Zixuan Li, Junyuan Tang, Kailun Su, Haoran Lu, Weijie Wan, Baijun Chen, et al. RoboDojo: A Unified Sim-and-Real Benchmark for Comprehensive Evaluation of Generalist Robot Manipulation Policies. *arXiv preprint arXiv:2607.04434*, 2026.

Dongzhou Cheng, Taoran Yi, Ye Fang, Xingwu Zhang, Fan Feng, Yixuan Li, Gengxiong Zhuang, Rongze Wang, Shuai Yang, Wei Song, Weizhi Xue, Minyan Wu, Jie Gui, Jiaqi Wang, and Tong Wu. In-Context Robot Learning with VLM Agents. *arXiv preprint arXiv:2609.19138*, 2026.

Zhili Cheng, Yuge Tu, Ran Li, Shiqi Dai, Jinyi Hu, Shengding Hu, Jiahao Li, Yang Shi, et al. EmbodiedEval: Evaluate Multimodal LLMs as Embodied Agents. *arXiv preprint arXiv:2501.11858*, 2025.

Seokju Cho, Ryo Hachiuma, Abhishek Badki, Hang Su, Byung-Kwan Lee, Chan Hee Song, Sifei Liu, Subhashree Radhakrishnan, Seungryong Kim, Yu-Chiang Frank Wang, and Min-Hung Chen. SpatialClaw: Rethinking action interface for agentic spatial reasoning. Preprint, 2026.

Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. SuperPoint: Self-supervised interest point detection and description. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 2018.

Junjia Du, Yadi Liu, Hongcheng Guo, Jiawei Wang, Haojian Huang, Yunyi Ni, and Zhoujun Li. DependEval: Benchmarking LLMs for Repository Dependency Understanding. In *Findings of the Association for Computational Linguistics: ACL 2025*, pp. 7150–7179. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.findings-acl.373.

Letian Fu, Justin Yu, Karim El-Refai, Ethan Kou, Haoru Xue, Huang Huang, Wenli Xiao, Guanzhi Wang, et al. CaP-X: A Framework for Benchmarking and Improving Coding Agents for Robot Manipulation. *arXiv preprint arXiv:2603.22435*, 2026.

Dibya Ghosh, Homer Rich Walke, Karl Pertsch, Kevin Black, Oier Mees, Sudeep Dasari, Joey Hejna, Tobias Kreiman, Charles Xu, Jianlan Luo, You Liang Tan, Lawrence Yunliang Chen, Quan Vuong, Ted Xiao, Pannag R Sanketi, Dorsa Sadigh, Chelsea Finn, and Sergey Levine. Octo: An Open-Source Generalist Robot Policy. In *Proceedings of Robotics: Science and Systems*, 2024. doi: 10.15607/RSS.2024.XX.090.

Pengyuan Guo, Zhonghao Mai, Zhengtong Xu, Kaidi Zhang, Quan Khanh Luu, Heng Zhang, Zichen Miao, Arash Ajoudani, Zachary Kingston, Qiang Qiu, and Yu She. PLanAR: Planning-Language-Grounded Agentic Reasoning for Robot Manipulation. *arXiv preprint arXiv:2602.01662*, 2026.

Yanjiang Guo, Yen-Jen Wang, Lihan Zha, and Jianyu Chen. DoReMi: Grounding Language Model by Detecting and Recovering from Plan-Execution Misalignment. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.

Haojian Huang, Harold Haodong Chen, Meng Luo, Junjia Du, Shanqing Xu, Ziheng Chen, Yanxiang Huang, Yinchuan Li, and Ying-Cong Chen. No Place to Hide: Benchmarking Video Hallucination with Background-Controlled Pairs. In *Computer Vision – ECCV 2026*, pp. 155–175, 2026a. doi: 10.1007/978-3-032-37461-5_9.

Haojian Huang, Zexi Li, Junhao Guo, Yehang Zhang, Wenxuan Peng, Bohan Zhou, Weilin Ruan, Leyi Wu, Chenxu Wang, Jianchong Su, Binghui Xie, Wosong Chen, Yingjie Xu, Tianhao Zhou, Suzeyu Chen, Pukun Zhao, Jiaqi He, Xinyi Li, Runze Li, Peiran Dong, Shaoxiang Dang, Jing Huang, Yingbing Chen, Yifan Chang, Tianyi Zhang, Shiyuan Deng, Haozhi Wang, Yangkai Wei, Wenqian Li, Han Yang, Kaiwen Zhou, Huaping Liu, James Cheng, Rui Shao, Donglin Wang, Yaochu Jin, Jianye Hao, Ying-Cong Chen, and Yinchuan Li. In-context learning for robots: Methods and applications. *arXiv preprint arXiv:2609.36012*, 2026b.

Haojian Huang, Chuanyu Qin, Yinchuan Li, and Yingcong Chen. Find, Fix, Reason: Context Repair for Video Reasoning. In *Proceedings of the International Conference on Machine Learning*, 2026c.

Haojian Huang, Jiahao Shi, Yinchuan Li, and Yingcong Chen. Affordance agent harness: Verification-gated skill orchestration. *arXiv preprint arXiv:2605.00663*, 2026d.

Wenlong Huang, Chen Wang, Ruohan Zhang, Yunzhu Li, Jiajun Wu, and Li Fei-Fei. VoxPoser: Composable 3D Value Maps for Robotic Manipulation with Language Models. In *Proceedings of The 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pp. 540–562. PMLR, 2023a.

Wenlong Huang, Fei Xia, Ted Xiao, Harris Chan, Jacky Liang, Pete Florence, Andy Zeng, Jonathan Tompson, Igor Mordatch, Yevgen Chebotar, Pierre Sermanet, Tomas Jackson,

Noah Brown, Linda Luu, Sergey Levine, Karol Hausman, and Brian Ichter. Inner Monologue: Embodied Reasoning through Planning with Language Models. In *Proceedings of The 6th Conference on Robot Learning*, volume 205 of *Proceedings of Machine Learning Research*, pp. 1769–1782. PMLR, 2023b.

Stephen James, Zicong Ma, David Rovick Arrojo, and Andrew J. Davison. RLBench: The Robot Learning Benchmark & Learning Environment. *arXiv preprint arXiv:1909.12271*, 2019.

Peter Jansen, Marc-Alexandre Côté, Tushar Khot, Erin Bransom, Bhavana Dalvi Mishra, Bodhisattwa Prasad Majumder, Oyvind Tafjord, and Peter Clark. DISCOVERYWORLD: A Virtual Environment for Developing and Evaluating Automated Scientific Discovery Agents. *arXiv preprint arXiv:2406.06769*, 2024.

Yunfan Jiang, Agrim Gupta, Zichen Zhang, Guanzhi Wang, Yongqiang Dou, Yanjun Chen, Li Fei-Fei, Anima Anandkumar, Yuke Zhu, and Linxi Fan. VIMA: Robot Manipulation with Multimodal Prompts. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 14975–15022. PMLR, 2023.

Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, et al. OpenVLA: An Open-Source Vision-Language-Action Model. *arXiv preprint arXiv:2406.09246*, 2024.

Chengshu Li, Ruohan Zhang, Josiah Wong, Cem Gokmen, Sanjana Srivastava, Roberto Martín-Martín, Chen Wang, Gabrael Levine, et al. BEHAVIOR-1K: A Human-Centered, Embodied AI Benchmark with 1,000 Everyday Activities and Realistic Simulation. *arXiv preprint arXiv:2403.09227*, 2024a.

Manling Li, Shiyu Zhao, Qineng Wang, Kangrui Wang, Yu Zhou, Sanjana Srivastava, Cem Gokmen, Tony Lee, et al. Embodied Agent Interface: Benchmarking LLMs for Embodied Decision Making. *arXiv preprint arXiv:2410.07166*, 2024b.

Qiang Li, Qingsen Yan, Haojian Huang, Peng Wu, Haokui Zhang, and Yanning Zhang. Text-Visual Semantic Constrained AI-Generated Image Quality Assessment. In *Proceedings of the 33rd ACM International Conference on Multimedia*, 2025. doi: 10.1145/3746027.3755471.

Zexi Li, Yehang Zhang, Haojian Huang, Bohan Zhou, Wenqian Li, Chenxu Wang, Yifan Chang, Yangkai Wei, Tianyi Zhang, Ying-Cong Chen, Kaiwen Zhou, Yinchuan Li, and James Cheng. RACaP: Agentic Reasoning, Acting, and Coding as Policies for Evolvable Robot Learning. *arXiv preprint arXiv:2609.29394*, 2026a.

Zexi Li, Yehang Zhang, Wenqian Li, Haojian Huang, Chenxu Wang, Shiyuan Deng, Yangkai Wei, Tianyi Zhang, Binghui Xie, Bohan Zhou, Yifan Chang, Kaiwen Zhou, Ying-Cong Chen, James Cheng, and Yinchuan Li. Robo-Harness K1: Harnessing Robot-Use Agents via Perception Augmentation. *arXiv preprint arXiv:2609.29389*, 2026b.

Erich Liang, Roma Bhattacharjee, Sreemanti Dey, Rafael Moschopoulos, Caitlin Wang, Michel Liao, Grace Tan, Andrew Wang, et al. InFlux: A Benchmark for Self-Calibration of Dynamic Intrinsics of Video Cameras. *arXiv preprint arXiv:2510.23589*, 2025.

Jacky Liang, Wenlong Huang, Fei Xia, Peng Xu, Karol Hausman, Brian Ichter, Pete Florence, and Andy Zeng. Code as Policies: Language Model Programs for Embodied Control. *arXiv preprint arXiv:2209.07753*, 2022.

Philipp Lindenberger, Paul-Edouard Sarlin, and Marc Pollefeys. LightGlue: Local feature matching at light speed. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.

Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. LIBERO: Benchmarking Knowledge Transfer for Lifelong Robot Learning. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Shengbang Liu, Zhengye Du, Zhilong Wan, Honghao Su, Chenxiang Xia, Chang Ge, Jinyang Xiao, Nan Wang, Zhipeng Hu, Huining Yuan, Shu'ang Yu, Yue Chen, Tianxing Chen, Chuankang Li, Maoqing Yao, Wenbo Ding, Yu Wang, Guanghui Ren, and Chao Yu. RoboValue: A Fine-Grained Sim-and-Real Benchmark for Unified Evaluation of Robotic Value Models. *arXiv preprint arXiv:2610.26663*, 2026.

Keyuan Luo, Yi Xie, Wanyun Ling, Yuanzhe Li, Ziyue Li, and Zhuoqi Zeng. Beyond Success Rate: Interventional Evaluation of Memory in Vision-Language-Action Systems. *arXiv preprint arXiv:2610.23372*, 2026.

Wufei Ma, Haoyu Chen, Guofeng Zhang, Yu-Cheng Chou, Jieneng Chen, Celso M de Melo, and Alan Yuille. 3DSRBench: A Comprehensive 3D Spatial Reasoning Benchmark. *arXiv preprint arXiv:2412.07825*, 2024.

Arjun Majumdar, Anurag Ajay, Xiaohan Zhang, Pranav Putta, Sriram Yenamandra, Mikael Henaff, Sneha Silwal, Paul Mcvay, Oleksandr Maksymets, Sergio Arnaud, Karmesh Yadav, Qiyang Li, Ben Newman, Mohit Sharma, Vincent Berges, Shiqi Zhang, Pulkit Agrawal, Yonatan Bisk, Dhruv Batra, Mrinal Kalakrishnan, Franziska Meier, Chris Paxton, Alexander Sax, and Aravind Rajeswaran. OpenEQA: Embodied Question Answering in the Era of Foundation Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 16488–16498, 2024.

Oier Mees, Lukas Hermann, Erick Rosete-Beas, and Wolfram Burgard. CALVIN: A Benchmark for Language-Conditioned Policy Learning for Long-Horizon Robot Manipulation Tasks. *arXiv preprint arXiv:2112.03227*, 2021.

Austin Myers, Ching L. Teo, Cornelia Fermüller, and Yiannis Aloimonos. Affordance Detection of Tool Parts from Geometric Features. In *IEEE International Conference on Robotics and Automation*, 2015.

Soroush Nasiriany, Abhiram Maddukuri, Lance Zhang, Adeet Parikh, Aaron Lo, Abhishek Joshi, Ajay Mandlekar, and Yuke Zhu. RoboCasa: Large-Scale Simulation of Everyday Tasks for Generalist Robots. *arXiv preprint arXiv:2406.02523*, 2024.

Fei Ni, Min Zhang, Pengyi Li, Yifu Yuan, Lingfeng Zhang, Yuecheng Liu, Peilong Han, Longxin Kou, et al. Embodied Arena: A Comprehensive, Unified, and Evolving Evaluation Platform for Embodied AI. *arXiv preprint arXiv:2509.15273*, 2025.

Sangwu Park, Wonjoong Kim, Yeonjun In, and Chanyoung Park. Do VLAs Really Follow Instructions? Diagnosing How-Level Instruction Following in Vision-Language-Action Models. *arXiv preprint arXiv:2610.21251*, 2026.

Xavier Puig, Kevin Ra, Marko Boben, Jiaman Li, Tingwu Wang, Sanja Fidler, and Antonio Torralba. VirtualHome: Simulating Household Activities via Programs. *arXiv preprint arXiv:1806.07011*, 2018.

Yu Qi, Haibo Zhao, Ziyu Guo, Siyuan Ma, Ziyan Chen, Yaokun Han, Renrui Zhang, Zitiantao Lin, et al. Dissecting Embodied Abilities in Multimodal Language Models through Skill-level Evaluation and Diagnosis. *arXiv preprint arXiv:2510.08759*, 2025.

Shengyi Qian and David F. Fouhey. Understanding 3D Object Interaction from a Single Image. *arXiv preprint arXiv:2305.09664*, 2023.

Hanoona Rasheed, Mohammed Irfan Kurpath, Bin Ren, Hisham Cholakkal, Fahad Shahbaz Khan, and Salman Khan. Hard vision, easy vision: What GPT-6 Astra reveals across computer vision. *arXiv preprint arXiv:2609.35718*, 2026.

Weilin Ruan, Bohan Zhou, Jinghao Li, Yangkai Wei, Wenqian Li, Peiran Dong, Yingjie Xu, Leyi Wu, Haojian Huang, Binghui Xie, Han Yang, Kaiwen Zhou, Yinchuan Li, Ying-Cong Chen, and James Cheng. Anvil: Scaling Embodied Spatial Intelligence through Knowledge Recirculation. *arXiv preprint arXiv:2610.21882*, 2026.

Pierre Sermanet, Tianli Ding, Jeffrey Zhao, Fei Xia, Debidatta Dwibedi, Keerthana Gopalakrishnan, Christine Chan, Gabriel Dulac-Arnold, Sharath Maddineni, Nikhil J Joshi, Pete Florence, Wei Han, Robert Baruch, Yao Lu, Suvir Mirchandani, Peng Xu, Pannag Sanketi, Karol Hausman, Izhak Shafran, Brian Ichter, and Yuan Cao. RoboVQA: Multimodal Long-Horizon Reasoning for Robotics. *arXiv preprint arXiv:2311.00899*, 2023.

Mohit Shridhar, Jesse Thomason, Daniel Gordon, Yonatan Bisk, Winson Han, Roozbeh Mottaghi, Luke Zettlemoyer, and Dieter Fox. ALFRED: A Benchmark for Interpreting Grounded Instructions for Everyday Tasks. *arXiv preprint arXiv:1912.01734*, 2019.

Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Côté, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. ALFWorld: Aligning Text and Embodied Environments for Interactive Learning. *arXiv preprint arXiv:2010.03768*, 2020.

Mohit Shridhar, Lucas Manuelli, and Dieter Fox. CLIPort: What and Where Pathways for Robotic Manipulation. *arXiv preprint arXiv:2109.12098*, 2021.

Ruiyang Si, Jianxin Bi, Shunyu Yang, Rui Ni, Wenbo Huang, Qiang Wang, Shulong Jiang, Duomin Wang, Xiuyu Li, Haiwen Feng, Zhen Dong, and Daquan Zhou. Fewer Tokens, Better Action: GPT-6 Astra Robot Agents with 14% Higher Success Rate but 65% Fewer Tokens. *arXiv preprint arXiv:2610.01939*, 2026.

Ishika Singh, Valts Blukis, Arsalan Mousavian, Ankit Goyal, Danfei Xu, Jonathan Tremblay, Dieter Fox, Jesse Thomason, and Animesh Garg. ProgPrompt: Generating Situated Robot Task Plans using Large Language Models. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 11523–11530, 2023. doi: 10.1109/ICRA48891.2023.10161317.

Chan Hee Song, Valts Blukis, Jonathan Tremblay, Stephen Tyree, Yu Su, and Stan Birchfield. RoboSpatial: Teaching Spatial Understanding to 2D and 3D Vision-Language Models for Robotics. *arXiv preprint arXiv:2411.16537*, 2024.

Stone Tao, Fanbo Xiang, Arth Shukla, Yuzhe Qin, Xander Hinrichsen, Xiaodi Yuan, Chen Bao, Xinsong Lin, et al. ManiSkill3: GPU Parallelized Robotics Simulation and Rendering for Generalizable Embodied AI. *arXiv preprint arXiv:2410.00425*, 2024.

Alexander Veicht, Paul-Edouard Sarlin, Philipp Lindenberger, and Marc Pollefeys. GeoCalib: Single-image calibration with geometric optimization. In *European Conference on Computer Vision*, 2024.

Hanqing Wang, Shaoyang Wang, Yiming Zhong, Zemin Yang, Jiamin Wang, Zhiqing Cui, Jiahao Yuan, Yifan Han, Mingyu Liu, and Yuexin Ma. Affordance-R1: Reinforcement Learning for Generalizable Affordance Reasoning in Multimodal Large Language Model. *arXiv preprint arXiv:2508.06206*, 2025a.

Qineng Wang, Baiqiao Yin, Pingyue Zhang, Jianshu Zhang, Kangrui Wang, Zihan Wang, Jieyu Zhang, Keshigeyan Chandrasegaran, et al. MindCube: Spatial Mental Modeling from Limited Views. *arXiv preprint arXiv:2506.21458*, 2025b.

Ruoyao Wang, Peter Jansen, Marc-Alexandre Côté, and Prithviraj Ammanabrolu. ScienceWorld: Is your Agent Smarter than a 5th Grader? *arXiv preprint arXiv:2203.07540*, 2022.

Zixuan Wang, Jia-Ning Li, Yinzhou Tang, Yanzhao Yu, Biao Wu, Meng Fang, Xin Li, and Yiwu Zhong. Cracking Mobile Manipulation: From Failure Diagnosis to Agentic Robot System. *arXiv preprint arXiv:2610.19600*, 2026.

Dongming Wu, Yanping Fu, Saike Huang, Yingfei Liu, Fan Jia, Nian Liu, Feng Dai, Tiancai Wang, Rao Muhammad Anwer, Fahad Shahbaz Khan, and Jianbing Shen. Rag-net: Large-scale reasoning-based affordance segmentation benchmark towards general grasping. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 11980–11990, October 2025.

Leyi Wu, Yifan Zhao, Jinjie Zhang, Suzeyu Chen, Wosong Chen, Zhifei Chen, Tianshuo Xu, Qingchun He, Hongxin Hu, Haojian Huang, Yangkai Wei, Wenqian Li, Yinchuan Li, and Ying-Cong Chen. RoboStressBench: Benchmarking VLM Robustness to Physical Visual Stress in Embodied Scenes. *arXiv preprint arXiv:2606.00828*, 2026.

Runsen Xu, Weiyao Wang, Hao Tang, Xingyu Chen, Xiaodong Wang, Fu-Jen Chu, Matt Feiszli, and Kevin J. Liang. Multi-SpatialMLLM: Multi-Frame Spatial Understanding with Multi-Modal Large Language Models. *arXiv preprint arXiv:2505.17015*, 2025.

Jihan Yang, Shusheng Yang, Anjali W. Gupta, Rilyn Han, Li Fei-Fei, and Saining Xie. Thinking in Space: How Multimodal Large Language Models See, Remember, and Recall Spaces. *arXiv preprint arXiv:2412.14171*, 2024.

Rui Yang, Hanyang Chen, Junyu Zhang, Mark Zhao, Cheng Qian, Kangrui Wang, Qineng Wang, Teja Venkat Koripella, et al. EmbodiedBench: Comprehensive Benchmarking Multi-modal Large Language Models for Vision-Driven Embodied Agents. *arXiv preprint arXiv:2502.09560*, 2025a.

Sihan Yang, Runsen Xu, Yiman Xie, Sizhe Yang, Mo Li, Jingli Lin, Chenming Zhu, Xiaochen Chen, et al. MMSI-Bench: A Benchmark for Multi-Image Spatial Intelligence. *arXiv preprint arXiv:2505.23764*, 2025b.

Shiduo Zhang, Zhe Xu, Peiju Liu, Xiaopeng Yu, Yuan Li, Qinghui Gao, Zhaoye Fei, Zhangyue Yin, et al. VLABench: A Large-Scale Benchmark for Language-Conditioned Robotics Manipulation with Long-Horizon Reasoning Tasks. *arXiv preprint arXiv:2412.18194*, 2024.

Yehang Zhang, Haojian Huang, Yifan Chang, Jianchong Su, Bohan Zhou, Yingjie Xu, Wosong Chen, Tianhao Zhou, Chenxu Wang, Tianyi Zhang, Yangkai Wei, Wenqian Li, Shiyuan Deng, Yinchuan Li, Ying-Cong Chen, and Zexi Li. World Action Agent: Harnessing VLMs for Robot Manipulation via World Action Rehearsal. *arXiv preprint arXiv:2609.29964*, 2026a.

Yehang Zhang, Jianchong Su, Haojian Huang, Yifan Chang, Tianhao Zhou, Xinli Xu, Yingjie Xu, Yinchuan Li, Zexi Li, and Ying-Cong Chen. WorldLines: Benchmarking and Modeling Long-Horizon Stateful Embodied Agents. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*, 2026b.

Zixin Zhang, Kanghao Chen, Hanqing Wang, Hongfei Zhang, Harold Haodong Chen, Chenfei Liao, Litao Guo, and Ying-Cong Chen. A4-Agent: An agentic framework for zero-shot affordance reasoning. *arXiv preprint arXiv:2512.14442*, 2025.

Pukun Zhao, Longxiang Wang, Chen Chen, Peicheng Wang, Fanqing Zhou, Runze Li, and Haojian Huang. STAR: Mitigating Cascading Errors in Spatial Reasoning via Turn-point Alignment and Segment-level DPO. In *IEEE International Conference on Multimedia and Expo (ICME)*, 2026a.

Pukun Zhao, Longxiang Wang, Miaowei Wang, Chen Chen, Fanqing Zhou, and Haojian Huang. EvoEmpirBench: Dynamic Spatial Reasoning with Agent-ExpVer. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pp. 36564–36572, 2026b. doi: 10.1609/aaai.v40i43.40979.

Bohan Zhou, Xingbei Chen, Emily Huang, Weilin Ruan, Haojian Huang, Yehang Zhang, Zexi Li, Wenqian Li, Qize Yu, Zetian Song, Leyi Wu, Jinghao Li, Mingxuan Song, Xinrun Xu, Zongyang Qiu, Yangkai Wei, Tianyi Zhang, Kaiwen Zhou, Yinchuan Li, and James Cheng. RoboAware: Learning to Coordinate Embodied Skills from Counterfactual Outcomes. *arXiv preprint arXiv:2610.11480*, 2026a.

Enshen Zhou, Yibo Li, Jingkun An, Jiayuan Zhang, Shanyu Rong, Mengzhen Liu, Yi Han, Yuheng Ji, Huajie Tan, Jiawei He, Pengwei Wang, Zhongyuan Wang, Cheng Chi, Lu Sheng, and Shanghang Zhang. Towards spatial trace with reasoning in vision-language models for robotics. *arXiv preprint arXiv:2512.13660*, 2026b.

Brianna Zitkovich, Tianhe Yu, Sichun Xu, Peng Xu, Ted Xiao, Fei Xia, Jialin Wu, Paul Wohlhart, Stefan Welker, Ayzaan Wahid, Quan Vuong, Vincent Vanhoucke, Huong Tran, Radu Soricut, Anikait Singh, Jaspiar Singh, Pierre Sermanet, Pannag R. Sanketi, Grecia Salazar, Michael S. Ryoo, Krista Reymann, Kanishka Rao, Karl Pertsch, Igor Mordatch, Henryk Michalewski, Yao Lu, Sergey Levine, Lisa Lee, Tsang-Wei Edward Lee, Isabel Leal, Yuheng Kuang, Dmitry Kalashnikov, Ryan Julian, Nikhil J. Joshi, Alex Irpan, Brian Ichter, Jasmine Hsu, Alexander Herzog, Karol Hausman, Keerthana Gopalakrishnan, Chuyuan Fu, Pete Florence, Chelsea Finn, Kumar Avinava Dubey, Danny Driess, Tianli Ding, Krzysztof Marcin Choromanski, Xi Chen, Yevgen Chebotar, Justice Carbajal, Noah Brown, Anthony Brohan, Montserrat Gonzalez Arenas, and Kehang Han. RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control. In *Proceedings of The 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pp. 2165–2183. PMLR, 2023.

Appendix: Supporting Evidence and Protocols

Reading guide. Appendices A–B define task construction, interfaces, and scoring. Appendix C expands Table 2, including source-composition sensitivity and dispersion for Opus 5.5 and DeepSeek-1017. Appendix D follows the five domains of Section 4.3, from full task comparisons to Manipulation's auxiliary metric definitions. Appendix E specifies recorded execution settings and the fixed-input multi-round review study, and Appendix F illustrates recorded actions and terminal outcomes, with additional examples across all five capabilities in Appendix H. Appendix I analyzes shared failures, resources, and recovery using the complete Planning records and linked execution traces. Appendix G gives the evaluation scales and public resources behind Table 1.

A Benchmark Construction and Source Inventory

Embodied Agent Arena contains 1,000 cases from 32 established sources and our GeoProbe benchmark (Table A1). A case is one query or episode; multiple queries can share a scene. Benchmark-specific allocation uses seed 42 and targets diagnostic coverage of the five capabilities in Section 3.1.

Table A1. The unified 1,000-case arena, drawn from 32 established sources and our GeoProbe benchmark. Detailed source allocations follow below.

Domain	Cases	Share (%)	Task coverage
Geometry	370	37.0	Calibration, pose, scale, tracing
Spatial Reasoning	220	22.0	Relations, views, video
Affordance	70	7.0	Functional parts, contact
Task Planning	157	15.7	Household, navigation, science
Manipulation	183	18.3	Robot operations, sequences
Total	1,000	100.0	660 fixed-input; 340 interactive

Query vocabulary. Figure 2a counts each term once per query, balances weights across source groups and domains, and includes English translations of GeoProbe queries. GeoProbe's ten protocols form one source group.

Geometry (370 cases). InFlux-Real (Liang et al., 2025) contributes 30 single-frame intrinsic-estimation cases, of which four have invalid reference annotations and are excluded from continuous-error calculations. Map-free (Arnold et al., 2022) contributes 30 relative-pose image pairs with known camera intrinsics. MultiSPA-derived uses Map-free images and poses with the camera-motion question formulation associated with Multi-SpatialMLLM (Xu et al., 2025): 12 angle, ten distance, and eight displacement-vector questions. This local adaptation retains the Map-free image and pose references. The pose transformation and camera-center displacement use distinct coordinate conventions and reference targets.

GeoProbe comprises ten diagnostic protocols: 168 queries over 130 unique RGB images, predominantly rendered in Blender, with four real-photo displacement cases. Shared scenes separate calibration, pose, depth, displacement, and scale requirements (Table A2).

TraceSpatial-Bench (Zhou et al., 2026b), introduced with RoboTracer, adds paired 2D and 3D queries on 56 scenes. These 112 Geometry cases predict offline object trajectories, scored for reference agreement and geometric constraints.

Table A2. GeoProbe's ten diagnostic protocols, with shared images and scenes.

Diagnostic	Cases	Diagnostic	Cases
Object displacement, fixed camera	12	Full camera intrinsics	24
Object displacement, small view change	12	Focal length, centered principal point	12
Object displacement, large view change	12	Relative camera pose	24
Real tabletop displacement	4	Sparse depth	12
Object dimensions	12	Reference-based scale	44

Spatial Reasoning (220 cases). 3DSRBench (Ma et al., 2024) contributes 50 relation questions; 28 flip-tagged entries were displayed unflipped. Scores use corrected references for five affected answers under the unflipped presentation. MindCube (Wang et al., 2025b) contributes 50 multi-view cases: 39 among-view, six around-view, and five rotation. VSI-Bench (Yang et al., 2024) contributes 50 video questions: nine absolute-distance, nine object-size, fourteen relative-direction, six relative-distance, five counting, four room-size, two appearance-order, and one route-planning. Eight frames form its initial observation. RoboSpatial-Home (Song et al., 2024) contributes ten cases each for compatibility, configuration, and context. BOP-ASK (Bhat et al., 2025) contributes twelve relative-position and eight relative-depth questions. MMSI-Bench (Yang et al., 2025b) contributes 20 cross-image questions. Together these comprise 183 categorical, 27 numerical, and ten point-set questions, scored as defined in Appendix B.3.

Affordance (70 cases). UMD (Myers et al., 2015) contributes 30 views spanning five tool classes and six functions: grasp, cut, scoop, contain, pound, and support. Multiple views can depict the same object. The 20 ReasonAff-style cases adapt instruction-conditioned queries motivated by ReasonAff (Wang et al., 2025a) to 3DOI (Qian & Fouhey, 2023) interaction annotations. Another 20 RAGNet-3DOI cases use explicit action-conditioned queries (Wu et al., 2025). The two subsets use disjoint images. These images and annotations support native point and region evaluation.

Task Planning (157 cases). ALFRED (Shridhar et al., 2019) and ALFWorld (Shridhar et al., 2020) contribute 20 visual cases each, covering placement, heating, cooling, cleaning, containers, multiple objects, and illumination. ScienceWorld (Wang et al., 2022) contributes 73 text-based cases across eighteen topics. VirtualHome (Puig et al., 2018) contributes 21 symbolic programs in seven scenes. The composition analysis removes these 94 cases together. DiscoveryWorld (Jansen et al., 2024) contributes ten tasks with structured observations. HumanCLAW contributes thirteen Habitat navigation and sitting episodes: five sofa, three toilet, two chair, two plant, and one television target. Observations are visual, textual, or symbolic.

Table A3. Manipulation tasks across 13 sources, with identical assignments for all agents.

Benchmark	Cases	Task coverage
CALVIN (Mees et al., 2021)	10	Ten sequences, each with five ordered subgoals
CLIPort (Shridhar et al., 2021)	18	Eighteen task and color-generalization configurations
RLBench (James et al., 2019)	10	Ten task configurations
RoboCasa (Nasiriany et al., 2024)	10	Seven atomic and three composite tasks
RoboCasa365	10	Three atomic and seven composite tasks
RoboWits	10	Ten official task configurations
RoboTwin 2.0 (Chen et al., 2025a)	50	Fifty distinct manipulation tasks
VIMA-Bench (Jiang et al., 2023)	10	Six placement-generalization and four new-task cases
VLABench (Zhang et al., 2024)	25	8 ID, 3 category, 4 commonsense, 7 semantic, 3 texture
ManiSkill (Tao et al., 2024)	10	Nine binary tabletop tasks; one quadruped reward task
LIBERO-PRO (Fu et al., 2026)	10	Ten object, spatial, goal, and multi-object configurations
robosuite	5	Two lift seeds; stacking, restacking, and wiping
BEHAVIOR-1K (Li et al., 2024a)	5	Five multi-object household activities

Manipulation (183 cases). Table A3 lists 182 binary tasks and one continuous-reward task. LIBERO-PRO, robosuite, and BEHAVIOR-1K use benchmark-specific code interfaces (Fu et al., 2026). RoboCasa365 uses variations designated for pretraining in the source inventory. Appendix B.1 specifies the native assistance available to agents.

B Evaluation Protocol and Reproducibility

B.1 Observations and Model-Directed Interaction

Geometry, Spatial Reasoning, and Affordance inspect fixed observations; Planning and Manipulation interact with an environment. All domains except Manipulation use a persistent Python runtime; Manipulation uses its native agent loop. Both enforce episode-level budgets and termination. Variables and environment state persist between code rounds, and images produced by a cell reach the next model request after that cell ends.

Table A4. Model-visible observations, actions, and feedback. Private scoring annotations are unavailable to the agent.

Domain	Observation support	Actions and feedback
Geometry	Image or image pair; intrinsics for Map-free	One final numerical submission
Spatial Reasoning	Images, crops, and video-frame queries	Evidence inspection and answer submission
Affordance	Image and interaction instruction	Point and box submission; polygons in the added experiment
Task Planning	Images, text, or symbolic state	Navigation, interaction, programs, public progress
Manipulation	Protocol-specific scene/state observations and native grounding helpers	Motion or skill calls; execution and task checks

ALFRED and visual ALFWorld expose observation, search, navigation, interaction, and memory primitives. CLIPort and VIMA expose instance inspection and pick-and-place actions. RoboTwin provides actor localization, pose inspection, planning probes, and execution. LIBERO-PRO, robosuite, and BEHAVIOR-1K expose benchmark-specific observation and action interfaces; RoboCasa includes fixture and contact localization. The model makes decisions using these native helpers; the baseline adds no specialist perception or verification pipeline. Geometry requests direct visual estimates and excludes pose-solver implementation.

A runtime exception can be followed by another code round if budget remains; prior state changes are not automatically undone. Model requests, code rounds, primitive calls, and environment steps count different units. Stored visualization frames are also distinct from images actually sent to the model. Qualitative figures sample the recorded camera streams.

B.2 Models and Execution Budgets

Table A5. Models and configured identifiers for all ten agents in the five domains.

Model label	Identifier
GPT-6 Astra	`gpt-6-astra`
GPT-6 Sol	`gpt-6-sol`
Claude Fable 5.1	`claude-fable-5-1`
Claude Opus 5.5	`claude-opus-5-5`
DeepSeek-V4.1-Flash-1017	`deepseek-flash`
Gemini 3.8 Flash	`gemini-3.8-flash`
Gemini 4 Argon	`gemini-4-argon`
Qwen 3.8 Max	`qwen3.8-max`
Qwen 3.5 397B-A17B	`qwen3.5-397b-a17b`
Qwen 3.5 27B	`qwen3.5-27b`

Gemini 3.8 Flash and Gemini 4 Argon are labeled Gemini 3.8 and Gemini 4 in compact figures and tables. Their recorded API identifiers are `gemini-3.8-flash` and `gemini-4-argon`, respectively. Appendix J details Gemini 4 Argon's coverage and execution settings.

DeepSeek-1017 (DS-1017 in compact figures and tables) denotes the 17 October 2026 release DeepSeek-V4.1-Flash-1017, served by `deepseek-flash`; this alias is not an immutable model snapshot. Its evaluation was recorded on 18–19 October 2026. DeepSeek-1017 Spatial runs use no YOLOE, and 3DSRBench inputs use the corrected image flips and matching native labels. The original fixed-input and Planning configurations use temperature zero. Opus 5.5 uses the recorded identifier above; its Manipulation reports retain their own source-specific execution settings. Baseline request limits are 1,800 output tokens and 360 seconds; fixed-input profiles also allow up to 16,384 tokens and 900 seconds. On the configured CLI interface, reasoning effort is maximum for Sol and high for Astra and Fable; API requests omit this parameter. Astra's first repetition on the original 90-case Geometry core uses the high-effort CLI configuration. Manipulation's request settings and recorded execution limits are specified in Appendix E.1.

Table A6. Baseline episode budgets. Tokens accumulate within each case; rounds are code interactions, not repetitions. Dashes denote unspecified fields. Appendix B.4 lists fixed-input settings.

Entry	Code rounds	Tokens	Env. steps	Primitives	Episode seconds	Task seconds
Geometry	12	120k	–	–	–	960
Spatial: non-VSI	6	96k	–	–	900	960
Spatial: VSI	8	240k	–	–	1,200	1,500
ALFRED, ALFWorld	8	320k	120	540	1,200	1,500
DiscoveryWorld	8	320k	100	400	1,200	1,500
HumanCLAW	101	2,000k	100	400	10,800	11,000
ScienceWorld	8	320k	100	500	1,200	1,500
VirtualHome	8	320k	16	120	1,200	1,500
Affordance	12	120k	–	–	–	960

Execution and native-call time limits are clipped by remaining budgets. Appendix E.1 gives Manipulation's recorded execution settings.

B.3 Scoring and Repeated Evaluation

Submission and termination. A final submission commits a fixed-input prediction. Interactive runs end when the agent stops, the environment reaches a terminal condition, or a source-specific round, token, action, or time limit is reached. The evaluator scores the submitted output or recorded state independently of the agent's declaration. A complete episode includes all its internal reasoning, actions, and recovery; these steps are not separate attempts.

Task scores and aggregation. Categorical questions use answer accuracy and interactive tasks use native goal success. Invalid or absent predictions remain in success denominators. Invalid reference annotations are excluded before scoring. Continuous errors average valid predictions with compatible metrics and units within each run. Repeated metrics then weight the run-level scores equally, with valid coverage reported separately. Numerical Spatial Reasoning uses exact-match success, $|\hat{y} - y| < 10^{-6}$, alongside continuous errors; this differs from VSI-Bench's official mean relative accuracy. Point-set questions retain their local set criterion.

Pass@1 and repeated scores. For N assigned cases and R repetitions, let $s_{ir} \in \{0, 1\}$ denote success on case i in run r. The average single-episode success rate is

$$\mathrm{Pass@1} = \frac{1}{R} \sum_{r=1}^{R} \left(\frac{1}{N} \sum_{i=1}^{N} s_{ir} \right). \tag{2}$$

This estimates success in one complete episode under the specified condition. Fresh agent state is used for formal repetitions. For any per-run score M_r, we compute

$$\bar{M} = \frac{1}{R} \sum_{r=1}^{R} M_r, \qquad v_M = \frac{1}{R-1} \sum_{r=1}^{R} (M_r - \bar{M})^2, \qquad \mathrm{SD}_M = \sqrt{v_M}. \tag{3}$$

Tables using $\bar{M} \pm \mathrm{SD}_M$ report variation between run-level scores. Variance is supplied separately in squared metric units. Figure 4 averages valid-case errors within each run and then weights the recorded runs equally. Its coverage sensitivity instead fixes the cases valid in every shared repetition for each model pair; those common-cohort estimates are reported separately. Figure 5a–b uses per-run categorical accuracies; panel c uses full-source native Pass@1. The original pairwise gaps average accuracy differences over shared run indices, with the comparator fixed per subtask. Run indices align reporting repetitions; seeds and tool configurations can differ. Run-to-run SD requires $R > 1$.

Scalar estimates and depth. For J valid targets, $e_j = \hat{y}_j - y_j$ gives $\mathrm{MAE} = J^{-1} \sum_j |e_j|$ and $\mathrm{RMSE} = \sqrt{J^{-1} \sum_j e_j^2}$. Absolute relative error is $\mathrm{AbsRel} = J^{-1} \sum_j |e_j|/|y_j|$ on nonzero targets, and $\mathrm{MAPE} = 100\,\mathrm{AbsRel}$ is expressed in percent. Size and focal-length estimates use absolute and relative errors. Depth additionally reports RMSE in meters. Depth errors are computed over positive finite predictions within each image, then averaged across images, with valid-point coverage reported. The fraction within 10% relative error and $\delta_1 = J_{\mathrm{all}}^{-1} \sum_j \mathbf{1}[\max(\hat{y}_j/y_j, y_j/\hat{y}_j) < 1.25]$ use all queried points as their denominator; invalid predictions contribute zero. Ordinal accuracy is the fraction of eligible point pairs with the correct depth ordering, with invalid predictions counted as incorrect. Scale alignment removes the global depth factor.

Displacement and camera pose. For a nonzero scalar or vector target y, motion relative error is $\|\hat{y} - y\|_2 / \|y\|_2$. The MultiSPA-derived tolerance is 20%; a zero target requires zero absolute error. GeoProbe object-displacement diagnostics separately test movement status, direction, absolute distance error, and direction agreement with distance error at most 1 cm. Stationary cases are evaluated without division by zero. Camera translation error is $e_t = \|\hat{\mathbf{t}} - \mathbf{t}\|_2$, reported

in centimeters. Rotation error is $e_R = \arccos[(\mathrm{tr}(R^\mathsf{T}\hat{R}) - 1)/2]$, in degrees, clipping the argument to $[-1, 1]$. GeoProbe reports component errors and joint success within 2° and 2 cm.

Camera calibration. Focal MAPE averages the absolute relative errors of f_x and f_y per case. Principal-point error is $e_c = \|\hat{\mathbf{c}} - \mathbf{c}\|_2$ in pixels, also normalized by the image diagonal D. Joint calibration requires both focal components to satisfy 10% relative error and $e_c \le 0.02D$. Main-table pose errors, complementary continuous metrics, and Figure 4 use equal-run means and run SD, as specified above.

Spatial tracing. For each case, predicted and reference paths are independently resampled to $L = 100$ equally spaced arc-length positions. Trajectory error is $\mathrm{RMSE}_{\mathrm{trace}} = \sqrt{L^{-1}\sum_{\ell=1}^{L}\|\hat{\mathbf{p}}_\ell - \mathbf{p}_\ell\|_2^2}$. We average this per-case value over valid scenes. The 2D metric uses normalized image coordinates; the 3D metric uses meters. The two metrics retain their respective units. The supplied variance is the population variance of the 56 per-scene RMSE values for a single run.

Offline 3D success requires valid initiation, target arrival, and collision checks to pass together. The predicted start must lie within 20 cm of the object point cloud or the reference start. At least one of the final three predicted points must lie within 20 cm of the target 3D box; for two-point paths, only the endpoint is checked. Translating the object point cloud to each checked predicted position must yield at most 20% overlap with the environment. Collision is checked at the sampled positions. Missing or invalid trajectories count as failures. 2D tracing reports continuous error. Qwen-27B trajectory outputs undergo format normalization that preserves all predicted coordinate values.

Native affordance grounding. For a directly predicted bounding box B and ground-truth mask G, the box interior is rasterized and $\mathrm{IoU}_{B,G} = |B \cap G|/|B \cup G|$. Contact validity is $\mathbf{1}[p \in G]$ for the submitted point p, with a reference-box fallback only where a mask is unavailable. A separate diagnostic uses $\mathrm{IoU}_{\mathrm{box}} = |B \cap B_G|/|B \cup B_G|$ against the reference box B_G. Strict box success requires $\mathrm{IoU}_{\mathrm{box}} \ge 0.5$; contact validity is scored independently. The source field named `giou` contains ordinary box–mask IoU, not generalized IoU. The separate polygon protocol rasterizes the directly submitted polygons into a region P and scores $|P \cap G|/|P \cup G|$. It adds no external segmenter or iterative segmentation refinement; Appendix C.5 gives output validation and coverage.

Resource accounting. Inference tokens, action calls, model rounds, and elapsed time measure different aspects of execution. The CALVIN recovery analysis counts episodes from the adopted records; detailed case traces report actions and elapsed time where available. Request, trial, and native-call time limits are specified separately in Appendix E.1.

B.4 Repetition Protocol and Result Accounting

Repetitions. In the original seven-agent cohort, non-tracing Geometry, Spatial Reasoning, and Affordance use five repetitions per case for six models and three for Fable. Opus 5.5 uses five runs for non-tracing Geometry, Spatial Reasoning, and point-and-box Affordance. DeepSeek-1017 uses five runs for non-tracing Geometry, Spatial Reasoning, and point-and-box Affordance. Tracing, Planning, Manipulation, and direct-polygon prediction use one run per assignment for all models. Run-to-run SD is reported only for repeated measurements. Manipulation's observation schedules appear in Appendices E.1 and C.4.

Additional execution configurations. Fixed-input evaluation also uses per-case limits of up to 48 code rounds and 480k tokens for Geometry and Affordance, 24 rounds and 384k tokens

for non-VSI Spatial Reasoning, and 32 rounds and 960k tokens for VSI. Task time limits reach 7,200 seconds, or 10,800 seconds for VSI. Budgets vary by source and execution configuration.

Result selection. In the original cohort, fixed-input and Planning trial slots use the first valid outcome, independent of correctness. Manipulation uses its fixed adopted evaluation records. The original Baseline CSV covers 182 binary assignments. Table 2 retains the original seven-model Baseline population and adds the adopted Opus 5.5, DeepSeek-1017, and Gemini 4 Argon results. Shared source interfaces and task denominators do not imply identical observation schedules or execution ceilings across batches. The recorded differences are specified in Appendix E.1; later logs do not replace the retained Baseline scores. Appendix E.1 defines the recorded settings and missing-record accounting.

Coverage and scoring denominators. Table A7 reports the original cohort and DeepSeek-1017 coverage. Opus 5.5 metric-specific valid counts are in Table A12; no whole-domain valid count is inferred from overlapping summary metrics. Opus 5.5 HumanCLAW uses the final completed record for each of the 13 assignments: navigation succeeds at 20 cm proximity, while sitting requires the native interaction condition. Two RLBench exceptions and one RoboTwin exception count as unsuccessful assignments; Opus 5.5 Manipulation remains 87/182 (47.8%). InFlux excludes four invalid references. Binary criteria retain all assigned tasks. Reported success requires the native evaluator, rather than an agent declaration. Superseded HumanCLAW attempts are not counted as additional assignments. Manipulation covers 182 binary tasks, including all five BEHAVIOR-1K assignments per model; AnymalC-Spin is excluded from this binary comparison. Continuous metrics retain finite valid predictions.

Table A7. Original seven agents and DeepSeek-1017 (DS-1017). Valid submission coverage for fixed-input domains and recorded terminal outcomes for Planning. Assigned denominators include repetitions and invalid-reference slots; invalid references are excluded from metric scoring. Manipulation uses the fixed allocation in Table A3.

Domain	Astra	Sol	Fable	Q-Max	Gemini 3.8	Q-397B	Q-27B	DS-1017
Geometry	1382/1402	1382/1402	819/886	1382/1402	1314/1402	1382/1402	1376/1402	1356/1402
Spatial Reasoning	1060/1100	1100/1100	586/660	1097/1100	1077/1100	1033/1100	982/1100	954/1100
Affordance	350/350	350/350	208/210	350/350	335/350	349/350	349/350	348/350
Task Planning	157/157	157/157	157/157	157/157	157/157	157/157	157/157	157/157

Manipulation diagnostics. CALVIN reports average consecutive goals completed. Other auxiliary metrics average task-normalized native criteria or control evidence over fixed applicable subsets. They differ in their scoring items and evidence time windows; Appendix D.5.1 defines the aggregation, coverage, and missing-record treatment.

Main-table metric scope. Geometry rotation and translation average valid predictions from the 30 Map-free and 24 GeoProbe pose cases within each run, in degrees and centimeters; the main table reports the mean and sample SD across runs. Spatial Pass@1 covers all 220 cases; numerical AbsRel covers 27, and categorical accuracy separately covers 183. Original Affordance box–mask IoU and contact validity cover all three sources, with 70 cases. Polygon IoU evaluates the same allocation under the direct-polygon protocol. Planning uses terminal success across 157 cases for every model. Manipulation uses task-weighted native success across 182 binary assignments; Tables A9 and A11 report each source separately.

Dispersion and valid subsets. Continuous metrics first average valid predictions within each run, then report the equal-weight mean and sample SD across runs (Equation 3). This SD

measures variation between repeated evaluations, not variation among individual cases. Common-valid-cohort checks appear in Appendices D.1 and D.2.

C Complete Model Comparisons

This section supports Section 4.2 at three resolutions: complementary native metrics, differences among source benchmarks, and the metric profiles shown in Figure 3. Task-level explanations follow in Appendix D.

C.1 Metrics and Source-Level Results

Table A8 gives complementary native metrics. Tables A9 and A11 resolve the results by source, using the denominators and repetitions in Appendix B.4. These breakdowns distinguish broad domain performance from strengths concentrated in particular task collections. **Reading the metric families.** GeoProbe focal MAPE combines full-intrinsics and focal-only queries; displacement MAE combines its four camera/environment conditions. Scale MAPE combines object dimensions and numerical reference-scale queries, while qualitative scale judgments retain Pass@1. TraceSpatial reports 2D and 3D errors separately because their coordinates and units differ. Each metric retains the target's physical interpretation.

Table A8. Complementary native metrics across four domains; DS-1017 abbreviates DeepSeek-V4.1-Flash-1017. Repeated metrics show the mean above ± run SD (five runs; three for Fable). Each run averages valid continuous predictions or scores all assigned binary cases; runs receive equal weight. Polygon IoU uses a separate single run for all ten models. Planning uses all 157 cases for every model. N counts eligible cases. GP denotes GeoProbe; Manipulation source results appear in Tables A9 and A11.

Metric	N	Astra	Sol	Fable	Q-Max	Gemini 3.8	Q-397B	Q-27B	Opus 5.5	DS-1017	Gemini 4
Geometry (370 cases)											
InFlux focal MAPE (%)↓	26	**53.02** ±2.15	**47.79** ±2.00	**52.65** ±2.22	**55.50** ±1.04	**52.40** ±3.40	**67.82** ±1.82	**65.86** ±2.08	**60.27** ±1.49	**62.92** ±1.57	**34.84** ±1.63
InFlux principal error (px)↓	26	**3.82** ±0.00	**3.82** ±0.00	**3.82** ±0.00	**3.82** ±0.00	**3.88** ±0.09	**3.82** ±0.00	**3.82** ±0.00	**3.82** ±0.00	**3.55** ±0.46	**3.82** ±0.00
Angle relative L2↓	12	**0.760** ±0.130	**0.915** ±0.442	**1.318** ±0.046	**1.410** ±0.281	**0.961** ±0.083	**1.298** ±0.172	**2.037** ±0.331	**1.166** ±0.113	**1.497** ±0.352	**0.735** ±0.046
Distance relative L2↓	10	**0.294** ±0.024	**0.418** ±0.130	**0.785** ±0.115	**0.560** ±0.090	**0.419** ±0.077	**0.411** ±0.059	**0.331** ±0.095	**0.455** ±0.069	**0.461** ±0.196	**0.313** ±0.017
Vector relative L2↓	8	**0.425** ±0.092	**0.591** ±0.155	**0.625** ±0.094	**0.454** ±0.122	**0.643** ±0.071	**1.018** ±0.080	**1.031** ±0.060	**0.516** ±0.042	**0.743** ±0.254	**0.529** ±0.120
Map-free rotation (°)↓	30	**30.08** ±12.15	**46.60** ±2.25	**50.81** ±8.99	**35.11** ±6.25	**45.19** ±7.64	**63.03** ±1.61	**64.94** ±6.89	**40.95** ±8.10	**52.68** ±6.56	**31.75** ±3.52
Map-free translation (cm)↓	30	**188.98** ±65.21	**265.85** ±10.41	**248.74** ±50.12	**243.06** ±18.41	**272.95** ±59.81	**430.82** ±18.13	**374.96** ±22.44	**243.13** ±35.82	**324.19** ±46.48	**185.06** ±14.78
Map-free direction (°)↓	30	**26.43** ±12.73	**48.91** ±2.14	**54.20** ±7.11	**38.92** ±5.54	**50.72** ±14.17	**95.15** ±6.59	**78.78** ±9.15	**44.29** ±9.06	**64.63** ±6.45	**27.47** ±2.95
GP focal MAPE (%)↓	36	**6.92** ±4.90	**9.57** ±1.25	**31.77** ±1.05	**14.67** ±2.72	**26.59** ±4.73	**33.39** ±2.99	**38.56** ±2.95	**16.48** ±6.13	**33.99** ±5.33	**28.39** ±0.91
GP principal error (px)↓	24	**51.95** ±8.15	**110.62** ±18.22	**62.81** ±8.54	**78.41** ±13.77	**66.83** ±9.70	**58.23** ±1.02	**59.05** ±1.01	**60.46** ±4.13	**75.40** ±9.96	**56.05** ±1.89
Depth MAE (m)↓	12	**0.307** ±0.089	**0.633** ±0.068	**0.416** ±0.088	**0.490** ±0.042	**0.587** ±0.067	**0.934** ±0.072	**0.945** ±0.043	**0.510** ±0.295	**0.688** ±0.089	**0.927** ±0.024
Depth RMSE (m)↓	12	**0.338** ±0.087	**0.753** ±0.086	**0.522** ±0.093	**0.598** ±0.041	**0.697** ±0.064	**1.053** ±0.080	**1.059** ±0.044	**0.599** ±0.302	**0.814** ±0.122	**0.973** ±0.024
Depth AbsRel↓	12	**0.104** ±0.029	**0.229** ±0.027	**0.142** ±0.027	**0.163** ±0.013	**0.192** ±0.021	**0.312** ±0.028	**0.292** ±0.013	**0.169** ±0.108	**0.229** ±0.025	**0.299** ±0.006
Displacement MAE (cm)↓	40	**0.585** ±0.092	**0.973** ±0.056	**1.488** ±0.163	**1.515** ±0.160	**2.278** ±0.170	**2.480** ±0.071	**2.935** ±0.082	**1.164** ±0.475	**1.934** ±0.318	**1.309** ±0.050
GP rotation (°)↓	24	**1.17** ±0.38	**2.54** ±1.02	**2.80** ±0.94	**2.21** ±0.18	**3.10** ±0.48	**7.55** ±0.77	**5.26** ±0.97	**3.05** ±0.96	**3.75** ±0.57	**2.71** ±0.36
GP translation (cm)↓	24	**5.36** ±2.11	**9.50** ±2.71	**9.24** ±2.05	**6.99** ±0.96	**7.85** ±0.90	**23.59** ±1.80	**12.05** ±0.52	**11.13** ±1.48	**14.52** ±1.80	**10.14** ±0.98

Complementary metrics, continued.

Metric	N	Astra	Sol	Fable	Q-Max	Gemini 3.8	Q-397B	Q-27B	Opus 5.5	DS-1017	Gemini 4
Scale MAPE (%)↓	44	1.90 ±0.28	2.84 ±0.21	3.76 ±0.05	3.77 ±0.16	3.93 ±0.46	16.07 ±1.73	16.45 ±0.48	3.91 ±2.74	7.62 ±1.13	2.21 ±0.09
Scale judgment Pass@1 (%)↑	12	100.0 ±0.0	100.0 ±0.0	100.0 ±0.0	100.0 ±0.0	100.0 ±0.0	100.0 ±0.0	93.3 ±7.0	100.0 ±0.0	98.3 ±3.7	100.0 ±0.0
2D trace RMSE↓	56	0.120	0.132	0.118	0.118	0.127	0.155	0.284	0.180	0.141	0.107
3D trace RMSE (m)↓	56	0.382	0.463	0.435	0.469	0.426	0.629	0.682	0.683	0.470	0.310
3D constraint success (%)↑	56	33.9	21.4	25.0	17.9	37.5	8.9	7.1	8.9	14.3	42.9
Spatial Reasoning (220 cases)											
Native Pass@1 (%)↑	220	69.4 ±1.0	60.7 ±2.5	62.0 ±4.6	68.9 ±1.1	65.4 ±1.3	50.6 ±1.8	48.0 ±0.9	67.5 ±5.1	51.6 ±1.5	71.6 ±0.7
Categorical Pass@1 (%)↑	183	82.2 ±0.6	71.3 ±3.0	72.7 ±5.2	81.2 ±1.3	76.7 ±1.9	60.0 ±2.2	56.8 ±0.7	79.2 ±5.6	61.3 ±1.7	84.2 ±1.3
Numerical AbsRel↓	27	0.214 ±0.016	0.273 ±0.016	0.185 ±0.011	0.206 ±0.033	0.161 ±0.028	0.251 ±0.027	0.288 ±0.023	0.184 ±0.040	0.309 ±0.035	0.169 ±0.023
Affordance (70 cases)											
BBox–GT box IoU↑	70	0.390 ±0.005	0.366 ±0.012	0.386 ±0.014	0.382 ±0.022	0.324 ±0.098	0.146 ±0.015	0.021 ±0.008	0.432 ±0.021	0.278 ±0.015	0.484 ±0.012
BBox–GT mask IoU↑	70	0.286 ±0.010	0.264 ±0.010	0.284 ±0.013	0.272 ±0.017	0.251 ±0.081	0.116 ±0.014	0.013 ±0.006	0.311 ±0.020	0.217 ±0.011	0.356 ±0.009
Polygon–GT mask IoU↑	70	0.379	0.325	0.328	0.307	0.291	0.092	0.051	0.397	0.249	0.441
BBox success@0.5 (%)↑	70	46.6 ±1.3	43.7 ±1.6	46.2 ±0.8	45.7 ±1.7	33.1 ±11.8	5.4 ±3.7	0.3 ±0.6	51.4 ±3.0	28.9 ±3.7	56.3 ±2.2
Contact-point validity (%)↑	70	60.3 ±2.6	52.0 ±3.6	57.1 ±2.5	52.9 ±4.2	44.6 ±13.0	32.6 ±4.7	6.3 ±1.6	61.1 ±1.9	50.9 ±4.2	59.7 ±1.2
Task Planning (157 cases)											
Native Pass@1 (%)↑	157	80.3	44.6	63.7	35.7	48.4	17.8	19.1	48.4	31.2	70.1

Table A9. Source results for the original seven agents and DeepSeek-1017 (DS-1017); Opus 5.5 appears in Table A11. Spatial/Affordance: mean above ± run SD (%); Planning: solved/assigned cases; Manipulation: success (%). N counts eligible cases.

Source	N	Astra	Sol	Fable	Q-Max	Gemini 3.8	Q-397B	Q-27B	DS-1017
Spatial Reasoning: categorical accuracy (%)									
3DSRBench	50	73.6 ±2.6	71.2 ±1.8	72.7 ±3.1	74.0 ±3.7	70.0 ±2.4	69.2 ±3.6	57.6 ±2.6	61.6 ±7.1
BOP-ASK	20	96.0 ±4.2	87.0 ±4.5	88.3 ±2.9	92.0 ±2.7	97.0 ±2.7	85.0 ±3.5	84.0 ±4.2	88.0 ±7.6
MindCube	50	90.8 ±1.1	68.0 ±7.3	79.3 ±8.1	90.8 ±2.7	89.2 ±3.0	46.0 ±2.8	50.4 ±4.1	54.4 ±4.3
MMSI-Bench	20	77.0 ±5.7	55.0 ±11.7	66.7 ±7.6	55.0 ±8.7	56.0 ±9.6	35.0 ±9.4	29.0 ±7.4	53.0 ±5.7
RoboSpatial	20	85.0 ±3.5	81.0 ±8.9	88.3 ±2.9	88.0 ±5.7	76.0 ±4.2	90.0 ±3.5	92.0 ±5.7	87.0 ±5.7
VSI-Bench	23	72.2 ±7.3	70.4 ±5.7	36.2 ±28.9	83.5 ±8.4	65.2 ±5.3	44.3 ±6.4	39.1 ±3.1	37.4 ±5.0
Affordance: strict box success at IoU ≥ 0.5 (%)									
ReasonAff-style	20	19.0 ±5.5	6.0 ±4.2	16.7 ±7.6	13.0 ±5.7	5.0 ±0.0	0.0 ±0.0	0.0 ±0.0	3.0 ±4.5
RAGNet-3DOI	20	14.0 ±4.2	15.0 ±5.0	13.3 ±2.9	7.0 ±2.7	6.0 ±2.2	4.0 ±2.2	1.0 ±2.2	5.0 ±6.1
UMD	30	86.7 ±3.3	88.0 ±4.5	87.8 ±5.1	93.3 ±2.4	70.0 ±27.0	10.0 ±7.8	0.0 ±0.0	62.0 ±6.9
Task Planning: terminal success (single run)									
ALFRED	20	15/20	3/20	7/20	3/20	2/20	1/20	0/20	1/20
ALFWorld	20	14/20	4/20	6/20	2/20	4/20	3/20	4/20	4/20
DiscoveryWorld	10	10/10	4/10	7/10	3/10	5/10	1/10	1/10	2/10
HumanCLAW	13	3/13	2/13	1/13	0/13	3/13	0/13	0/13	2/13
ScienceWorld	73	64/73	38/73	59/73	29/73	45/73	11/73	14/73	28/73
VirtualHome	21	20/21	19/21	20/21	19/21	17/21	12/21	11/21	12/21

Source-level comparisons, continued.

Source	N	Astra	Sol	Fable	Q-Max	Gemini 3.8	Q-397B	Q-27B	DS-1017
Total	157	126/157	70/157	100/157	56/157	76/157	28/157	30/157	49/157
Manipulation: task success (%)									
CALVIN	10	30.0	0.0	0.0	0.0	0.0	0.0	0.0	10.0
CLIPort	18	61.1	27.8	33.3	22.2	27.8	22.2	16.7	33.3
RLBench	10	60.0	20.0	10.0	0.0	20.0	0.0	10.0	20.0
RoboCasa	10	10.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RoboCasa365	10	0.0	0.0	10.0	0.0	0.0	0.0	0.0	0.0
RoboTwin 2.0	50	38.0	10.0	14.0	4.0	8.0	6.0	8.0	4.0
RoboWits	10	10.0	10.0	10.0	10.0	20.0	0.0	10.0	20.0
VIMA-Bench	10	40.0	30.0	30.0	10.0	20.0	30.0	30.0	60.0
VLABench	25	72.0	52.0	60.0	20.0	36.0	24.0	24.0	36.0
ManiSkill	9	55.6	0.0	–	0.0	11.1	0.0	0.0	11.1
LIBERO-PRO	10	40.0	30.0	60.0	10.0	40.0	0.0	10.0	50.0
robosuite	5	80.0	80.0	100.0	60.0	100.0	40.0	20.0	80.0
BEHAVIOR-1K	5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total	182	41.8	19.8	24.7†	9.3	18.7	9.9	11.0	20.9

†See Appendix E.1 for coverage and aggregation.

C.2 Sensitivity to Source Composition

Table A10. Task-composition analysis (success %). Spatial excludes VSI-Bench from the categorical set (160 remaining cases); Planning removes ScienceWorld and VirtualHome; Affordance removes UMD and uses strict box success. All columns retain the assigned-case denominator of the indicated subset. Spatial and Affordance show mean ± run SD; Planning uses one run.

Agent	Spatial without VSI	Planning without text/symbolic	Affordance without UMD
Astra	83.6±1.2	**66.7**	16.5±1.4
Sol	71.4±3.3	**20.6**	10.5±2.7
Fable	77.9±3.1	**33.3**	15.0±5.0
Qwen-Max	80.9±1.4	**12.7**	10.0±3.5
Gemini 3.8	78.4±2.2	**22.2**	5.5±1.1
Qwen-397B	62.2±1.6	**7.9**	2.0±1.1
Qwen-27B	59.4±0.9	**7.9**	0.5±1.1
Opus 5.5	81.4±6.9	**28.6**	19.0±4.9
DeepSeek-1017	64.8±1.6	**14.3**	4.0±3.8

Removing ScienceWorld and VirtualHome changes Astra's Planning success from 80.3% to 66.7%, and Qwen-Max's from 35.7% to 12.7%. The same table isolates categorical Spatial Reasoning without VSI-Bench and strict affordance-box success without UMD. These subset comparisons measure sensitivity to the task mixture.

Equal source weighting. To measure sensitivity to source allocation in the original seven-agent cohort, we average complete source scores with equal weight, $M_{\text{source}} = B^{-1} \sum_{b=1}^{B} M_b$, retaining every assigned case within each source. Across all six Spatial sources, Astra scores 71.9% versus Qwen-Max's 69.1%; across all three Affordance sources, contact validity is 54.2% versus Fable's 50.9%; across all six Planning sources, success is 75.2% versus Fable's 53.1%. These are the highest non-Astra macro scores within the original seven-agent cohort. The ranking therefore persists under both case weighting and equal source weighting. These

analyses use the complete domain allocations, separate from the 46-case multi-round review diagnostic in Section 4.4.

C.3 Opus 5.5 Repetitions, Dispersion, and Coverage

Opus 5.5 has five runs on 258 non-tracing Geometry cases, 220 Spatial Reasoning cases, and 70 point-and-box Affordance cases (2,740 assigned records). Four invalid InFlux references are excluded from eligible scoring. Repeated results use the equal-weight mean of per-run metrics and sample SD (divisor four). Tracing, direct polygons, Planning, and Manipulation each retain one run. Table A12 gives metric-specific coverage.

Planning and Manipulation. Planning includes all 157 assignments. The five non-HumanCLAW sources contribute 73 successes, and HumanCLAW adds 3/13, for 76/157 (48.4%). On HumanCLAW, Opus 5.5 succeeds on 3/5 navigation tasks and 0/8 sitting tasks. It finds the target and reaches within 20 cm on 7/13 cases, including 4/8 sitting tasks; proximity does not establish interaction success. Opus 5.5 Manipulation retains its adopted 87/182 result, counting two RLBench exceptions and one RoboTwin exception as unsuccessful assignments. Its CALVIN reports use automatic per-round RGB and the recorded ceilings verified in Appendix E.1. The supplemental recordings in Appendix H illustrate individual episodes; source reports supply their execution configurations.

Sequential progress and final conditions. Opus 5.5 completes at least one through five consecutive CALVIN goals on 80%, 70%, 40%, 40%, and 10% of the ten assignments. Summing these survival rates gives 2.40 consecutive goals on average. On the five ManiSkill diagnostic tasks, it satisfies all final goal conditions on PickCube, PlaceSphere, and StackCube, and none on PegInsertionSide or PlugCharger; their mean condition score is 60%. These recorded diagnostics supplement the main evaluation; Opus 5.5 is outside the paired observation experiment. Opus 5.5 3D tracing has 8.9% joint success across 56 assignments, alongside the 0.683 m RMSE in Table A8.

Table A11. Opus 5.5 source results. Planning includes all six sources. Manipulation retains all binary assignments, including execution errors. Spatial values cover all questions in each source and show mean ± run SD across five repetitions.

Domain	Source	Assigned	Success (%)
Spatial	3DSRBench	50	76.8 ± 9.9
Spatial	BOP-ASK	20	95.0 ± 0.0
Spatial	MindCube	50	85.2 ± 6.7
Spatial	MMSI-Bench	20	59.0 ± 8.2
Spatial	RoboSpatial-Home	30	61.3 ± 4.5
Spatial	VSI-Bench	50	36.4 ± 1.7
Planning	ALFRED	20	15.0
Planning	ALFWorld	20	45.0
Planning	DiscoveryWorld	10	30.0
Planning	ScienceWorld	73	58.9
Planning	VirtualHome	21	71.4
Planning	HumanCLAW	13	23.1
Planning	All assigned	157	48.4
Manipulation	CALVIN	10	10.0
Manipulation	CLIPort	18	61.1

Domain	Source	Assigned	Success (%)
Manipulation	RLBench	10	30.0
Manipulation	RoboCasa	10	0.0
Manipulation	RoboCasa365	10	10.0
Manipulation	RoboWits	10	10.0
Manipulation	RoboTwin 2.0	50	46.0
Manipulation	VIMA-Bench	10	100.0
Manipulation	VLABench	25	76.0
Manipulation	ManiSkill	9	55.6
Manipulation	LIBERO-PRO	10	80.0
Manipulation	Robosuite	5	100.0
Manipulation	BEHAVIOR-1K	5	0.0
Manipulation	All assigned	182	47.8

Table A12. Opus 5.5 repeated metrics as mean ± run SD. Coverage counts valid continuous predictions or all assigned binary predictions across five runs.

Metric	Mean ± SD	Scored/eligible
InFlux focal MAPE	60.27±1.49	130/130
InFlux principal error	3.82±0.00	130/130
Angle relative L2	1.166±0.113	60/60
Distance relative L2	0.455±0.069	50/50
Vector relative L2	0.516±0.042	40/40
Map-free rotation	40.95±8.10	142/150
Map-free translation	243.13±35.82	142/150
Map-free direction	44.29±9.06	142/150
GP focal MAPE	16.48±6.13	180/180
GP principal error	60.46±4.13	120/120
Depth MAE	0.510±0.295	60/60
Depth RMSE	0.599±0.302	60/60
Depth AbsRel	0.169±0.108	60/60
Displacement MAE	1.164±0.475	200/200
GP rotation	3.05±0.96	120/120
GP translation	11.13±1.48	120/120
Scale MAPE	3.91±2.74	220/220
Scale judgment Pass@1	100.0±0.0	60/60
Numerical AbsRel	0.184±0.040	130/135
BBox–GT box IoU	0.432±0.021	350/350
BBox–GT mask IoU	0.311±0.020	350/350
BBox success@0.5	51.4±3.0	350/350
Contact-point validity	61.1±1.9	350/350
Spatial Native Pass@1	67.5±5.1	1100/1100
Categorical Pass@1	79.2±5.6	915/915

C.4 DeepSeek-1017: Protocol, Dispersion, and Task Outcomes

DeepSeek-1017 denotes DeepSeek-V4.1-Flash-1017, the 17 October 2026 release served by `deepseek-flash`. Its main evaluation contains 3,192 assigned case–run slots covering all 1,000 cases; the added polygon experiment contributes 70 separate slots. The supplied archive fixes one adopted record per task and run; superseded transport and adapter attempts remain provenance records rather than additional trials. Non-tracing Geometry, Spatial Reasoning, and point-and-box Affordance have five runs; tracing, Planning, and Manipulation have one. Four invalid InFlux references are excluded from eligible scoring. Failed or absent binary

predictions retain their assigned denominator; continuous errors use finite valid predictions. Five-run SDs measure variation between run means.

Observations and execution budgets. Spatial runs use no YOLOE; 3DSRBench's corrected flipped inputs use matching native labels. In Manipulation, DeepSeek-1017 uses native tools and receives current RGB views at every decision, completing 38/182 tasks. Its source-specific round, token, and in-episode time ceilings appear in Table A15. Original Affordance uses native points and rectangles. A separate direct-polygon run scores all 70 assigned cases, yielding 0.249 IoU; its 67 valid outputs and three zero-scored failures are detailed in Appendix C.5.

Local evidence and complete task outcomes. The source breakdown in Table A14 connects aggregate performance to task demands. DeepSeek-1017 succeeds on 2/5 HumanCLAW navigation tasks and 0/8 sitting tasks. It solves 49/157 Planning assignments, falling to 9/63 when text/symbolic ScienceWorld and VirtualHome are removed. CALVIN averages 1.40 consecutive goals, while only 1/10 full five-goal sequences succeeds. In the five-task ManiSkill diagnostic, none of the final condition sets is fully satisfied. BEHAVIOR-1K has no complete success in five assignments: pre-termination logs confirm localization in four and a grasp candidate in three. Only stages confirmed before termination enter the diagnostic.

Table A13. DeepSeek-1017 mean ± run SD and metric-specific coverage. Continuous metrics average valid predictions within each run; binary rates retain all assigned cases. Polygon, Planning, and Manipulation use one run.

Metric	Mean	Run SD	Scored/eligible
Pose rotation (°)	30.2	3.9	259/270
Pose translation (cm)	182.0	29.0	259/270
Spatial Pass@1 (%)	51.6	1.5	1100/1100
Numerical AbsRel	0.309	0.035	120/135
Box–mask IoU	0.217	0.011	348/350
Contact validity (%)	50.9	4.2	350/350
Polygon–mask IoU	0.249	—	70/70
Planning success (%)	31.210	—	157/157
Manipulation success (%)	20.879	—	182/182

Table A14. DeepSeek-1017 native source outcomes. Spatial includes all question types; Affordance uses the original point-or-box criterion. Manipulation measures complete-task success.

Domain	Source	Cases	Success (%)
Spatial	3DSRBench	50	61.6±7.1
Spatial	BOP-ASK	20	88.0±7.6
Spatial	MMSI-Bench	20	53.0±5.7
Spatial	MindCube	50	54.4±4.3
Spatial	RoboSpatial-Home	30	58.0±3.8
Spatial	VSI-Bench	50	20.0±2.0
Affordance	RAGNet-3DOI	20	39.0±7.4
Affordance	ReasonAff-style	20	15.0±3.5
Affordance	UMD	30	86.0±6.4
Planning	ALFRED	20	5.0
Planning	ALFWorld	20	20.0
Planning	DiscoveryWorld	10	20.0
Planning	HumanCLAW	13	15.4
Planning	ScienceWorld	73	38.4

Domain	Source	Cases	Success (%)
Planning	VirtualHome	21	57.1
Manipulation	BEHAVIOR-1K	5	0.0
Manipulation	CALVIN	10	10.0
Manipulation	CLIPort	18	33.3
Manipulation	LIBERO-PRO	10	50.0
Manipulation	ManiSkill	9	11.1
Manipulation	RLBench	10	20.0
Manipulation	RoboCasa	10	0.0
Manipulation	RoboCasa365	10	0.0
Manipulation	RoboTwin2	50	4.0
Manipulation	RoboWits	10	20.0
Manipulation	Robosuite	5	80.0
Manipulation	VIMA-Bench	10	60.0
Manipulation	VLABench	25	36.0

Table A15. DeepSeek-1017 Manipulation ceilings. Tokens are cumulative; minutes are in-episode wall-clock limits. Process wrappers allow additional startup and teardown time. Each assigned task is one episode. ManiSkill success counts nine binary tasks; its additional locomotion case has a continuous reward.

Source	Model rounds	Tokens	Minutes
CALVIN	40	1280k	120
BEHAVIOR-1K	24	640k	180
LIBERO-PRO	24	640k	180
Robosuite	24	640k	180
CLIPort	24	960k	60
ManiSkill	32	640k	60
RLBench	16	640k	60
RoboCasa	24	1120k	60
RoboCasa365	24	1120k	60
RoboTwin2	16	640k	60
RoboWits	24	640k	60
VIMA-Bench	16	640k	60
VLABench	16	640k	60

Table A16. DeepSeek-1017 execution stages. Fractions count confirmed events per assignment, with zero credit for unverified events. BEHAVIOR-1K uses pre-termination evidence.

Source	Diagnostic	Confirmed / assigned
CALVIN	All five ordered goals completed	1/10
ManiSkill	All final conditions on the five-task subset	0/5
RoboCasa	Approach; grasp and lift; stable release	3/4; 3/4; 0/4
RoboCasa365	Approach; grasp and lift; stable release	3/7; 3/7; 0/7
BEHAVIOR-1K	Localization; grasp candidate	4/5; 3/5
	Arrival; holding; held motion; release	0/5 each

C.5 Direct Polygon Grounding

The ten-model Affordance experiment retains the same 70 images and interaction instructions, with 20 RAGNet, 20 ReasonAff-style/3DOI, and 30 UMD cases. Each model directly predicts polygons in image coordinates alongside its point and bounding box. Rasterizing the polygons

yields a predicted mask for IoU and Dice; no SAM, detector, or iterative segmentation refinement is added. The separate polygon run tests a more flexible output representation on the same tasks. Table 2 compares box and polygon overlap, while Table A8 reports point validity.

Coverage and aggregation. Table A17 reports single-run polygon scores by source. Invalid polygons and absent final submissions score zero. The original point/box pass flag is not treated as polygon success. No run-to-run SD is inferred from one run. DeepSeek-1017 yields 67 valid polygons: two outputs have vertices outside the image, and one case reaches the 12-round limit without a submission. All three contribute zero to the 70-case mean (24.92%). Its existing thinking-enabled, high-effort setting uses a 1,800-token visible-response ceiling; the 120k-token case budget also counts reasoning. The other limits remain 12 rounds and 900 seconds per case. Each of the 70 cases contributes one adopted attempt.

Table A17. Polygon–mask IoU (%), with one run on 20 RAGNet, 20 ReasonAff-style, and 30 UMD cases. Invalid or absent submissions score zero; all 70 cases enter the overall mean.

Model	RAGNet	ReasonAff	UMD	Overall
Astra	19.19	10.60	68.52	37.88
Sol	13.22	7.40	62.17	32.54
Fable	12.95	17.00	56.51	32.77
Qwen-Max	5.64	7.01	63.28	30.74
Gemini 3.8	10.92	4.21	57.71	29.06
Qwen-397B	5.86	2.45	16.02	9.24
Qwen-27B	3.98	0.30	9.05	5.10
Opus 5.5	18.55	13.29	71.49	39.74
DeepSeek-1017	8.67	9.22	46.21	24.92
Gemini 4 Argon	14.03	30.71	73.08	44.10

Gemini 4 Argon has the highest aggregate at 44.10%, followed by Opus 5.5 at 39.74% and Astra at 37.88%; all three score far lower on the interaction-conditioned scenes than on UMD tools (Figure 6b).

C.6 Interpreting the Capability Profiles

Figure 3 places complementary metrics on fixed visualization scales. Geometry is $50[\exp(-r/45) + \exp(-t/150)]$, where r and t are the mean per-run rotation error in degrees and translation error in centimeters. Spatial is $[p + 100\exp(-a/0.25)]/2$, where p is native Pass@1 (%) and a is numerical AbsRel. Affordance is $[100u + c]/2$, where u is original box–mask IoU and c is contact validity (%). Planning and Manipulation use their task success rates (%). G, S, A, P, and M denote these five domains. These monotone scales summarize complementary metrics; they are not native task tolerances or success probabilities.

Opus 5.5 includes all 157 Planning assignments, scoring 76/157 (48.4%), and all 182 adopted Manipulation assignments. The Affordance index uses box–mask IoU and contact validity from Table A8; polygon IoU is reported as an additional metric.

DeepSeek-1017 uses the same transformations, giving G/S/A/P indices of 40.4/40.4/36.3/31.2 and a Manipulation score of 20.9. Its Manipulation result uses the recorded per-round RGB schedule. Gemini 4 Argon gives 57.8/61.2/47.7/70.1/44.0 and also uses per-round RGB in Manipulation. Panel a plots all ten profiles, and panel b reports their domain indices separately.

D Task Requirements and Model Comparisons

These diagnostics explain the task requirements behind Section 4.3 and Figures 4–8. Focused comparisons examine Astra against the original seven-agent cohort, while task tables include DeepSeek-1017 where matching records are available. Appendix C.3 provides Opus 5.5 results. Each domain connects the aggregate score to specific estimation, grounding, or execution conditions; Appendix B.4 defines repetitions and valid coverage.

D.1 Geometry: Task Requirements and Source Conditions

Task-level comparisons. Table A18a supplies the absolute errors behind Figure 4a. Its relative reduction is $100(1 - e_{\text{Astra}}/e_{\text{peer}})$, using the lowest-error non-Astra peer in the original cohort for each target. Equal-run means retain task-specific units: MAPE for focal length, size, and height, and $100 \times \text{AbsRel}$ for depth. Astra leads the five GeoProbe targets; Sol leads on InFlux focal estimation.

Coverage sensitivity. Table A18b restricts each Astra–peer comparison to cases valid in every shared run. All six comparison directions remain unchanged. The ranking is thus stable on the common valid cohort.

Table A18. Geometry errors (%), mean ± run SD; lower is better. (a) Eight agents: the original seven and DeepSeek-1017 (DS-1017). GP denotes GeoProbe. (b) Comparisons on common-valid cases and runs.

(a) All valid cases

Task	Astra	Sol	Fable	Q-Max	Gemini 3.8	Q-397B	Q-27B	DS-1017
GP focal-only	6.2±2.1	11.2±5.5	29.5±1.8	13.8±3.1	23.7±12.9	38.6±3.2	38.4±5.0	38.1±10.5
InFlux focal	53.0±2.1	47.8±2.0	52.6±2.2	55.5±1.0	52.4±3.4	67.8±1.8	65.9±2.1	62.9±1.6
GP depth	10.4±2.9	22.9±2.7	14.2±2.7	16.3±1.3	19.2±2.1	31.2±2.8	29.2±1.3	22.9±2.5
GP object size	6.3±1.0	8.5±0.7	7.6±0.2	9.6±0.8	8.3±0.3	11.4±0.6	9.4±0.3	8.9±0.7
GP ref. height	0.31±0.11	0.90±0.36	2.31±0.08	1.69±0.72	2.73±0.75	17.55±1.22	13.69±0.86	5.71±1.49
GP height ratio	0.32±0.15	0.95±0.41	2.23±0.17	2.17±0.22	2.67±0.28	20.35±0.87	17.08±2.70	8.35±5.02

(b) Matched common-valid cohorts

Task	Cases	Rounds	Peer	Astra	Peer
GP focal-only	12	5	Sol	6.21±2.10	11.16±5.47
InFlux focal	26	5	Sol	53.02±2.15	47.79±2.00
GP depth	9	3	Fable	10.16±3.49	14.22±2.88
GP object size	12	3	Fable	6.44±1.36	7.63±0.18
GP ref. height	12	5	Sol	0.31±0.11	0.90±0.36
GP height ratio	12	5	Sol	0.32±0.15	0.95±0.41

Camera motion and real photographs. The rendered displacement subsets share 12 object interventions while changing the camera condition. Table A19 compares Astra with Sol, the lowest-MAE original non-Astra peer under each displacement condition. The fixed-camera advantage persists when the two stationary Blender cases are excluded. Under the native direction-and-1 cm criterion, both agents achieve 96.7% fixed-camera success: the continuous error distinguishes precision that this tolerance does not resolve.

The four real-photo pairs form a separate tabletop subset. Their displacement labels are supplied annotations, without independent camera or metric calibration. The Blender comparison uses different scenes and object motions.

Table A19. Displacement MAE (cm), mean ± run SD. Sol is the strongest original non-Astra comparator; DeepSeek-1017 is added under the same conditions. Moving-only excludes stationary cases.

Condition	Cases	Astra	Sol	DeepSeek-1017
Blender: fixed camera	12	0.240±0.109	0.467±0.267	1.517±0.105
Blender: fixed, moving-only	10	0.288±0.131	0.560±0.321	1.780±0.076
Blender: small view change	12	0.567±0.158	0.678±0.152	2.022±0.346
Blender: large view change	12	1.125±0.144	1.958±0.295	2.492±0.908
Real photographs	4	0.050±0.068	0.415±0.236	1.250±0.234

Coupled motion and metric scale reveal distinct estimation errors. Table A20 separates the GeoProbe pose modes: mixed rotation and translation produce 14.16 cm translation error and zero joint success, despite much lower errors on simpler motions. The best non-Astra translation estimate on the mixed subset is Qwen-Max at 16.77 cm; none of the original seven agents achieves joint success on this subset. Translation-only cases have zero target rotation. Depth also separates ordering from metric scale: Astra's ordinal accuracy is 96.3%, whereas raw AbsRel is 0.104; oracle scale alignment reduces it to 0.038. Alignment uses reference depth.

Explicit scale references and regular rendered geometry support these estimates.

Table A20. GeoProbe camera-pose modes. Errors and joint success show mean ± run SD. Joint success uses 2°/2 cm tolerances; failed predictions remain in its denominator.

Motion	Model	Cases	Rotation (°)	Translation (cm)	Joint success (%)
Stationary	Astra	4	0.00±0.00	0.00±0.00	100.0±0.0
	DeepSeek-1017	4	0.00±0.00	0.00±0.00	100.0±0.0
Rotation only	Astra	6	0.35±0.78	1.00±2.24	96.7±7.5
	DeepSeek-1017	6	5.43±1.57	13.82±5.69	13.3±13.9
Translation only	Astra	6	0.00±0.00	1.55±1.59	76.7±25.3
	DeepSeek-1017	6	1.18±0.91	11.03±3.22	20.0±13.9
Rotation + translation	Astra	8	3.26±0.75	14.16±3.94	0.0±0.0
	DeepSeek-1017	8	6.36±0.54	25.16±2.73	0.0±0.0

TraceSpatial-Bench: path matching and boundary constraints. Astra has lower 3D path RMSE than Gemini 3.8 (0.382 versus 0.426 m), but fewer jointly valid paths (19/56 versus 21/56; Table A8). Among Astra's trajectories, 27 fail initiation, 24 fail arrival, and ten fail collision checks; these sets overlap. Of its 37 joint failures, 36 violate initiation or arrival, while only one satisfies both boundaries and fails collision avoidance alone. This supports the boundary-anchoring diagnosis in Figure 4c. The offline tolerances and sampled-position collision rule are defined in Appendix B.3.

D.2 Spatial Reasoning: Cross-Source Task Contrasts

Complete task coverage. Table A22 covers all 43 groups across six sources. Source categories are retained; MindCube uses task axes and question templates, prioritizing composed rotation and turn–translation before individual motion and relation queries.

Figure labels and comparisons. Figure 5a uses six groups: BOP-ASK left/right (12 cases), MindCube object relations (12) and camera shift (13), 3DSRBench actor-relative side (7) and object facing (6), and VSI hard direction (4). Panel b uses BOP closer/farther (5/3), 3DSR camera/object distance (10/2), and VSI relative distance (6). Every model uses the same case IDs in each group. Values average five run accuracies, except Fable's three, and retain invalid predictions in the denominator. The former "Gemini" peer in the original diagnostic is Gemini 3.8 Flash, verified

from its source records; Gemini 4 Argon is a separate row. The released CSV gives all 110 cell means, run SDs, and denominators. These small task groups locate observed strengths and failures; they are not additional independent benchmarks.

Shared-round gaps. The original diagnostic below selects one task per source and its highest-mean original non-Astra peer; Figure 5c shows native source scores for all ten agents. Gaps and SD then use shared run indices (Table A21; protocol in Appendix B.3). On object-facing queries, Fable's three-run mean is 72.2% versus Astra's 50.0%; Astra averages 56.7% across all five runs.

Table A21. Original source-specific Astra–peer comparisons. Accuracy and gaps use shared available runs and are in percentages and percentage points. Run-index alignment follows Appendix B.3.

Task	Peer	Rounds	Astra	Peer	Gap ± SD
MMSI camera–region	Fable	3	100.0	77.8	+22.2±19.2
VSI medium direction	Qwen-Max	5	100.0	83.3	+16.7±11.8
BOP-ASK left/right	Gemini 3.8	5	95.0	96.7	−1.7±7.0
MindCube camera shift	Qwen-Max	5	78.5	89.2	−10.8±14.0
3DSR object facing	Fable	3	50.0	72.2	−22.2±9.6
RoboSpatial configuration	Q-397B	5	78.0	100.0	−22.0±4.5

Numerical error and acceptance are complementary. The numerical rows in Table A22 use the exact-match criterion from Appendix B.3. Table A23a instead measures continuous error on valid predictions. Fixing Astra's cohort across runs preserves low object-size AbsRel (0.091), while absolute-distance coverage is only 25/45 submissions (Table A23b). The table reports error together with valid-prediction coverage.

Table A22. All Spatial task groups, with DeepSeek-1017 (DS-1017) added to the original cohort. Entries are mean ± run SD (%); bold means mark the highest score in each row, including ties. Astra's column is shaded. N counts distinct cases; each assigned case remains in the success denominator.

Task	N	**Astra**	Sol	Fable	Q-Max	Gemini 3.8	Q-397B	Q-27B	DS-1017
3DSRBench									50 cases
World height	4	60.0±13.7	50.0±0.0	**66.7**±14.4	55.0±11.2	25.0±0.0	50.0±0.0	50.0±0.0	25.0±0.0
Above / below	7	**77.1**±7.8	60.0±12.0	71.4±0.0	**77.1**±12.8	74.3±12.0	74.3±6.4	68.6±12.0	60.0±21.2
Camera depth	10	90.0±7.1	98.0±4.5	93.3±5.8	90.0±0.0	92.0±4.5	96.0±5.5	**100.0**±0.0	92.0±8.4
Next to	1	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	**60.0**±54.8	0.0±0.0	0.0±0.0
Object distance	2	70.0±27.4	80.0±27.4	83.3±28.9	**100.0**±0.0	**100.0**±0.0	50.0±0.0	80.0±27.4	50.0±0.0
Parallel / perpendicular	1	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	20.0±44.7	**100.0**±0.0	80.0±44.7	40.0±54.8
Same/different facing	4	55.0±11.2	50.0±0.0	50.0±0.0	50.0±0.0	50.0±0.0	**75.0**±17.7	35.0±13.7	40.0±13.7
Object-facing side	6	56.7±9.1	53.3±13.9	**72.2**±9.6	66.7±0.0	66.7±0.0	36.7±18.3	23.3±9.1	56.7±19.0
In front of actor	4	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	95.0±11.2	90.0±13.7
Actor left/right	7	**74.3**±6.4	71.4±14.3	61.9±8.2	71.4±10.1	65.7±7.8	45.7±12.0	5.7±7.8	**74.3**±6.4
Object viewpoint	4	**70.0**±11.2	**70.0**±11.2	50.0±0.0	60.0±13.7	**70.0**±11.2	**70.0**±11.2	65.0±28.5	30.0±11.2
BOP-ASK									20 cases
Closer to camera	5	96.0±8.9	96.0±8.9	86.7±11.5	**100.0**±0.0	96.0±8.9	80.0±14.1	84.0±8.9	84.0±16.7
Farther from camera	3	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	66.7±0.0	40.0±14.9	86.7±18.3
Left / right	12	95.0±4.6	80.0±4.6	86.1±4.8	86.7±4.6	**96.7**±4.6	91.7±0.0	95.0±4.6	90.0±7.0
MindCube									50 cases
Behind observer	7	97.1±6.4	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	88.6±12.0	80.0±16.3	65.7±16.3	**100.0**±0.0
Camera translation	13	78.5±10.0	64.6±8.8	56.4±16.0	**89.2**±6.9	84.6±5.4	20.0±6.9	35.4±12.9	27.7±14.0
Imagined observer	2	**90.0**±22.4	70.0±27.4	50.0±0.0	80.0±27.4	70.0±27.4	30.0±27.4	30.0±27.4	60.0±22.4
Object relations	12	**96.7**±4.6	63.3±12.6	91.7±8.3	85.0±3.7	95.0±7.5	31.7±7.0	35.0±3.7	45.0±9.5
Composed rotation	5	**100.0**±0.0	96.0±8.9	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	96.0±8.9	96.0±8.9
Turn then translate	11	**90.9**±6.4	43.6±7.6	75.8±10.5	**90.9**±6.4	87.3±10.4	49.1±13.8	58.2±10.4	47.3±4.1

Table A22 continued. Values are mean ± run SD (%).

Task	N	**Astra**	Sol	Fable	Q-Max	Gemini 3.8	Q-397B	Q-27B	DS-1017
MMSI									20 cases
Attribute (Appr.)	2	**90.0**±22.4	70.0±27.4	50.0±0.0	60.0±22.4	50.0±0.0	70.0±27.4	40.0±22.4	60.0±41.8
Attribute (Meas.)	2	**100.0**±0.0	50.0±0.0	66.7±57.7	40.0±22.4	50.0±0.0	0.0±0.0	40.0±22.4	80.0±27.4
MSR	2	40.0±41.8	30.0±27.4	**50.0**±0.0	40.0±22.4	30.0±27.4	30.0±44.7	40.0±22.4	**50.0**±0.0
Motion (Cam.)	1	80.0±44.7	80.0±44.7	**100.0**±0.0	**100.0**±0.0	60.0±54.8	0.0±0.0	0.0±0.0	60.0±54.8
Motion (Obj.)	1	**100.0**±0.0	80.0±44.7	**100.0**±0.0	80.0±44.7	80.0±44.7	80.0±44.7	20.0±44.7	80.0±44.7
Camera–camera relation	3	**100.0**±0.0	73.3±27.9	**100.0**±0.0	66.7±0.0	73.3±14.9	26.7±14.9	33.3±0.0	80.0±18.3
Camera–object relation	1	0.0±0.0	40.0±54.8	**66.7**±57.7	0.0±0.0	60.0±54.8	40.0±54.8	0.0±0.0	20.0±44.7
Camera–region relation	3	**100.0**±0.0	60.0±14.9	77.8±19.2	66.7±0.0	73.3±14.9	66.7±23.6	53.3±18.3	46.7±18.3
Object–object relation	1	**100.0**±0.0	80.0±44.7	**100.0**±0.0	60.0±54.8	20.0±44.7	80.0±44.7	60.0±54.8	80.0±44.7
Object–region relation	4	**50.0**±0.0	30.0±11.2	25.0±25.0	45.0±20.9	**50.0**±30.6	5.0±11.2	0.0±0.0	15.0±22.4
RoboSpatial									30 cases
Placement compatibility	10	92.0±4.5	92.0±8.4	**96.7**±5.8	90.0±0.0	64.0±8.9	80.0±7.1	90.0±7.1	92.0±13.0
Configuration	10	78.0±4.5	70.0±10.0	80.0±0.0	86.0±11.4	88.0±4.5	**100.0**±0.0	94.0±8.9	82.0±8.4
Free-space point set	10	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0
VSI-Bench									50 cases
Appearance order	2	40.0±22.4	80.0±27.4	50.0±50.0	**100.0**±0.0	40.0±41.8	40.0±22.4	80.0±27.4	10.0±22.4
Absolute distance	9	2.2±5.0	2.2±5.0	3.7±6.4	0.0±0.0	**6.7**±6.1	0.0±0.0	2.2±5.0	0.0±0.0
Object count	5	32.0±11.0	20.0±14.1	**40.0**±20.0	36.0±16.7	36.0±16.7	28.0±11.0	12.0±11.0	16.0±8.9
Direction: easy	4	**100.0**±0.0	85.0±22.4	33.3±28.9	**100.0**±0.0	85.0±13.7	30.0±20.9	45.0±11.2	50.0±17.7
Direction: hard	4	45.0±11.2	30.0±27.4	25.0±43.3	**65.0**±28.5	60.0±22.4	30.0±32.6	5.0±11.2	0.0±0.0
Direction: medium	6	**100.0**±0.0	73.3±9.1	38.9±38.5	83.3±11.8	66.7±11.8	40.0±14.9	20.0±13.9	43.3±14.9
Relative distance	6	50.0±20.4	**76.7**±22.4	38.9±9.6	**76.7**±9.1	56.7±14.9	60.0±19.0	53.3±13.9	46.7±21.7
Object size	9	4.4±6.1	**22.2**±7.9	11.1±0.0	13.3±5.0	11.1±7.9	2.2±5.0	8.9±5.0	6.7±6.1
Room area	4	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0
Route planning	1	**100.0**±0.0	**100.0**±0.0	33.3±57.7	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0

Table A23. VSI-Bench numerical estimation. (a) AbsRel, equal-run mean ± SD on each model's valid predictions; lower is better. (b) Astra's coverage and fixed-cohort sensitivity: common cases have valid predictions in every recorded round.

(a) Numerical error across agents

Task	Astra	Sol	Fable	Q-Max	Gemini 3.8	Q-397B	Q-27B	DS-1017
Object count	0.35±0.05	0.40±0.11	0.20±0.11	0.23±0.11	0.31±0.08	0.41±0.06	0.42±0.04	0.34±0.09
Absolute distance	0.28±0.11	0.42±0.05	0.28±0.07	0.26±0.02	0.12±0.03	0.29±0.09	0.31±0.08	0.48±0.09
Object size	0.09±0.01	0.09±0.02	0.10±0.01	0.12±0.02	0.05±0.01	0.13±0.02	0.19±0.02	0.14±0.07
Room area	0.24±0.03	0.19±0.03	0.21±0.04	0.26±0.07	0.27±0.04	0.22±0.05	0.26±0.10	0.35±0.12

(b) Astra coverage and common-cohort error

Task	Cases	Valid / assigned	Common cases	Common AbsRel
Object count	5	25/25	5	0.350±0.053
Absolute distance	9	25/45	2	0.279±0.083
Object size	9	42/45	8	0.091±0.017
Room area	4	20/20	4	0.240±0.029

Other task requirements. RoboSpatial separates placement compatibility (92.0%), configuration (78.0%), and free-space point sets (0.0%) for Astra. All agents in this table fail the point-set criterion. Compatibility and configuration judgments succeed more often than point-set predictions.

D.3 Affordance: Contact and Region Extent

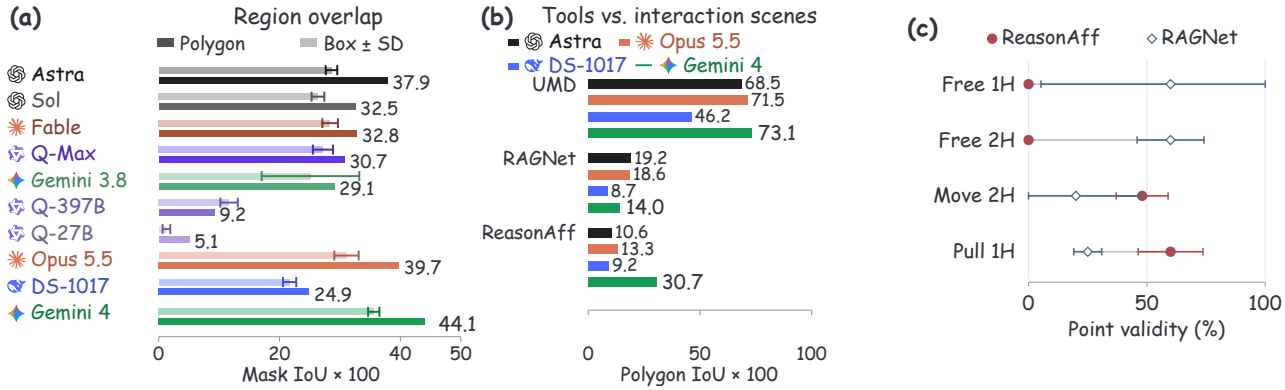


Figure A1. Flexible outlines help, but interaction scenes remain difficult. (a) Aggregate mask IoU for native boxes and direct polygons across all ten agents. Box whiskers show sample SD over five runs (three for Fable); polygon bars use one separate run on the same 70 cases. (b) Polygon IoU on tools (UMD) and interaction-conditioned scenes (RAGNet and ReasonAff) for Astra, Opus 5.5, DeepSeek-1017, and Gemini 4 Argon. (c) Astra's original contact validity by action, with run SD; 1H/2H denotes one/two hands. Invalid or absent polygon outputs score zero. Appendix C.5 defines the polygon protocol.

From valid contact to functional-region extent. Table A24 reports all fifteen action/hand groups for Astra and DeepSeek-1017, including zero-scoring conditions. UMD contact validity averages 96.7% for Astra and 94.4% for Fable. Cutting combines 100% valid contacts and 96% strict box success with box–mask IoU of 0.361; the box IoU is 0.694. This separates successful localization from the limitations of rectangular mask coverage. Containment adds a same-representation comparison: Astra's strict box success is 44%, versus Qwen-Max's 96% (Table A25).

Table A24. Astra and DeepSeek-1017 Affordance metrics, mean ± run SD. IoU compares the native box to the GT mask; strict box success compares it to the GT box at IoU ≥ 0.5. Point and box success are percentages.

Action / hands	Model	Cases	Box–mask IoU	Point (%)	Box (%)
RAGNet-3DOI					
free movement / two hands	Astra	2	0.046±0.063	20.0±27.4	20.0±27.4
	DeepSeek-1017	2	0.179±0.070	50.0±0.0	30.0±27.4
free / one hand	Astra	1	0.384±0.351	60.0±54.8	60.0±54.8
	DeepSeek-1017	1	0.241±0.291	40.0±54.8	20.0±44.7
free / two hands	Astra	5	0.249±0.070	60.0±14.1	36.0±16.7
	DeepSeek-1017	5	0.083±0.030	28.0±11.0	4.0±8.9
pull / one hand	Astra	12	0.008±0.002	25.0±5.9	0.0±0.0
	DeepSeek-1017	12	0.053±0.019	41.7±8.3	0.0±0.0
ReasonAff-style					
free movement / two hands	Astra	5	0.270±0.055	48.0±11.0	48.0±11.0
	DeepSeek-1017	5	0.026±0.028	0.0±0.0	0.0±0.0
free / one hand	Astra	4	0.000±0.000	0.0±0.0	0.0±0.0
	DeepSeek-1017	4	0.035±0.040	0.0±0.0	0.0±0.0
free / two hands	Astra	4	0.001±0.002	0.0±0.0	0.0±0.0
	DeepSeek-1017	4	0.004±0.008	0.0±0.0	0.0±0.0
pull / one hand	Astra	4	0.098±0.033	60.0±13.7	0.0±0.0
	DeepSeek-1017	4	0.086±0.047	55.0±11.2	0.0±0.0
push / one hand	Astra	3	0.318±0.221	46.7±29.8	46.7±29.8
	DeepSeek-1017	3	0.146±0.201	26.7±27.9	20.0±29.8

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Action / hands	Model	Cases	Box–mask IoU	Point (%)	Box (%)
UMD					
contain	Astra	5	0.472±0.067	100.0±0.0	44.0±16.7
	DeepSeek-1017	5	0.593±0.045	96.0±8.9	84.0±16.7
cut	Astra	5	0.361±0.009	100.0±0.0	96.0±8.9
	DeepSeek-1017	5	0.223±0.019	48.0±30.3	52.0±17.9
grasp	Astra	5	0.491±0.007	100.0±0.0	100.0±0.0
	DeepSeek-1017	5	0.377±0.063	92.0±11.0	64.0±29.7
pound	Astra	5	0.637±0.009	100.0±0.0	100.0±0.0
	DeepSeek-1017	5	0.490±0.034	96.0±8.9	44.0±16.7
scoop	Astra	5	0.504±0.054	80.0±0.0	80.0±0.0
	DeepSeek-1017	5	0.337±0.059	76.0±16.7	52.0±17.9
support	Astra	5	0.638±0.017	100.0±0.0	100.0±0.0
	DeepSeek-1017	5	0.466±0.056	88.0±17.9	76.0±16.7

Contact, extent, and shape give different comparisons. Astra leads box–mask IoU on UMD cutting, grasping, scooping, and support, while Sol leads containment and pounding. In the interaction-conditioned sources, one-handed pulling has zero strict box success for every original model despite nonzero contact validity. Table A25 identifies the best peer separately for each reported criterion. ReasonAff and RAGNet retain their original scenes and interaction annotations.

Table A25. Metric-specific Affordance peers. Values are mean ± run SD, using the repetition counts in Appendix B.4. Peers are selected from the original cohort; DS-1017 adds DeepSeek-1017 under the same criterion.

Task / criterion	Astra	Original peer	Model	DS-1017
UMD cutting: box–mask IoU	0.361±0.009	0.341±0.022	Fable	0.223±0.019
UMD cutting: strict box (%)	96.0±8.9	100.0±0.0	Fable	52.0±17.9
UMD containment: box–mask IoU	0.472±0.067	0.673±0.069	Sol	0.593±0.045
UMD containment: strict box (%)	44.0±16.7	96.0±8.9	Qwen-Max	84.0±16.7
ReasonAff pulling: point (%)	60.0±13.7	50.0±0.0	Fable	55.0±11.2
RAGNet pulling: point (%)	25.0±5.9	31.7±7.0	Qwen-Max	41.7±8.3
ReasonAff two-hand movement: point (%)	48.0±11.0	26.7±11.5	Fable	0.0±0.0
RAGNet two-hand movement: point (%)	20.0±27.4	50.0±0.0	Sol / Fable	50.0±0.0

D.4 Task Planning: Goal Families and Terminal States

Compound goals and source-specific requirements. Table A26 covers all 56 native families across the six sources. Each row compares Astra with the highest-success original peer for that family. Figure A2a compares five models by source; panels b–c hold Fable fixed for household and scientific goal comparisons. Heating and dual placement test state change and multiple-object goal retention; ScienceWorld distinguishes experimental topics and instrument use. This preserves the task-specific advantages and reversals behind the aggregate Planning lead.

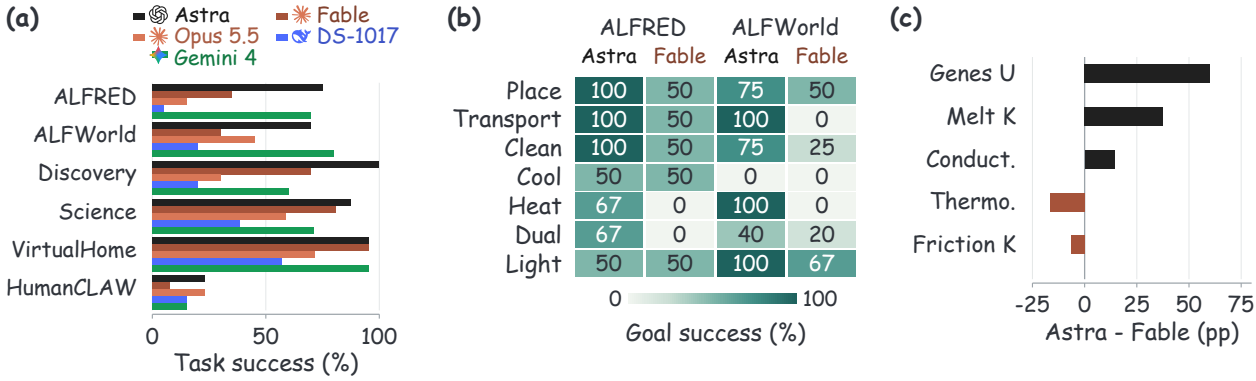

Figure A2. Planning success varies across sources and goals. (a) Source-level success for Astra, Fable, Opus 5.5, DeepSeek-1017, and Gemini 4 Argon. DeepSeek-1017 and Gemini 4 Argon each complete 2/13 HumanCLAW tasks, both navigation. Opus 5.5 completes 3/13 HumanCLAW tasks, all navigation; none of its eight sitting tasks succeeds. (b) Astra/Fable success on seven household goal families. (c) Their five nonzero ScienceWorld gaps; thirteen families tie, and K/U denotes known/unknown conditions. Each assignment uses one run. Appendix D.4 gives the task-family breakdown; Table A11 gives Opus 5.5 coverage.

Table A26. Task Planning by native goal family, with one episode per assigned case. Entries are solved/assigned cases. One highest-success peer from the original cohort is shown per row, with DeepSeek-1017 (DS-1017) added separately; * denotes a tie with another peer. Source-level results appear in Table A9.

Goal family	Astra	Original peer	Model	DS-1017
ALFRED				
Illumination	1/2	1/2	Fable*	0/2
Placement	4/4	2/4	Sol*	0/4
Container transport	2/2	1/2	Fable*	1/2
Cleaning	2/2	1/2	Sol*	0/2
Cooling	2/4	2/4	Fable	0/4
Heating	2/3	0/3	Sol*	0/3
Dual placement	2/3	0/3	Sol*	0/3
ALFWorld				
Illumination	3/3	2/3	Fable*	2/3
Placement	3/4	2/4	Sol*	1/4
Container transport	2/2	1/2	Gemini 3.8*	1/2
Cleaning	3/4	1/4	Sol*	0/4
Cooling	0/1	0/1	Sol*	0/1
Heating	1/1	0/1	Sol*	0/1
Dual placement	2/5	1/5	Sol*	0/5
DiscoveryWorld				
Archaeology dating	1/1	1/1	Sol*	0/1
Combinatorial chemistry	1/1	1/1	Sol*	0/1
It's not rocket science	1/1	1/1	Fable	0/1
Lost in translation	1/1	1/1	Fable	0/1
Plant nutrients	1/1	0/1	Sol*	0/1
Discovery feed	1/1	1/1	Sol*	1/1
Doors	1/1	1/1	Qwen-Max	0/1
Navigation in a house	1/1	1/1	Fable*	0/1
Pick and give	1/1	1/1	Sol*	1/1
Space sick	1/1	1/1	Fable	0/1
HumanCLAW				
Navigate to contact	3/5	3/5	Gemini 3.8	2/5
Sit on target	0/8	0/8	Sol*	0/8
ScienceWorld				
Find animal	3/3	3/3	Fable	0/3

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Goal family	Astra	Original peer	Model	DS-1017
Find living thing	2/2	2/2	Sol*	2/2
Find non living thing	3/3	3/3	Sol*	3/3
Find plant	3/3	3/3	Sol*	2/3
Grow fruit	2/3	2/3	Fable	0/3
Inclined plane determine angle	2/2	2/2	Fable	2/2
Inclined plane friction named surfaces	15/16	16/16	Fable	8/16
Inclined plane friction unnamed surfaces	1/1	1/1	Fable*	0/1
Lifespan longest lived	2/2	2/2	Sol*	2/2
Lifespan longest lived then shortest lived	1/1	1/1	Sol*	1/1
Lifespan shortest lived	1/1	1/1	Sol*	1/1
Measure melting point known substance	8/8	6/8	Gemini 3.8	5/8
Measure melting point unknown substance	1/1	1/1	Sol*	0/1
Mendelian genetics known plant	1/1	1/1	Sol*	1/1
Mendelian genetics unknown plant	3/5	1/5	Qwen-Max*	0/5
Test conductivity	6/7	6/7	Gemini 3.8	0/7
Test conductivity of unknown substances	6/8	6/8	Fable	1/8
Use thermometer	4/6	5/6	Fable*	0/6
VirtualHome				
Grab	1/1	1/1	Sol*	0/1
Greet	1/1	1/1	Sol*	1/1
Lie	1/1	1/1	Sol*	1/1
Navigation or observation	1/1	1/1	Sol*	1/1
Read	2/2	2/2	Fable*	0/2
Rinse	1/1	1/1	Sol*	0/1
Sit	3/3	3/3	Sol*	1/3
Switchoff	1/1	1/1	Sol*	1/1
Switchon	7/7	7/7	Sol*	5/7
Touch	1/1	1/1	Sol*	1/1
Type	1/1	0/1	Sol*	0/1
Wash	0/1	1/1	Sol*	1/1

Attempted actions and terminal goals. All eleven Astra household failures exhaust their assigned budget. ScienceWorld's nine failures have scored task-failure outcomes with no recorded parser-no-match, invalid-action, or code-exception events; negative score sentinels are not physical-progress measures. HumanCLAW identifies a different limit (Table A27): six Astra sitting episodes find the target, five approach within 20 cm and issue a sit action, yet none succeeds. Fable records no target-finding success on these episodes. The shared zero terminal rate conceals different progress before the required body–object condition.

D.5 Manipulation: Progress and Complete Goals

Table A28 expands the adopted Manipulation results to all thirteen sources and ten agents. Astra records 76 successes out of 182 assignments; Fable records 45, with nine missing ManiSkill records receiving no success credit. Table A29 retains the available original-cohort auxiliary metrics. Later-model diagnostics appear in Appendices C.4 and J.

D.5.1 CALVIN Progress and Recovery

The analysis uses the same ten sequence IDs for all ten models, giving 100 single-run episodes. The first seven models use the frozen original CALVIN records; Opus 5.5, DeepSeek-1017, and

Table A27. HumanCLAW sitting diagnostics for eight assigned cases per model. Counts pool independently recorded conditions. † denotes lower bounds because process fields are missing for two Gemini 3.8 cases; dashes denote unavailable process metrics. Terminal success retains all eight assignments. Opus 5.5 uses its final single-run records; near-plus-sit action counts are not supplied. DeepSeek-1017 adds 0/8 terminal sitting success; comparable process counts are not supplied.

Model	Process coverage	Target found	Within 20 cm	Near + sit issued	Success
Astra	8/8	6	5	5	0
Sol	8/8	3	1	1	0
Fable	8/8	0	0	0	0
Qwen-Max	0/8	–	–	–	0
Gemini 3.8	6/8	3†	3†	3†	0
Q-397B	0/8	–	–	–	0
Q-27B	0/8	–	–	–	0
Opus 5.5	8/8	4	4	–	0
DeepSeek-1017	–	–	–	–	0

Table A28. Adopted Manipulation successes across thirteen sources and ten agents. Headers give assigned tasks; missing Fable ManiSkill records (–) receive no success credit.

Model	CALVIN	CLIPort	RLBench	RoboCasa	RC365	RoboTwin	RoboWits
Assigned	10	18	10	10	10	50	10
Astra	3	11	6	1	0	19	1
Sol	0	5	2	0	0	5	1
Fable	0	6	1	0	1	7	1
Q-Max	0	4	0	0	0	2	1
Gemini 3.8	0	5	2	0	0	4	2
Q-397B	0	4	0	0	0	3	0
Q-27B	0	3	1	0	0	4	1
Opus 5.5	1	11	3	0	1	23	1
DS-1017	1	6	2	0	0	2	2
Gemini 4	4	7	8	2	1	14	1

Model	VIMA	VLABench	ManiSkill	LIBERO	robosuite	BEHAVIOR
Assigned	10	25	9	10	5	5
Astra	4	18	5	4	4	0
Sol	3	13	0	3	4	0
Fable	3	15	–	6	5	0
Q-Max	1	5	0	1	3	0
Gemini 3.8	2	9	1	4	5	0
Q-397B	3	6	0	0	2	0
Q-27B	3	6	0	1	1	0
Opus 5.5	10	19	5	8	5	0
DS-1017	6	9	1	5	4	0
Gemini 4	9	16	5	8	5	0

Gemini 4 Argon use their adopted result archives. All report hashes are verified. The recomputed mean prefixes and full-sequence counts agree with the retained CALVIN aggregates. Figure 8a counts episodes with at least k verified consecutive goals, for $k = 1, \ldots, 5$.

Execution recovery. An execution-error episode contains a round with a code-execution failure, including syntax, interface-contract, or runtime/tool errors. Transport failures, timeouts, and budget guards are excluded. Resumed execution requires a later round marked `execution_ok=true`; completed recovery requires a later successful native task verification. An episode contributes once to each count, regardless of the number of errors. Thus, the three

Table A29. Auxiliary metrics for the adopted original seven-model Manipulation records. CALVIN reports average consecutive goals completed (0–5); other rows average task-normalized criteria or control evidence (%). Definitions and time windows appear in Table A31. – denotes unavailable records.

Benchmark	N	Astra	Sol	Fable	Q-Max	Gemini 3.8	Q-397B	Q-27B
CALVIN goals (0–5)	10	3.10	1.40	0.90	0.80	1.80	0.00	0.10
RoboCasa	4	–	–	–	–	–	–	–
RoboCasa365	7	–	–	–	–	–	–	–
ManiSkill conditions	5	60.00	0.00	–	0.00	0.00	0.00	0.00
LIBERO-PRO	10	20.00	20.00	30.00	5.00	20.00	0.00	5.00
robosuite	5	60.00	60.00	80.00	40.00	80.00	30.00	16.67
BEHAVIOR-1K control	5	13.33	16.67	30.00	16.67	6.67	6.67	6.67

completed recoveries are a subset of the 62 resumed episodes and the 65 error-containing episodes (Table A30). These counts describe the recorded settings in Appendix E.1.

Table A30. CALVIN: progress and recovery on ten assigned sequences per model. Error, Resumed, and Then complete are nested episode counts. Full success includes episodes with and without an error. Mean prefix retains the ten-task denominator.

Model	Mean prefix	Error	Resumed	Then complete	Full success
Astra	3.1	3	3	1	3
Sol	1.4	3	3	0	0
Fable	0.9	9	9	0	0
Q-Max	0.8	10	9	0	0
Gemini 3.8	1.8	8	8	0	0
Q-397B	0.0	2	2	0	0
Q-27B	0.1	7	6	0	0
Opus 5.5	2.4	9	9	1	1
DS-1017	1.4	10	10	1	1
Gemini 4	3.3	4	3	0	4
All episodes	–	65	62	3	9

Incomplete prefix evidence. One Qwen-Max episode has no verified prefix; five Qwen-397B episodes stop without a reported prefix. These six assignments receive no verified goal credit, preserving the original denominator rather than treating an unknown prefix as measured progress. The other 94 prefixes are explicit. Complete-sequence success requires the native verifier for every model. Per-episode hashes, error-round indices, prefix values, and aggregate counts are available in the released analysis data.

Metric definitions and coverage. The 183-case Manipulation inventory contains 182 binary tasks and one continuous-return AnymalC-Spin task. AnymalC-Spin is evaluated separately through continuous return. For each model and configuration, task success is $\mathrm{SR} = 100N^{-1}\sum_i s_i$, where $s_i = 1$ requires native acceptance. Runs without native acceptance receive zero success credit, including timeouts, execution or infrastructure failures, and missing records. Each task contributes one run to the assigned denominator.

CALVIN scores a five-goal sequence as successful only when all five goals are completed in order. Its average number of consecutive goals completed is $\bar{L} = 10^{-1}\sum_{i=1}^{10} L_i$, where $L_i \in \{0, \ldots, 5\}$ is the verified consecutive prefix. Unverified prefixes receive no goal credit; Appendix D.5.1 gives the case-level coverage.

For the other auxiliary metrics, case i has K_i scoring items and verified indicators z_{ij}. Its score is $u_i = K_i^{-1}\sum_{j=1}^{K_i} z_{ij}$; the reported mean is $100N^{-1}\sum_i u_i$. Failed or unknown items contribute

zero. This weights tasks equally even when K_i differs. The diagnostics summarize recorded control events on fixed task subsets using the scoring items and evidence windows in Table A31.

Table A31. Six auxiliary metrics, averaging task-normalized evidence within each subset.

Source	N	Scoring items	Evidence window
RoboCasa	4	Approach, grasp and lift, stable release	Any verified event during execution
RoboCasa365	7	Same three transport milestones	Any verified event during execution
ManiSkill	5	Two or three native conditions per task	Final verification state
LIBERO-PRO	10	Two task-specific conditions	Final state and limited process records
robosuite	5	Lifting: 2; stacking: 3; wiping: 4	Native success and reward-derived evidence
BEHAVIOR-1K	5	Six control-evidence items	Before the first episode-end signal

Transport control versus destination success. RoboCasa uses ArrangeBreadBasket, LoadDishwasher, PickPlaceToasterToCounter, and RecycleBottlesByType. RoboCasa365 uses BreadAndCheese, MakeBananaMilkshake, MicrowaveThawingFridge, MixCakeFrosting, PlaceMeatInMarinade, PlaceVegetablesEvenly, and RemoveCuttingBoardItems. Door, appliance, and navigation-only tasks are outside these subsets. Approach requires executed motion and confirmed arrival; grasp-and-lift requires both grasp and lift confirmation. Stable release requires prior grasp-and-lift of the same canonically named object, followed by confirmed release and stability. Each item scores at most once per case. These events need not cover every task object, and release does not establish the requested destination relation.

Table A32. ManiSkill's five-task diagnostic. Items are evaluated at final verification; grasp history is separate. The other four binary tasks remain in complete-task success.

Task	K_i	Scored conditions
PickCube	2	Object placed at its goal; placed and robot static. The suspended goal requires no release.
PlaceSphere	3	Sphere on its support; on support and released; both with the sphere static.
StackCube	3	Cube A on B; on B and released; both with cube A static.
PlugCharger	2	Goal distance ≤ 0.005 m; distance satisfied and angular error ≤ 0.2 rad.
PegInsertionSide	2	Peg-head $x \geq -0.015$ m in the hole frame; official insertion success.

LiftPegUpright, PullCube, PushCube, and RollBall contribute to the nine-task success rate but not this diagnostic. Dashes mark unavailable Fable baseline records in both sets.

Task-specific conditions in LIBERO-PRO and robosuite. LIBERO-PRO uses fixed init-0 instances: spatial 1/7, object 2/5, goal 1/7, libero_10 1/9, spatial_swap 0, and object_swap 4. Ordinary placement scores recorded grasp and the official target relation; stove opening scores operation of the correct control and an open stove. The two longer tasks score a cup inside a microwave followed by the door being closed, or one food item followed by both items in a basket. Missing grasp/control records score zero; for the longer tasks, necessary preceding relations are confirmed only on official success.

robosuite uses cube lifting with seeds 99/50, cube stacking and restacking with seed 14, and spill wiping with seed 48. Lifting scores recorded grasp and native success; grasp history is missing. Stacking uses the fixed local reward q: $q > 0.125$ supports a grasp/lift/stack condition, $q \geq 0.5$ supports the top cube being lifted, and native success supplies the third item. A restacking zero can mask intermediate events under the two-cube airborne rule; these items remain unknown and receive zero credit. Wiping checks 25%, 50%, 75%, and complete removal. Current records verify the first three only on native complete success, rather than interpreting total reward as a wiped fraction. Consequently, Fable's 100% complete-task success and 80% auxiliary score are compatible: missing intermediate evidence lowers the latter.

Table A33. BEHAVIOR-1K control evidence. Each item records one type of control event and scores once per case; terminal BDDL goal success is evaluated separately.

Item	Positive evidence
Localization	Finite coordinates or pose from public RGB-D localization or grasp sampling.
Candidate	A nonempty pregrasp or grasp candidate set.
Arrival	Measured arrival from approach or recovery feedback for the same arm.
Holding	An explicit in-hand check or holding confirmation inside checked placement.
Held motion	Associated measured motion of at least 5 cm while holding, or movement with a passed holding check inside checked placement.
Release	Holding, opening, then no longer holding; or release verified in checked placement.

The five tasks are outfit_a_basic_toolbox, can_meat, tidying_bedroom, putting_shoes_on_rack, and chop_an_onion. Records are linked by arm, name, and temporal order; physical-instance IDs are not consistently recorded. Object identity and inside/ontop relations are checked separately by the terminal goal evaluator. Held motion uses a 5 cm threshold. Control-event records cover four tasks for each of four models and the toolbox task for each of the other three. All retain the five-task denominator. Scoring uses the evaluated episodes up to their first end signal and requires direct evidence for each item. Fable's nine verified items yield $100 \times 9/(5 \times 6) = 30\%$, versus Astra's four items and 13.33%. RoboWits reports task success.

E Execution Settings and Multi-Round Review

Table A34. Fixed-input multi-round review on the shared 46-case subset. I/T/W counts improved/tied/worsened Geometry metric entries; SR_{50} requires box IoU≥ 0.5. Appendix E.2 specifies the cases and tools.

		Geometry	Spatial Reasoning	Affordance	
Model	Condition	I/T/W	Acc. (%)↑	Box IoU↑	SR_{50} (%)↑
Astra	Initial	–	**72.2**	0.2903	22.2
	Self-review	3/20/1	66.7	0.2914	22.2
	Crop review	4/19/1	66.7	0.2912	22.2
	Tool-assisted	5/15/4	**72.2**	**0.3174**	**33.3**
Qwen-27B	Initial	–	55.6	0.0336	0.0
	Self-review	6/15/3	50.0	**0.1524**	0.0
	Crop review	3/19/1	44.4	0.0937	0.0
	Tool-assisted	10/6/8	**61.1**	0.1452	**11.1**

E.1 Recorded Manipulation Settings

The main evaluation retains 182 binary assignments per model, with one adopted episode per assignment. Nine missing Fable ManiSkill records receive no success credit in the domain total and remain unavailable in source-level comparisons. The original seven-model aggregate includes the adopted LIBERO-PRO, robosuite, and BEHAVIOR-1K records. Table A28 lists all thirteen sources for all ten agents.

Observations and execution ceilings. All agents use native observations and tools. Each adopted batch retains its recorded image schedule and resource limits. CALVIN's original seven-model reports allow 640k tokens and 60 minutes; Fable allows 16 model rounds and the other six allow 20. The Opus 5.5, DeepSeek-1017, and Gemini 4 Argon reports allow 40 rounds,

1.28M tokens, and 120 minutes, with automatic per-round RGB. A model round may execute several native actions. The protocol audit links the adopted reports and their hashes.

The CALVIN progress and recovery analysis uses these 100 adopted episodes and reproduces their mean prefixes and full-sequence success counts. It describes recorded execution, rather than an intervention on observation frequency or budget. Other source-specific limits appear in Table A6, Table A15, and Table A39. Missing historical configuration fields remain unspecified. Tool calls, native physical steps, model rounds, and elapsed time are separate counters; video frame rate does not specify the model's observation frequency.

E.2 Fixed-Input Multi-Round Review

Multi-round review lets agents revisit their initial answers using the same task observations and scoring criteria. Self-review adds reconsideration, Crop review adds image crops, and Tool-assisted review adds specialist visual and geometric measurements. The three conditions permit up to two, two, and six additional responses, respectively, across Geometry, Spatial Reasoning, and Affordance. Table A34 reports results; sampling and tools are specified below.

Shared cases. Astra and Qwen-27B share 46 cases: 19 Geometry, 18 Spatial Reasoning, and nine Affordance. A 72-case pilot covers 24 benchmark/task strata selected by task category and SHA-256 ordering with seed 20260923, independently of scores and answers. It excludes the 60 exact cases from an earlier performance-selected study. One case per stratum forms a 24-case development set; withholding two more cases that share development scenes leaves the 46 evaluation cases. Some scenes recur, and some cases were inspected during development.

Table A35. Review conditions for the same cases and native scoring interfaces. Budgets are additional to native execution; response limits count output tokens.

Condition	Evidence available during review	Additional response budget
Initial	Original task observations	None
Self-review	Original observations and initial answer	Two responses, up to 1,800 tokens each
Crop review	Self-review evidence plus image crops	Two responses, up to 1,800 tokens each
Tool-assisted	RGB, crops, specialist measurements	Six responses, each up to 4,096 tokens; 16 requests

Evidence construction. Tool-assisted review adapts GCA's measurement and reference-frame formalization (Chen et al., 2025b) and SpatialClaw's inspectable execution (Cho et al., 2026). It formulates the measurement before consulting the candidate answer, then uses a persistent Python session. SAM 3 (Carion et al., 2025) supplies regions; SuperPoint (DeTone et al., 2018) and LightGlue (Lindenberger et al., 2023) supply correspondences; GeoCalib (Veicht et al., 2024) estimates calibration. Numerical routines compute relations and estimates. Finite-value and coordinate checks validate the computed outputs. Remaining semantic errors include selecting a whole-object mask for a functional part and ambiguous translation scale.

Observation and execution controls. Specialist tools receive only previously exposed RGB, without predicted depth, learned 3D reconstruction, private calibration, scoring annotations, or extra video frames. Public depth or masks originally supplied by a task remain available to the base model. Review reuses native candidates where available; one Astra Spatial case starts from its public observation because the candidate is missing. Temperature is zero, and total-token and wall-time accounting includes native execution. Incomplete runs may be retried; the first completed result is retained regardless of correctness, yielding one outcome per condition and case.

Outcome interpretation. Geometry counts improved, tied, and worsened results across 24 metric entries from 19 cases; one Qwen-27B crop entry is unavailable. Spatial accuracy retains 18 assigned cases, with missing answers counted as incorrect. Affordance reports box–box IoU and success at IoU≥ 0.5 over nine cases.

E.3 Native Capability and Complete Task Execution

Native capability concerns the decisions a model makes through an execution interface. Comparing robot-agent studies therefore requires identifying the task requirements, the competence supplied by the interface, and the resources available for completion. These distinctions contextualize the arena's findings without changing its central question: can the model complete the task through its available operations?

Context and supplied control competence. Robotic in-context learning uses demonstrations and interaction through context-conditioned policies, geometric transfer, world models, or skill- and agent-based control (Huang et al., 2026b). Fixed model weights can therefore coexist with substantial execution support. Learned skill selection (Ahn et al., 2022), programmatic composition of perception and control APIs (Liang et al., 2022), and spatial value maps (Huang et al., 2023a) assign different responsibilities to the model and executor. GPT-Policy (Cheng et al., 2026) conditions a fixed VLM on demonstrations, action references, and feedback, while a constrained controller checks and executes its requests. Our baseline leaves task interpretation and action selection to the model through source-native helpers; the supplementary review study separately introduces specialist measurements.

Task composition and execution support change the comparison. Our RoboCasa sample contains seven atomic and three composite tasks; RoboCasa365 contains three atomic and seven composite tasks (Appendix A). PyRUA-Lean (Si et al., 2026) evaluates 18 atomic and 32 composite RoboCasa365 task types at five seeds each, with 86.7% and 39.4% success. Its Target50 selection differs from our pretrain-designated instances. Removing its frozen RLDX-1 policy lowers atomic and composite success to 65.6% and 20.0%, while retaining programmatic composition of other primitives. Execution allowances also differ: our RoboCasa protocols use 24 model rounds and 60 minutes of trial time; PyRUA-Lean uses 40 calls for atomic tasks, 100 for composite tasks, and a two-hour watchdog. Both allow multiple actions per Python response. These settings evaluate different combinations of model reasoning, supplied control, and task demands.

More interaction helps when it supports effective control. GPT-Policy (Cheng et al., 2026) reports 0/3 towel-pickup successes without demonstrations, averaging 96.3 decisions and 24.6 minutes. Human video raises success to 2/3 and lowers those averages to 76.7 decisions and 18.9 minutes, including failed trials. Our CALVIN analysis distinguishes a further requirement: 62 error-containing episodes resume code execution, but only three complete the sequence (Appendix D.5.1). Restored executability and restored task progress must therefore be measured separately.

Precision must survive through termination. GPT-Policy's plug task requires the plug to remain fully seated after release. Robot video yields 0/3 successes; adding aligned action references yields 2/3 (Cheng et al., 2026). Our within-task diagnostics expose related final-state requirements: initiation or arrival fails in 36 of Astra's 37 unsuccessful traces, and five HumanCLAW episodes reach the seat and issue a sitting action without satisfying the sitting condition. The common issue is converting locally plausible decisions into the exact state the task requires. This is the ability the arena measures across native VLM agents, together with the interaction and time consumed before completion or termination.

F Case Studies

From spatial judgments to usable contacts. Figures A3 and A4 illustrate Section 4.3 with six cases from the first evaluation run.

① *Spatial references.* In these runs, Astra correctly judges placement compatibility but misidentifies object-facing direction and camera translation. ② *Contact versus coverage.* Astra identifies usable tool and furniture regions. For pulling, three models place valid contacts on the handle, but their boxes cover only a fraction of the reference door region.

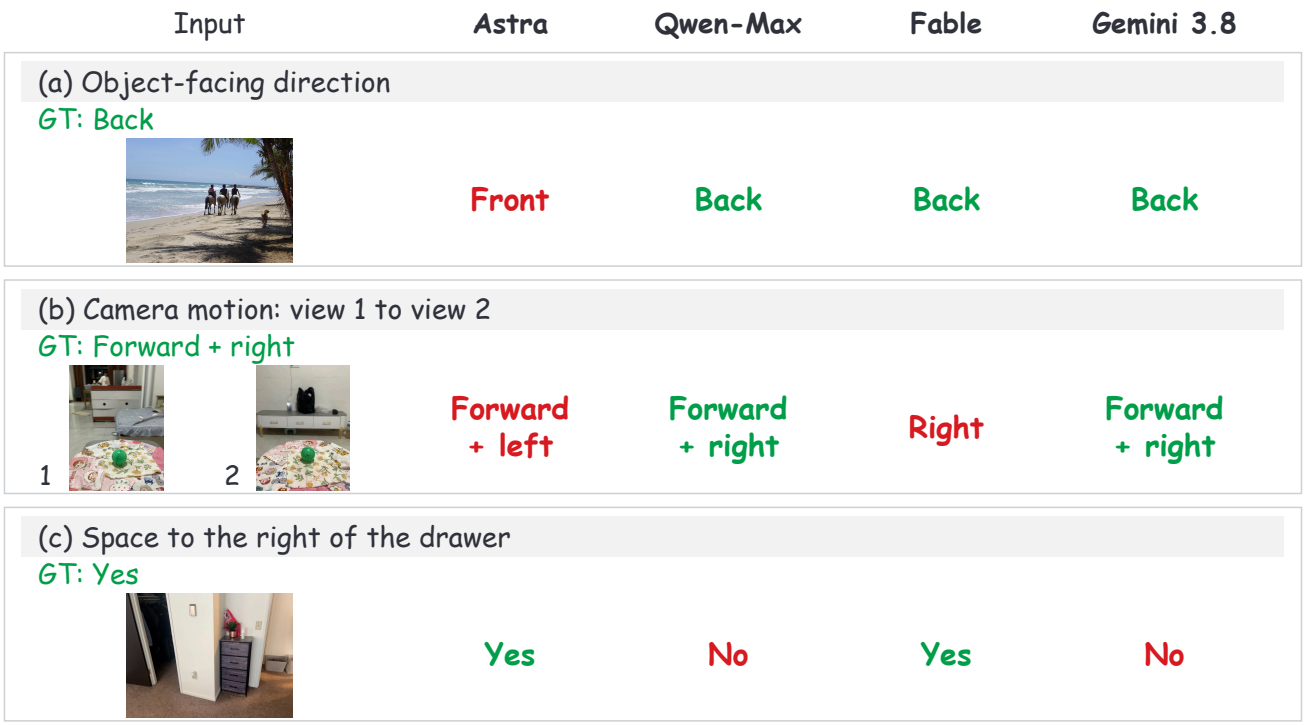


Figure A3. Spatial judgments from the first evaluation run: 3DSRBench, MindCube, and RoboSpatial. The questions concern the dog's facing direction, camera motion, and whether the plant fits beside the drawer. Green answers match the reference; red answers are incorrect.

Motion and task completion. Figure A5 contrasts visible motion with verified completion. Figure 8 complements these examples with the full CALVIN progress and recovery counts.

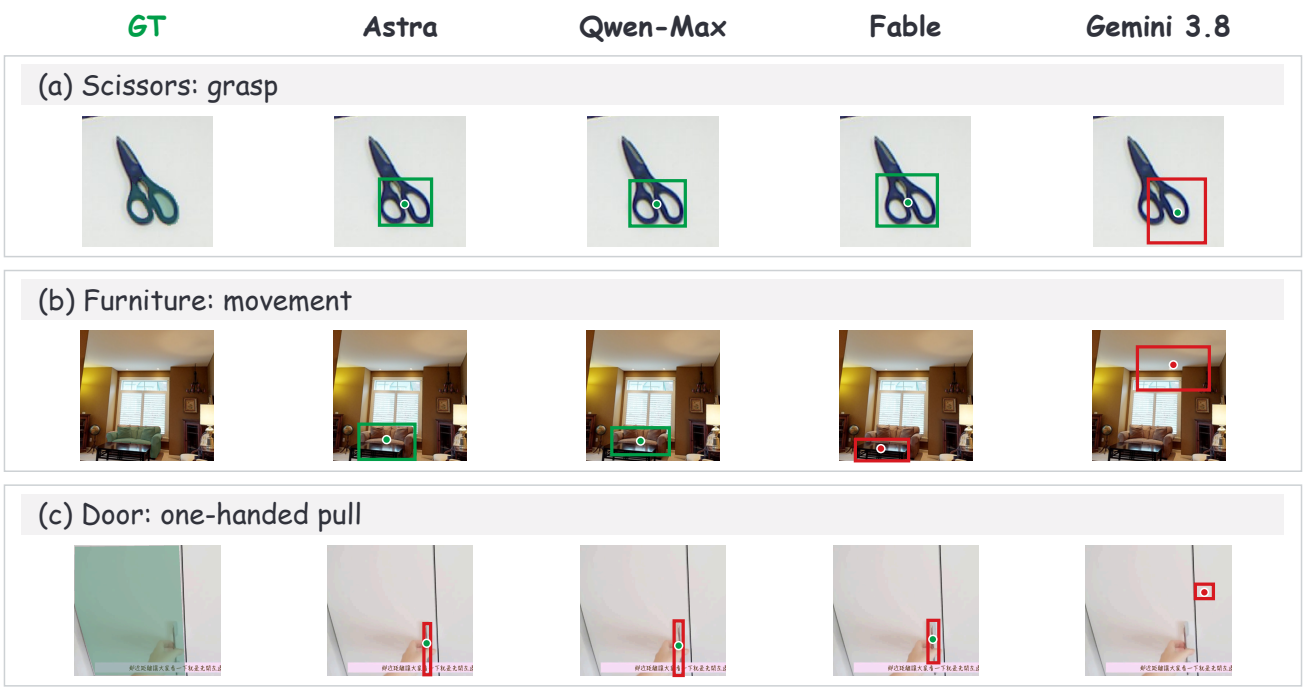


Figure A4. Affordance localization in UMD, ReasonAff, and RAGNet. Green points lie inside the reference mask; green boxes have mask IoU≥ 0.5. Red denotes failure. A valid contact can occupy a small part of the functional region, as the handle-versus-door example shows.

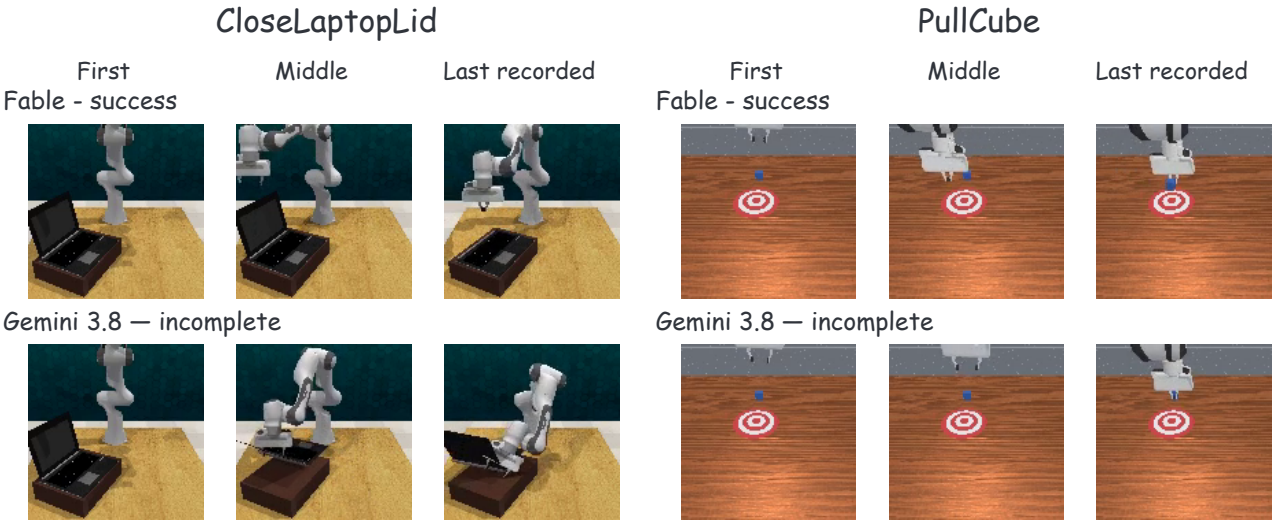


Figure A5. Archived execution contrasts under matched task settings and budgets. CloseLaptopLid uses RLBench's front camera; PullCube uses ManiSkill's base camera. Native records determine success. These are selected native-interface episodes.

G Evaluation Scales and Public Resources

Table A36 expands Table 1 with instance counts and released artifacts. Task categories, training demonstrations, and source datasets are excluded from the instance count. Our release provides a fixed 1,000-case index across five capability domains, together with observations, episode initializations, and evaluation code.

OpenEQA adds navigation to episodic QA in its active track; Memory denotes cross-session persistence. VLABench's non-interactive track scores action sequences from fixed multi-view images against reference graphs. Hard Vision, Easy Vision varies reasoning effort and tool use; its 55 source benchmarks do not specify an instance count.

Table A36. Reported evaluation populations and released resources. Counts follow the cited papers; artifact availability is checked on 3–4 October 2026. Task-type and source counts describe coverage rather than tested instances. Links lead to author-maintained resources.

Study	Reported evaluation population	Released evaluation resources
OpenEQA (2024)	1,636 QA entries; the 557 HM3D questions also support active navigation. GPT-4V's active evaluation uses a 184-question subset.	Baseline and scoring code; QA manifest and episode-history download instructions.
BEAR (2025)	4,469 multimodal QA entries; robot online validation is additional.	Agent code; benchmark data and scoring scripts.
EmbodiedBench (2025a)	1,128 testing instances in four environments.	Code and configurations; benchmark datasets.
EmbodiedEval (2025)	328 tasks in 125 scenes.	Code; 328-entry test set and scene assets.
Embodied Arena (2025)	22 source benchmarks; a combined evaluated-instance count is not reported.	Platform and configurations; source datasets are downloaded separately.
CaP-X (2026)	Core: seven task types, each with 100 trials per tier. Extensions include 30 LIBERO-PRO task types and two BEHAVIOR tasks with 25 trials each.	Code, environments, and evaluation scripts. The repository lists 39 task types; the paper describes 187 platform tasks.
VLABench (2024)	100 task categories: 60 primitive and 40 composite; episode counts depend on the experiment.	Code and evaluation tracks; fixed-image VLM evaluation data.
WorldLines (2026b)	310 memory QA entries and 21 planning cases; the 62-entry ablation uses a QA subset.	The paper supplies prompts and examples; an author code/data release was not located.
Hard/Easy Vision (2026)	55 benchmarks, 34 capabilities, nine areas; total and per-benchmark evaluated-instance counts are not reported.	Results and qualitative examples; the repository contains a README and overview image, without evaluation scripts or a test manifest.
Embodied Agent Arena	**1,000 cases** from 32 established sources and GeoProbe, including 168 GeoProbe cases.	Harness and evaluation code; case index, observations, and episode initializations.

H Additional Cases Across the Five Capabilities

Figures A6–A10 connect instructions to predictions and execution. The public case atlas covers 181 tasks across 43 benchmark entries and subsets, with ten-model results on each task. The execution explorer links 460 episode records in 46 same-task groups across all ten models, including 189 detailed action/feedback traces (Appendix I). The selection spans task types, model disagreements, and successful and unsuccessful outcomes. Recordings, observation replays, and scene-only context are labeled separately.

(a) GeoProbe | Real Object Motion

Track the green-handled glass jug relative to the table.

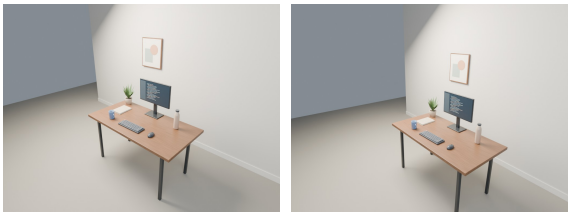
View 1 View 2

Model	Direction / cm
GPT-6 Astra	right / 2
GPT-6 Sol	right / 2
Fable 5.1	right / 4
Gemini 3.8 Flash	none / 0
Qwen 3.8 Max	right / 3
Qwen 3.5 397B	none / 0
Qwen 3.5 27B	left / 7
Opus 5.5	right / 2.5
DeepSeek-1017	up-right / 4

Reference: right, 2 cm

(b) GeoProbe | Camera Translation

Estimate camera B's position in the coordinate frame of camera A.



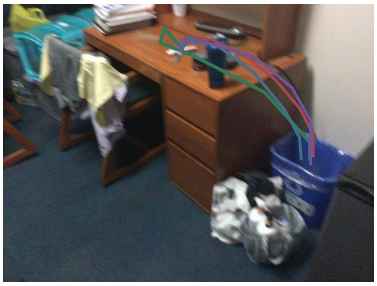
View 1 View 2

Model	Translation (cm)
GPT-6 Astra	[8, 0, -10]
GPT-6 Sol	[10, -10, 0]
Fable 5.1	[6, -3, 0]
Gemini 3.8 Flash	[25, 12, 0]
Qwen 3.8 Max	[10, 4, 0]
Qwen 3.5 397B	[15, 8, 5]
Qwen 3.5 27B	[8, 3, -12]
Opus 5.5	[15, -5, 0]
DeepSeek-1017	[6, -5, 12]

Reference translation: [12, 9, 0] cm

(c) TraceSpatial | 3D Trajectory

Predict a trajectory from the toothpaste to the blue waste bin.



GT Astra Fable Opus 5.5

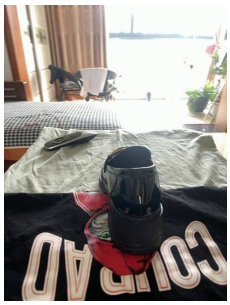
Image projection of 3D paths

Model	3D Fréchet (m)
GPT-6 Astra	0.525
GPT-6 Sol	0.634
Fable 5.1	0.826
Gemini 3.8 Flash	1.148
Qwen 3.8 Max	0.699
Qwen 3.5 397B	0.715
Qwen 3.5 27B	0.497
Opus 5.5	0.549
DeepSeek-1017	0.556

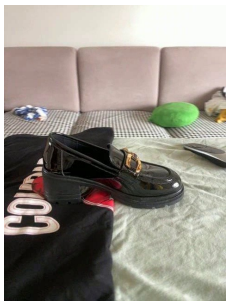
Figure A6. Geometry: displacement, reference frames, and trajectories. Selected first-run predictions. (a) The jug moves right by 2 cm relative to the table. (b) Camera translation uses camera A's frame, with reference $[12, 9, 0]$ cm; orientation is omitted. (c) TraceSpatial paths are projected into the image; the errors are 3D Fréchet distances in meters, distinct from aggregate trajectory RMSE. Reference, Astra, Fable, and Opus 5.5 paths are drawn, with all nine errors listed.

(a) MindCube | Viewpoint Change

In which direction did the camera move between these views?



View 1



View 2

Model	First-run answer
GPT-6 Astra	D
GPT-6 Sol	B
Fable 5.1	A
Gemini 3.8 Flash	A
Qwen 3.8 Max	A
Qwen 3.5 397B	C
Qwen 3.5 27B	A
Opus 5.5	A
DeepSeek-1017	A

Reference A: diagonally forward and right

(b) MMSI-Bench | Relative Direction

If the open cupboard faces north, where is the cooking area?

View 1

View 2

Model	First-run answer
GPT-6 Astra	A
GPT-6 Sol	A
Fable 5.1	A
Gemini 3.8 Flash	D
Qwen 3.8 Max	C
Qwen 3.5 397B	B
Qwen 3.5 27B	B
Opus 5.5	D
DeepSeek-1017	D

Reference A: northwest

(c) 3DSRBench | Object Orientation

Which side of the person is facing the camera?

Model	First-run answer
GPT-6 Astra	B
GPT-6 Sol	B
Fable 5.1	C
Gemini 3.8 Flash	B
Qwen 3.8 Max	B
Qwen 3.5 397B	C
Qwen 3.5 27B	B
Opus 5.5	C
DeepSeek-1017	C

Reference B: left

Figure A7. Spatial Reasoning: errors depend on the reference frame. Selected first-run answers without YOLOE. (a) MindCube asks about camera movement between two views: A is diagonally forward-right, B right, C left, and D diagonally forward-left. (b) MMSI-Bench displays two selected views from its multi-view input and asks for a direction anchored to the open cupboard: A is northwest, B southwest, C north, and D northeast. (c) 3DSRBench asks which side of the person faces the camera: A is front, B left, C back, and D right. Green letters match the source reference; red letters do not. Different agents succeed on different frame transformations.

Figure A8. Affordance: region shape and target identity are separate challenges. Direct polygon outputs on the same UMD, ReasonAff, and RAGNet tasks. Green fill is the reference mask; colored native contours are model predictions. Each row uses one shared crop for the reference, Astra, Fable 5.1, Opus 5.5, and DeepSeek-1017. IoU is computed over the full original image, not the crop. (a) Scissor outlines differ in extent and alignment. (b) Handle contours cover only part of the annotated pulling region. (c) Agents select different objects under the two-hand instruction. These are single-run examples; the project-page gallery provides all ten models, full images, and recorded polygon coordinates.

Boundary precision and task interpretation fail differently. On the scissors, Opus 5.5 reaches 0.627 IoU while DeepSeek-1017 reaches 0.128: both identify the tool, but their outlines recover different parts of its cutting region. On the RAGNet image, Opus 5.5 reaches 0.942 while the other three displayed models score zero, reflecting different target selections rather than a small boundary error.

A plausible contact is not a complete region. All four pulling predictions concentrate on the handle, while the annotation covers a larger region; IoU remains 0.096–0.140. This example explains why contact validity and region overlap measure different aspects of affordance, and why neither alone establishes successful physical execution.

(a) ALFWorld | Object and Light

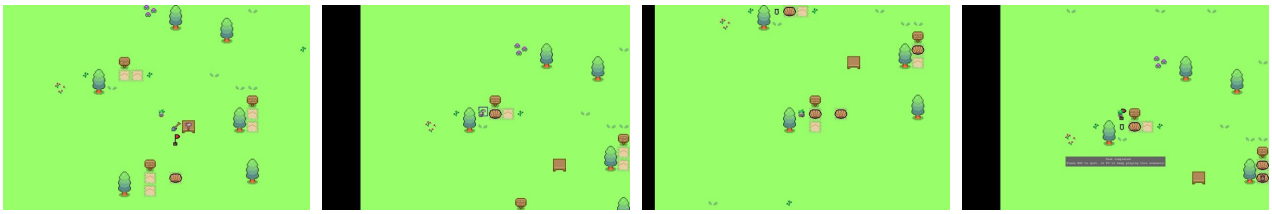
Turn on a lamp while holding the requested object.

obs. 1 obs. 3 obs. 6 obs. 8

Claude Opus 5.5 Reported success | 8 code rounds | 80.9 s

(b) DiscoveryWorld | Archaeology

Date the artifacts and flag the site containing the oldest one.

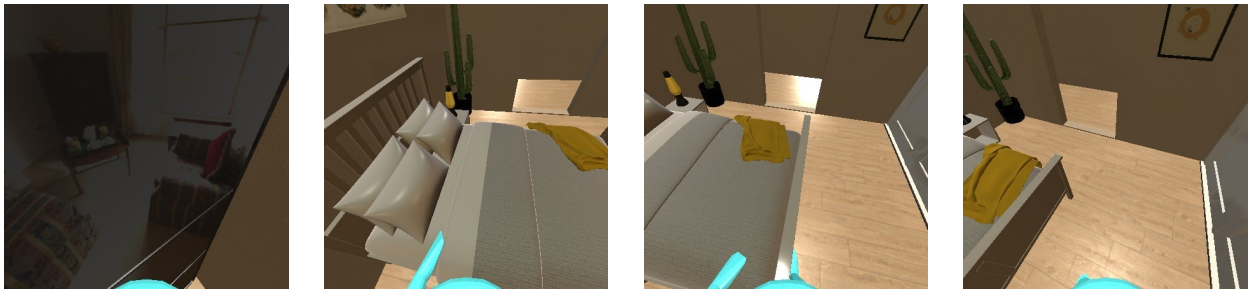


obs. 1 obs. 13 obs. 31 obs. 44

Claude Opus 5.5 Reported success | 7 code rounds | 167.9 s

(c) HumanCLAW | Sitting

Find the couch, approach until touching it, and sit.



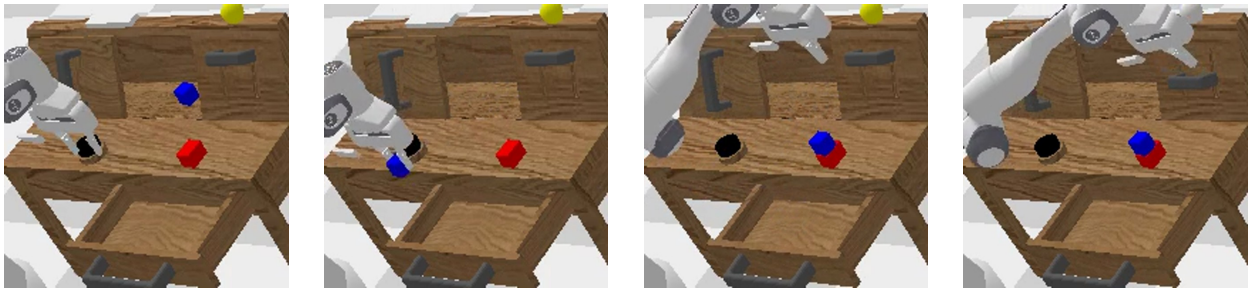
obs. 1 obs. 20 obs. 49 obs. 69

Claude Opus 5.5 Task failure | 70 code rounds | 933.6 s

Figure A9. Task Planning: a terminal episode need not be a completed task. Additional Opus 5.5 runs. (a) ALFWorld reports success after 8 code rounds under ceilings of 20 minutes, 8 rounds, 120 environment steps, 540 primitive calls, and 320k tokens. (b) DiscoveryWorld reports success after 7 rounds under 20 minutes, 8 rounds, 100 steps, 400 calls, and 320k tokens; its text/JSON interaction is illustrated with environment snapshots. (c) HumanCLAW stops after 70 rounds (933.6 s) without completing the couch-sitting task, under ceilings of 180 minutes, 101 rounds, 100 steps, 400 calls, and 2M tokens. The last case is a failed terminal state, not a wall-clock timeout. Frames are sampled saved observations, not continuous video.

(a) CALVIN | Sequence 0526

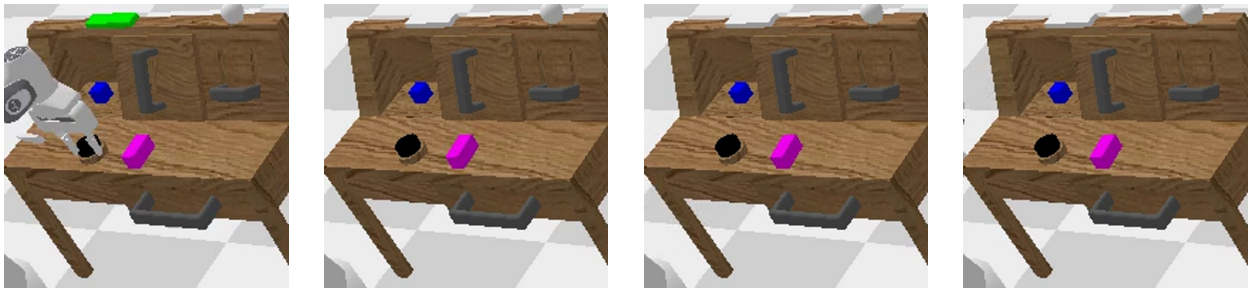
Complete the five instructed manipulation subgoals in sequence.



Claude Opus 5.5 Reported success | 40-round ceiling | 229.4 s

(b) CALVIN | Sequence 0557

Complete the five instructed manipulation subgoals in sequence.



Claude Opus 5.5 Round limit reached | 40 rounds | 978.6 s

(c) LIBERO-PRO | Mug and Microwave

Put the yellow-and-white mug in the microwave and close it.

Claude Opus 5.5 Round limit reached | 24 rounds | 1308.6 s

Figure A10. Manipulation: motion can continue without completing all goals. Additional Opus 5.5 runs through recorded source-native interfaces. (a) CALVIN sequence 0526 reports success; (b) sequence 0557 exhausts 40 model rounds. Both allow 7,000 primitive calls and 1.28M tokens. (c) LIBERO-PRO requires placing a mug inside the microwave and closing it; the run exhausts 24 model rounds, with ceilings of 256 primitive calls and 640k tokens. Outcomes come from the terminal reports. Frame labels index the encoded recording, whereas the seconds beside each outcome are wall-clock runtime. Different task instances and source interfaces are illustrated here, not a matched comparison of models.

H.1 Execution Across Models

Figure A11 pairs task instructions and chronological ALFRED frames with outcomes, model rounds, and elapsed runtime for the earlier nine-model cohort. Tasks are selected separately for each model; DeepSeek-1017 shares the spray-bottle task with Astra. The full ten-model CALVIN diagnostics appear in Figure 8.

Completion, interaction, and time. Table A37 groups identical task identifiers. Fable completes the RoboCasa drawer task in 20 rounds; four other agents exhaust their budgets at 24, spanning 215.5–3,223.2 s. On LIBERO-PRO, Qwen-397B exhausts its budget while three agents complete the placement. On ALFWorld, Sol succeeds in fewer rounds than Fable (6 versus 7), but takes more than twice as long (433.8 versus 197.5 s). Thus, neither interaction count nor elapsed time alone establishes complete, timely task execution.

Execution settings. ALFRED and ALFWorld use their recorded native interfaces. LIBERO-PRO and RoboCasa examples retain their individual source settings. These selected episodes illustrate execution; Table 2 aggregates the adopted evaluation assignments. Runtime includes model and tool execution. Budget exhaustion need not coincide with the model-round ceiling; the recorded termination is retained.

Table A37. Shared-task execution records. Rows within each block use the same task identifier and source interface. Rounds count recorded model responses; time is elapsed wall-clock runtime. ALFWorld allows 8 rounds, 120 environment steps, and 20 minutes. LIBERO-PRO and RoboCasa allow 24 rounds and 256 primitive calls, with token ceilings of 640k and 1.12M, respectively. These are selected episodes, not repeated-run averages.

Model	Rounds	Time (s)	Recorded outcome
ALFWorld: put a cleaned bar of soap on a counter			
GPT-6 Sol	6	433.8	Success
Claude Fable 5.1	7	197.5	Success
LIBERO-PRO: put the tomato sauce in the basket			
Claude Fable 5.1	16	1,553.5	Success
Gemini 3.8 Flash	15	429.0	Success
Qwen 3.5 397B-A17B	24	934.6	Budget exhausted
Claude Opus 5.5	20	2,376.8	Success
RoboCasa: open the left drawer			
GPT-6 Sol	24	735.5	Budget exhausted
Claude Fable 5.1	20	618.1	Success
Gemini 3.8 Flash	24	246.3	Budget exhausted
Qwen 3.8 Max	24	3,223.2	Budget exhausted
Qwen 3.5 27B	24	215.5	Budget exhausted

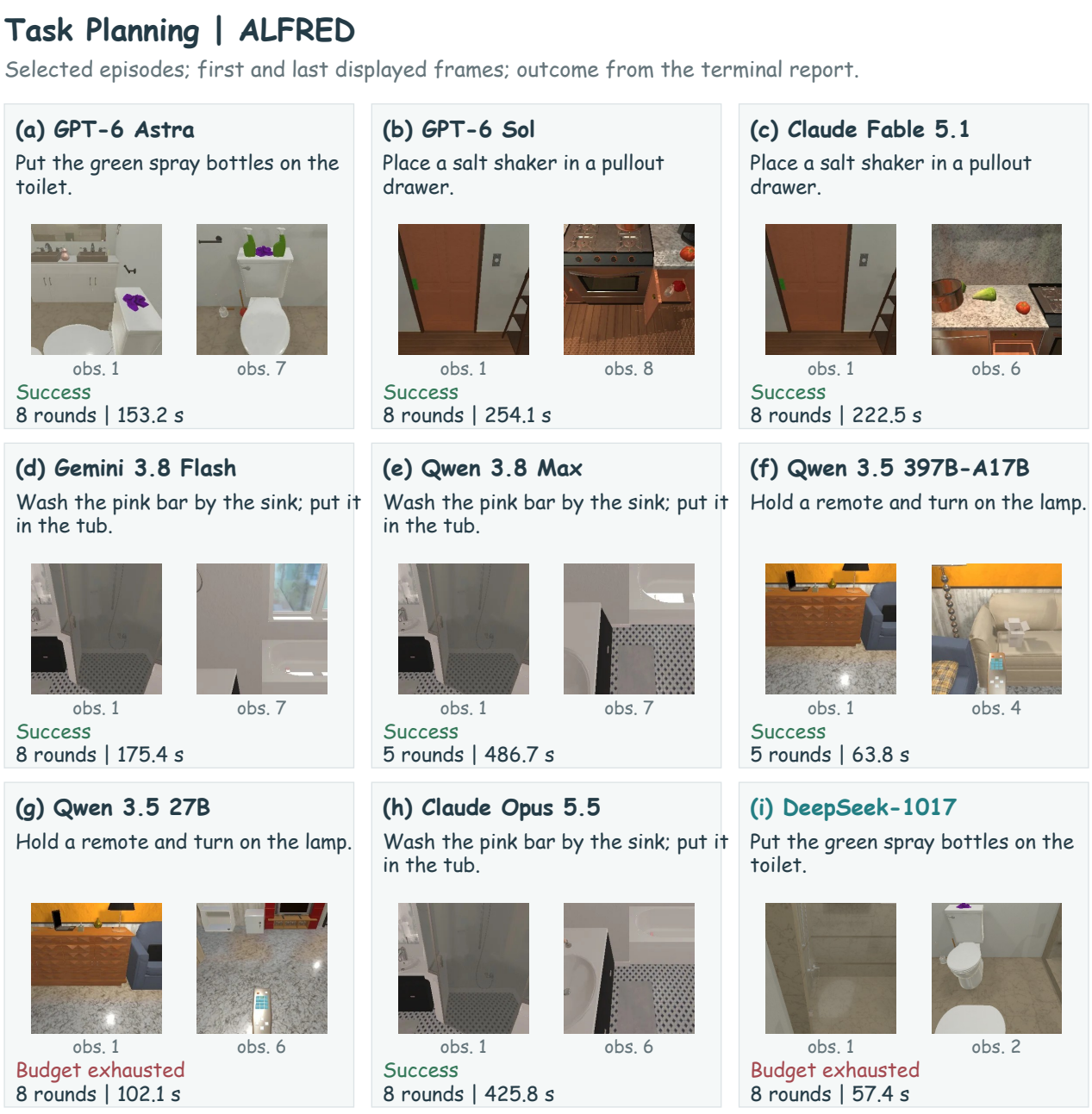


Figure A11. Nine-model task execution in ALFRED. Each panel shows the first and last displayed observations from an individual Baseline episode, with a shortened instruction, terminal outcome, rounds, and wall-clock runtime. All runs allow 8 model rounds, 120 environment steps, and 20 minutes. DeepSeek-1017 exhausts 8 rounds on the same spray-bottle task as Astra. The two Qwen models in panels (f,g) share the remote-and-lamp task: Qwen-397B reports success after 5 rounds, whereas Qwen-27B exhausts its budget at 8. Other panels cover separately selected tasks. Saved observations preserve temporal order with unequal intervals.

I From Recorded Actions to Complete Tasks

This analysis connects the aggregate scores to task overlap, resource allocation, and the state changes inside an episode. We use 1,570 adopted Planning records: one episode per model on the same 157 assignments for all ten models. The online analysis provides source filters, model-pair comparisons, and downloadable episode metrics. Its execution explorer links 460 episode records in 46 same-task groups, including 189 detailed action/feedback traces, to model decisions, primitive actions, returned feedback, and terminal outcomes.

I.1 Complementary Strengths and Shared Failures

Astra completes 126 tasks. Taking the union of all ten models' successes covers 142 tasks, adding 16 beyond Astra; 15 tasks remain unsuccessful for every model. Only eight tasks are completed by all ten (Figure A12a). The common failure set contains two ALFRED, two ALFWorld, nine HumanCLAW, and two ScienceWorld tasks. The HumanCLAW subset includes all eight sitting tasks and one television task. This overlap distinguishes tasks on which models complement one another from those on which model selection alone does not resolve the observed failure. This union describes complementarity after execution, without testing a model-selection policy.

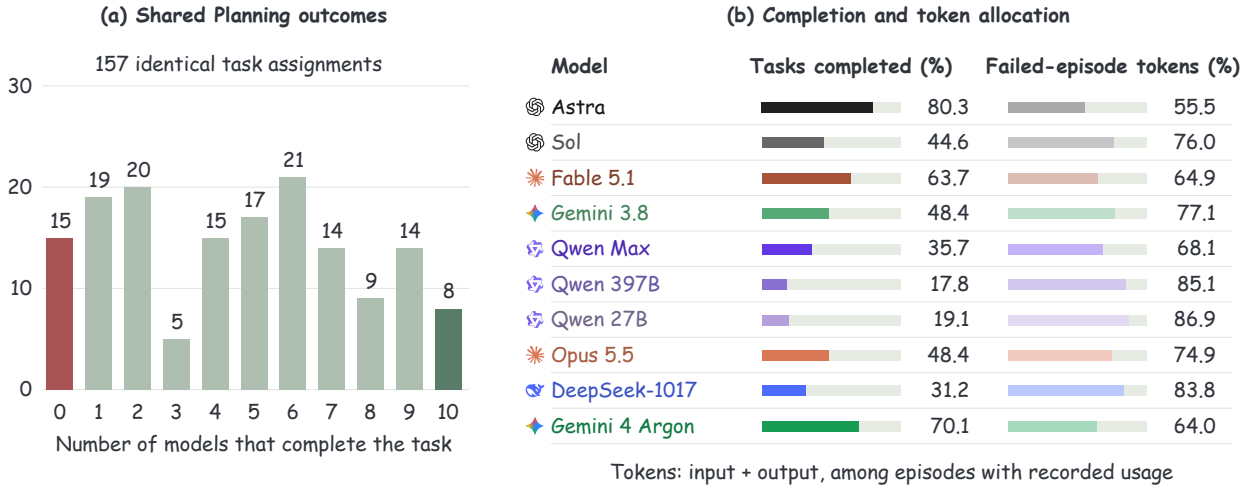


Figure A12. Shared Planning failures and the resources spent on them. (a) Number of tasks completed by exactly zero through ten models, over the same 157 assignments. (b) Completed-task percentage and the share of recorded tokens spent in unsuccessful episodes. Tokens sum recorded input and output usage; Table A38 gives usage coverage. All model names use the identifiers in Appendix B.2.

I.2 Completion, Latency, and Token Allocation

Unsuccessful episodes consume 55.5–86.9% of each model's recorded Planning tokens (Figure A12b). Astra completes 80.3% of its assignments, yet its 31 unsuccessful episodes account for 55.5% of recorded tokens. The mean unsuccessful episode uses 5.1 times as many tokens as the mean successful episode (591,162 versus 116,435 tokens). These attempts dominate token use despite being a minority of its tasks. The median episode duration ranges from 57.1 s for DeepSeek-1017 to 280.7 s for Sol (Table A38). Short duration has to be read with completion: it can describe an efficient solution or an early unsuccessful stop.

On the same HumanCLAW chair-navigation task, Gemini 4 Argon completes in 17 rounds and 437.3 s, Astra in 18 rounds and 470.7 s, and DeepSeek-1017 in 65 rounds but 405.7 s. Fewer decisions need not imply lower elapsed time. In the selected BEHAVIOR-1K toolbox task, none

Table A38. Planning resources and recovery over 157 tasks per model. Time is episode wall time in seconds; IQR spans the 25th–75th percentiles. Failed tokens is the percentage of recorded tokens spent in unsuccessful episodes; usage gives the number of episodes with token records, out of 157. Recovery counts completed episodes after an execution error, divided by episodes containing an error.

Model	Completed	Median (s)	IQR (s)	Failed tokens (%)	Usage	Recovery
Astra	126/157	231.7	153.2–361.8	55.5	157	1/1
Sol	70/157	280.7	199.9–427.1	76.0	157	1/6
Fable	100/157	109.5	62.0–201.4	64.9	157	6/25
Gemini 3.8	76/157	61.8	28.6–175.9	77.1	157	4/24
Qwen-Max	56/157	273.6	192.7–407.3	68.1	133	10/50
Qwen-397B	28/157	98.7	58.2–136.4	85.1	144	2/55
Qwen-27B	30/157	77.9	55.5–127.8	86.9	145	3/43
Opus 5.5	76/157	135.6	105.3–211.0	74.9	157	48/105
DeepSeek-1017	49/157	57.1	36.1–78.5	83.8	157	9/50
Gemini 4 Argon	110/157	159.5	51.5–612.1	64.0	157	16/28

of the ten models completes the goal. Gemini 4 Argon uses 603,393 tokens before the next round is blocked by the 640,000-token ceiling, after 37.5 minutes of a 180-minute limit. The stopping resource and the unmet goal conditions explain why the episode ends.

I.3 Recovery Matters When It Changes the Task State

Gemini 4 Argon completes 16 of 28 Planning episodes containing an execution error; Opus 5.5 completes 48 of 105. These counts distinguish encountering errors from recovering and finishing. Error records include malformed code and rejected API use in the evaluated interface. The Archaeology Dating task shows what successful recovery entails (Figure A13). Opus 5.5's first two rounds fail before any environment action. It subsequently collects the tools, excavates the sites, reads the artifact ages, and places the flag at the oldest site. The native score reaches 1.0 in round seven. Gemini 4 Argon encounters an error in round two, resumes progress in round three, and completes in round eight. DeepSeek-1017 reaches 0.5 by round five, but its last three rounds end in code errors without further environment steps. A corrected response advances the episode only when the environment state moves toward the goal.

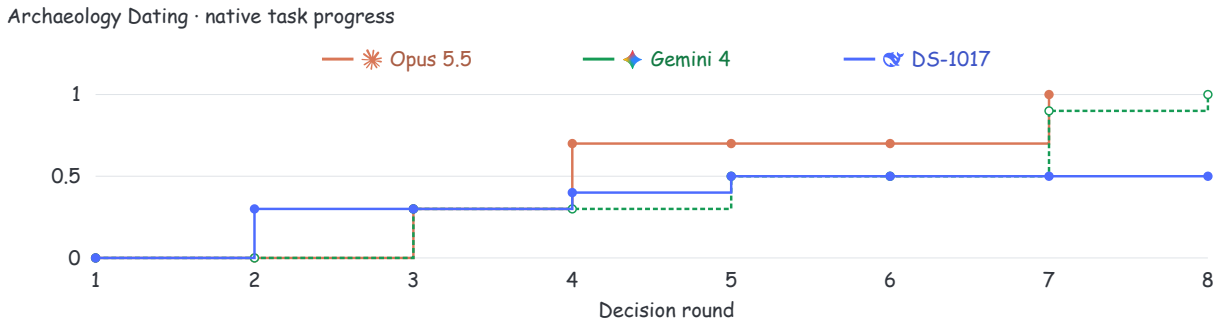


Figure A13. Recovery must change the task state. Archaeology Dating: Opus 5.5 and Gemini 4 Argon recover from early code errors and reach native score 1.0; DeepSeek-1017 stalls at score 0.5. Points are recorded decision rounds, with steps showing the latest native score.

I.4 Physical Activity and Verified Completion

CALVIN's adopted sequence records separate executable code from successful control (Figure 8). Among 65 episodes containing execution errors, 62 later run code successfully, but only three subsequently complete all five ordered goals. Table A30 gives the per-model counts.

Resuming interaction clears an immediate obstacle; the remaining sequence still requires preserving achieved conditions and satisfying the next one.

ALFRED separates activity from goal satisfaction: for cleaning a soap bar and placing it in the bathtub, Gemini 4 Argon satisfies all three conditions in six rounds; DeepSeek-1017 ends at one of three after eight. On RoboCasa's left-drawer task, Gemini 4 Argon finishes in 13 rounds despite an error in round ten, whereas Sol exhausts 24 rounds without the required state. On charger insertion, Gemini 4 Argon stops at the token limit after 23 native steps, still 15.9 cm from the goal. These executions retain their source settings; Table 2 reports the adopted aggregate population. A generalist must preserve achieved conditions, verify the remaining ones, and direct further interaction toward them.

I.5 Task Alignment and Measurement

Planning assignments are aligned by the shared task ID. HumanCLAW wrapper-specific IDs are checked against native task identity. The 157 adopted Gemini 4 Argon Planning reports and 109 gallery executions match the earlier result digests; historical retries are excluded. Episode time uses the reported total duration, or the campaign duration when supplied, otherwise the last recorded loop time. Token fractions exclude missing usage rather than substituting zero. Recovery requires native final success after an execution error; budget guards and timed-out rounds are excluded from the error count.

In the video explorer, code cells are matched to recorded rounds using timestamps and execution status. The Gemini 4 Argon import aligns 773 native code cells; four unmatched cells remain separately inspectable. The 189 detailed episodes contain 2,856 rounds and 17,130 retained primitive calls; the complete explorer also includes 271 round-summary records. Per-round curves show native scores or satisfied goal conditions where those quantities are recorded. Encoded-video time and episode wall time remain distinct: recordings illustrate motion, while trace timestamps locate decisions and measured progress. Each displayed comparison links to its original reports, task IDs, and execution settings.

Reading the broader case collection. Static panels show first-run predictions; spatial cases exclude YOLOE. In point-and-box panels, contact labels test reference-mask membership and box IoU compares rectangles. Direct-polygon panels compare rasterized predictions with reference masks (Appendix C.5). Execution panels retain their recorded source interfaces and observation schedules. Opus 5.5 CALVIN uses automatic per-round RGB and its recorded ceilings, as verified in Appendix E.1. Four earlier HumanCLAW recordings with conflicting labels and feedback, and the Qwen-397B proximity demonstration, are excluded from evaluation. Opus 5.5 HumanCLAW statistics use the final 13 records in Appendix C.3. Planning panels show ordered saved observations; DiscoveryWorld snapshots illustrate text/JSON interaction. Manipulation frame labels use encoded-video time.

J Gemini 4 Argon: Coverage and Execution Settings

Gemini 4 Argon uses `gemini-4-argon`, recorded on 20–21 October 2026. The release contains 3,192 adopted trial records covering all 1,000 arena cases, plus 70 polygon submissions. Non-tracing Geometry, Spatial Reasoning, and point/box Affordance use five repetitions; tracing, polygons, Planning, and Manipulation use one. Continuous metrics average valid outputs within each run and then weight the five runs equally. Spatial inputs use corrected 3DSRBench image flips and native labels, without YOLOE. Tables 2 and A8 include these results alongside the unchanged earlier models.

Gemini 4 Argon scores 71.6% ± 0.7 Spatial Pass@1, 0.356 ± 0.009 box–mask IoU, and 0.441 polygon IoU. Its 3D path RMSE is 0.310 m, with 24/56 joint successes (42.9%). Despite its leading trajectory precision, 32 of 56 paths violate at least one required constraint. In Planning it completes 110/157 tasks (70.1%): 14/20 ALFRED, 16/20 ALFWorld, 6/10 DiscoveryWorld, 2/13 HumanCLAW, 52/73 ScienceWorld, and 20/21 VirtualHome. Both HumanCLAW successes are navigation; all eight sitting tasks remain incomplete.

Manipulation uses current RGB at every decision and the source-specific budgets below. It completes 80/182 binary tasks (44.0%); Anymal's continuous-reward case remains separate. CALVIN averages 3.30 consecutive goals with 4/10 complete sequences. Figure 8 compares verified CALVIN progress across all ten models. The execution-log analysis in Appendix I covers all ten models.

Table A39. Gemini 4 Argon Manipulation: task success and per-episode ceilings. Tokens include reasoning and context; rounds are decision interactions. One episode per binary assignment.

Source	Success	SR (%)	Rounds	Tokens (k)	Time (s)
CALVIN	4/10	40.0	40	1,280	7,200
BEHAVIOR-1K	0/5	0.0	24	640	10,800
LIBERO-PRO	8/10	80.0	24	640	10,800
robosuite	5/5	100.0	24	640	10,800
CLIPort	7/18	38.9	24	960	3,600
ManiSkill	5/9	55.6	32	640	3,600
RLBench	8/10	80.0	16	640	3,600
RoboCasa	2/10	20.0	24	1,120	3,600
RoboCasa365	1/10	10.0	24	1,120	3,600
RoboTwin 2.0	14/50	28.0	16	640	3,600
RoboWits	1/10	10.0	24	640	3,600
VIMA	9/10	90.0	16	640	3,600
VLABench	16/25	64.0	16	640	3,600